EXAM PACK

EEE421

Solid State Devices



Exam Pack

Session 2019–20 *Solutions and formula sheet of the 4th Year 1st Semester Final Examination, 2023, and its two term tests*

IN THIS PACK

1 PAST PAPER SOLUTIONS
Question Paper Solutions 2019–20 — pages 3--108 — 106 pages

2 FORMULA SHEET
Formula Sheet — pages 109--119 — 11 pages

Tap a document to open it. Each one keeps its own cover, contents and links.

Prepared by Nur Mohammad

November 2026

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1 PAST PAPER SOLUTIONS
Question Paper Solutions 2019–20

pages 3--108

Question Paper 5
1(a) Three Classes of Solids 13
1(b) Atoms in a Face-Centered Cubic Lattice 15
1(c) Average Thermal Speed of Electrons and Holes 17
1(d) Schottky Barrier and Ohmic Contact 21
2(a) Carrier Concentrations in a Compensated Semiconductor 25
2(b) Hall Effect Measurement on a Silicon Sample . 29
2(c) Carrier Concentration Versus Temperature . . . 32
2(d) Energy Band Diagram of a Compensated Semiconductor 34
3(a) Density of States, Fermi Function and Carrier Distribution in n-Type Material 37
3(b) How the Barrier Potential Keeps the p and n Regions in Equilibrium 40
3(c) Dielectric Relaxation Time 42
3(d) Donor Concentration and Mobility of Compensated n-Type Silicon 45
4(a) Effect of Temperature and Doping on Mobility 49
4(b) Ambipolar Transport 52
4(c) Built-In Potential, Depletion Width and Peak Field at 300 K and 80 K 56
4(d) Time and Space Dependence of Excess Carriers 59
5(a) Steady-State Excess Carriers Generated at the Origin 63
5(b) Ideal PN Junction Current 66
5(c) Quasi-Fermi Energy Levels 70
5(d) Built-In Potential, Depletion Width and Peak Field Under Reverse Bias 73
6(a) Inversion Charge and Threshold Voltage of a Nanoscale MOSFET 76
6(b) Operating Mode of an NPN Transistor 80
6(c) Advantages of the MOSFET 82
6(d) Base Doping and Emitter Injection Efficiency . 83
7(a) Electron and Hole Concentrations from Charge Neutrality 87
7(b) Variation of the Fermi Level in an Extrinsic Semiconductor 90
7(c) Minimum Band Gap of a Designed n-Type Semiconductor 93
7(d) Occupation Probability at the Conduction Band Edge and the Carrier Concentrations 95
8(a) Steady-State Excess Carriers in an Infinite p-Type Crystal 98
8(b) The Ambipolar Transport Equation 101
8(c) Generation and Recombination Processes 105

2 FORMULA SHEET
Formula Sheet

pages 109--119

1 Crystals and thermal motion 112
2 Carriers in equilibrium 112
3 Dopants and charge neutrality 113
4 Transport 113
5 Excess carriers 114
6 PN junction 115
7 MOSFET and BJT 115
8 Key formulas 117
9 Symbol index 119

PAST PAPER SOLUTIONS

EEE421

Solid State Devices

Question Paper Solutions

Session 2019–20

4th Year 1st Semester Final Examination, 2023, with Term Tests 1 and 2

QUESTION MAP

Q1	Q2	Q3	Q4	Q5	Q6
25 marks	25 marks	25 marks	25 marks	25 marks	25 marks

Q7	Q8
20 marks	20 marks

EACH ANSWER

Theory → Answer → Keyword Explanation → Notes and Remarks → Final Concise Answer

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November 2026

BACK OF THE COVER

The paper at a glance

3 h	100	4 of 6	45 min
Total time	Total marks	Questions to answer	Per question

Time = marks × 1.8 min (180 min for 100 marks); a part with its own time (a term test) uses that. Tick the questions you will answer; each sub-part links to its solution.

Q	Sub-part	Marks	Time	Pick
Part-A · Answer any two of the following questions				
Q1	(a) Three Classes of Solids	6	11 min	☐
	(b) Atoms in a Face-Centered Cubic Lattice	6	11 min	
	(c) Average Thermal Speed of Electrons and Holes	7	13 min	
	(d) Schottky Barrier and Ohmic Contact	6	11 min	
Q2	(a) Carrier Concentrations in a Compensated Semiconductor	6	11 min	☐
	(b) Hall Effect Measurement on a Silicon Sample	8	14 min	
	(c) Carrier Concentration Versus Temperature	5	9 min	
	(d) Energy Band Diagram of a Compensated Semiconductor	6	11 min	
Q3	(a) Density of States, Fermi Function and Carrier Distribution in n-Type Material	8	14 min	☐
	(b) How the Barrier Potential Keeps the p and n Regions in Equilibrium	4	7 min	
	(c) Dielectric Relaxation Time	7	13 min	
	(d) Donor Concentration and Mobility of Compensated n-Type Silicon	6	11 min	
Part-B · Answer any two of the following questions				
Q4	(a) Effect of Temperature and Doping on Mobility	6	11 min	☐
	(b) Ambipolar Transport	7	13 min	
	(c) Built-In Potential, Depletion Width and Peak Field at 300 K and 80 K	6	11 min	
	(d) Time and Space Dependence of Excess Carriers	6	11 min	
Q5	(a) Steady-State Excess Carriers Generated at the Origin	7	13 min	☐
	(b) Ideal PN Junction Current	8	14 min	
	(c) Quasi-Fermi Energy Levels	4	7 min	
	(d) Built-In Potential, Depletion Width and Peak Field Under Reverse Bias	6	11 min	
Q6	(a) Inversion Charge and Threshold Voltage of a Nanoscale MOSFET	12	22 min	☐
	(b) Operating Mode of an NPN Transistor	6	11 min	
	(c) Advantages of the MOSFET	3	5 min	
	(d) Base Doping and Emitter Injection Efficiency	4	7 min	
Term Test 1 · 5 marks · 40 min				
Q7	(a) Electron and Hole Concentrations from Charge Neutrality	5	40 min	☐
	(b) Variation of the Fermi Level in an Extrinsic Semiconductor	5	40 min	
	(c) Minimum Band Gap of a Designed n-Type Semiconductor	5	40 min	
	(d) Occupation Probability at the Conduction Band Edge and the Carrier Concentrations	5	40 min	
Term Test 2 · 6 marks · 35 min				
Q8	(a) Steady-State Excess Carriers in an Infinite p-Type Crystal	6	35 min	☐
	(b) The Ambipolar Transport Equation	7	41 min	
	(c) Generation and Recombination Processes	7	41 min	

Formulas, symbols and constants follow Neamen, Semiconductor Physics and Devices (4th ed.).

Shahjalal University of Science and Technology

Department of Electrical and Electronic Engineering
4th Year 1st Semester Final Examination, 2023
Course Code: EEE 421, Session: 2019-2020
Course Title: Solid State Devices

Full Marks: 100 **Credit: 3.0** **Exam Duration: 3 Hours**

Part-A

*[Answer any **two** of the following questions]*

1. (a) Describe three classifications of solids-amorphous, polycrystalline, and single crystal. 6
   (b) The lattice constant of a face-centered cubic lattice is 0.425 nm. Determine the 6
      i. effective number of atoms per unit cell
      ii. volume density of atoms
   (c) If the free electrons in a semiconductor have effective mass $m_c = 0.1m_o$, where m_o is the mass of the electron in vacuum, what is their average speed at $T = 300K$? If the free holes in the same material have effective mass $m_v = 1.5m_o$, what is their average speed at the same temperature? Assume a Maxwell-Boltzmann distribution of the electron energies. 7
   (d) Explain Schottky barrier. How it differs from ohmic contact? Explain. 6

2. (a) Starting from charge neutrality condition, Find thermal equilibrium carrier concentration at compensated semiconductor in terms of N_a, N_d and n_i 6
   (b) Consider silicon at $T = 300$ K. A Hall effect device is fabricated with the following geometry: $d = 5 \times 10^{-3}$ cm, $W = 5 \times 10^{-2}$ cm and $L = 0.50$ cm. The electrical parameters measured are: $I_x = 0.50$ mA, $V_x = 1.25$ V, and $B_z = 650$ *gauss* $= 6.5 \times 10^{-2}$ *tesla*. The Hall field is $E_H = -16.5$ mV cm^{-1}. Determine: 8
      i. the Hall voltage
      ii. the conductivity type
      iii. the majority carrier concentration and
      iv. the majority carrier mobility
   (c) Using proper graph, explain how carrier concentration varies with temperature. 5
   (d) Sketch and briefly describe energy band diagram of compensated semiconductor. 6

3. (a) Sketch and explain density of states functions, Fermi distribution, and carrier concentration distribution curves of n-type material in a single graph. 8
   (b) Explain how barrier potential maintains equilibrium p, n regions. 4
   (c) Explain dielectric relaxation time. Find the equation for dielectric relaxation. 7
   (d) Consider a compensated n-type silicon at $T = 300$ K with a conductivity of ρ [*sic*: should be σ] $= 16\ (\Omega - \text{cm})^{-1}$ and an acceptor doping concentration of 10^{17} cm^{-3}. Determine the doner [*sic*: should be donor] concentration and the electron mobility. 6

Part-B

*[Answer any **two** of the following questions]*

4. (a) With proper graphs, explain how mobility is affected by temperature and doping concentration. 6

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(b) Define ambipolar transport. Find ambipolar transport equation. Hence show that diffuse and drift behavior of excess majority carrier is determined by excess minority carrier. 7

(c) For a pn junction with $N_a = 1.5 \times 10^{18}\text{cm}^{-3}$, $N_d = 1.5 \times 10^{16}\text{cm}^{-3}$ and $n_i = 2 \times 10^{10}\text{cm}^{-3}$ 6

i. calculate the built-in potential barrier V_{bi}
ii. calculate the depletion region with [*sic*: should be width] and E_{max} (Assume $\epsilon_s = 11.7\epsilon_o$) and
iii. what will be the width at liquid nitrogen temperature (80K).

(d) With proper graphs, explain time and spatial dependence of excess carrier concentration. 6

5. (a) Consider a p-type semiconductor that is homogeneous and infinite in extent. Assume a zero applied electric field. For a one-dimensional crystal, assume that excess carriers are being generated at $x = 0$ only. The excess carriers being generated at $x = 0$ will being [*sic*: should be will begin] diffusing in both the $-x$ and $+x$ directions. Calculate the steady-state excess carrier concentration as a function of x. 7

(b) Find ideal pn junction current and sketch ideal electron and hole current components. 8

(c) Consider an n-type semiconductor at $T = 300$ K with carrier concentrations of $n_o = 10^{15}$ cm^{-3}, $n_i = 10^{10}$ cm^{-3} and $p_o = 10^5$ cm^{-3}. In non-equilibrium, assume that the excess carrier concentrations are $\delta n = \delta p = 10^{-13}$ [*sic*: should be 10^{13}] cm^{-3}. Calculate the quasi-Fermi energy levels. 4

(d) An abrupt silicon pn junction has dopant concentrations of $N_a = 2 \times 10^{16}$ cm^{-3} and $N_d = 2 \times 10^{15}$ cm^{-3} at $T = 300$ K. Calculate: 6

i. V_{bi}
ii. W at $V_R = 0$ and $V_R = 8$ V and
iii. the maximum electric field in the space charge region at $V_R = 0$ and $V_R = 8$ V

6. (a) The body of a nanoscale N-channel MOSFET is doped at $N_A = 5^{18}$ [*sic*: should be 5×10^{18}] cm^{-3}. 12

i. Determine the maximum inversion-layer charge per unit area (Q_I) that can be reached so that the electric field in the oxide remains below $E_{ox} = 0.6$ V/nm. Neglect any interface-trap and gate-oxide charge.
ii. If the channel length and the channel width are $L = 50$ nm and $W = 500$ nm, respectively, determine how many electrons are creating this inversion layer.
iii. Calculate the threshold voltage of a MOSFET having gate-oxide thickness $t_{ox} = 1$ nm and flat-band voltage $V_{FB} = 1.0$ V.
iv. What gate voltage is needed to form the maximum inversion-layer charge?

(b) Determine the mode of operation of an NPN BJT with $\beta \approx 450$, if it is known that: 6

i. $V_{BE} = 0.7V$, $V_{CE} = 5.2V$
ii. $V_{BE} = 0.7V$, $V_{CE} = 0.2V$
iii. $V_{BE} = 0.8V$, $V_{BC} = 0.3V$
iv. $V_{BE} = -0.8V$, $V_{BC} = 0.7V$
v. $I_C = 455mA$, $I_B = 1mA$
vi. $I_C = 455mA$, $I_E = 502mA$

(c) Mention some advantages of using MOSFET. 3

(d) Does the doping level in the base of a BJT affect the emitter efficiency γ_E? If so, why? 4

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Some useful equations

$f_F(E) = \frac{1}{1 + exp\left(\frac{E-E_F}{kT}\right)}$	$g_c(E) = \frac{4\pi(2m_n^*)^{3/2}}{h^3}\sqrt{E - E_c}$
$\epsilon_{Si} = 11.7\epsilon_0$	$\frac{m^*}{m_0} = 0.26(P\ at\ Si)$
$n_0 = N_c\ exp\left[\frac{-(E_c - E_F)}{kT}\right]$	$n_0 = n_i\ exp\left[\frac{E_F - E_{Fi})}{kT}\right]$
$n_i^2 = N_c N_v exp\left[-\frac{E_c - E_v}{kT}\right] = N_c N_v exp\left[-\frac{E_g}{kT}\right]$	$n_d = \frac{N_d}{1 + \frac{1}{2}exp\left(\frac{E_d - E_F}{kT}\right)}$
$n_0 = \frac{N_d - N_a}{2} + \sqrt{\frac{(N_d - N_a)^2}{2} + n_i^2}$	$E_F - E_{Fi} = kT\ ln\left(\frac{n_0}{n_i}\right)$
$J = en\mu_n E + ep\mu_p E + eD_n\Delta n - eD_p\Delta p$	$W = \left\{\frac{2\epsilon_s V_{bi}}{e}\left[\frac{N_a + N_d}{N_a N_d}\right]\right\}^{\frac{1}{2}}$
$D_p\frac{\partial^2(\delta p)}{\partial x^2} - \mu_p\left(E\frac{\partial(\delta p)}{\partial x} + p\frac{\partial E}{\partial x}\right) + g_p - \frac{p}{\tau_{pt}} = \frac{\partial(\delta p)}{\partial t}$ $D_n\frac{\partial^2(\delta n)}{\partial x^2} + \mu_n E\frac{\partial(\delta n)}{\partial x} + g' - \frac{\delta}{\tau_{n0}} = \frac{\partial(\delta n)}{\partial t}$	$\frac{\partial p}{\partial t} = -\frac{\partial F_p^+}{\partial x} + g_p - \frac{p}{\tau_{pt}}$ $V_{bi} = \frac{kT}{e}ln\left(\frac{N_a N_d}{n_i^2}\right)$
$\delta p(x,t) = \frac{e^{-\frac{t}{\rho_{p0}}}}{(4\pi D_p t)^{1/2}}exp\left[\frac{-(x - \mu_p E_0 t)^2}{4D_p t}\right]$	$x_n = \left\{\frac{2\epsilon_s V_{bi}}{e}\left[\frac{N_a}{N_d}\right]\left[\frac{1}{N_a + N_d}\right]\right\}^{1/2}$
$E_{max} = -\frac{e}{\epsilon_s}\left\{\frac{2\epsilon_s V_{bi}}{e}\frac{N_a N_d}{N_a + N_d}\right\}^{\frac{1}{2}}$	$C' = \left\{\frac{e\epsilon_s N_a N_d}{2(V_{bi} + V_R)(N_a + N_d)}\right\}^{\frac{1}{2}}$
$\delta n_p(x) = n_p(x) - n_{p0} = n_{p0}\left[exp\left(\frac{eV_a}{kT}\right) - 1\right]exp\left(\frac{x_p + x}{L_n}\right)$	$p_n(x_n) = p_{n0}\ exp\left(\frac{eV_a}{kT}\right)$
$J = J_p(x_n) + J_n(-x_p) = \left[\frac{eD_p p_{n0}}{L_p} + \frac{eD_n n_{p0}}{L_n}\right]\left[exp\left(\frac{eV_a}{kT}\right) - 1\right]$	
$\delta n_B(x) \approx \frac{n_{B0}}{x_B}\left\{\left[exp\left(\frac{eV_{BE}}{kT}\right) - 1\right](x_B - x) - x\right\}$	$\delta p_C(x'') \equiv p_{C0}\ exp\left(-\frac{x''}{L_C}\right)$
$\delta p_E(x') \approx \frac{p_{E0}}{x_E}\left[exp\left(\frac{eV_{BE}}{kT}\right) - 1\right](x_E - x')$	

Term Test 1 (its questions (a) to (d) are 7(a) to 7(d))

[EEE 421 (2019-20) TT1, Time: 40 minutes, Marks: $5 \times 4 = 20$*]*

7. (a) Starting from charge neutrality condition, derive expressions for n_0 and p_0 in terms of N_a, N_d, and n_i. 5

   (b) With proper equation and graph, explain variation of Fermi energy level in extrinsic semiconductor. 5

   (c) A special semiconductor material is to be "designed". The semiconductor is to be n-type and doped with $1 \times 10^{15} cm^{-3}$ donor atoms. Assume complete ionization and assume $N_a = 0$. The effective density of states functions are given by $N_c = N_v = 1.5 \times 10^{19} cm^{-3}$ and are independent of temperature. A particular semiconductor device fabricated with this material requires the electron concentration to be no greater than $1.01 \times 10^{15} cm^{-3}$ at $T = 400\ K$. What is the minimum value of the bandgap energy? 5

   (d) (i) Calculate the probability of an electron to be occupied at the lowest energy state of the conduction band of silicon when the Fermi energy is 0.3 eV below the conduction band at room temperature. (ii) Calculate the thermal electron concentration n_0. (iii) What

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is the thermal hole concentration p_0? (Room temperature N_c is 2.8×10^{19} cm^{-3}, n_i is 1.5×10^{10} cm^{-3}) 5

Term Test 2 (its questions (a) to (c) are 8(a) to 8(c))

[EEE 421 (2019-20) TT2, Time: 35 minutes, Marks: 6+7+7 = 20]

8. (a) Consider a p-type semiconductor that is homogeneous and infinite in extent. Assume a zero applied electric field. For a one-dimensional crystal, assume that excess carriers are being generated at $x = 0$ only. The excess carriers being generated at $x = 0$ will begin diffusing in both the $-x$ and $+x$ directions. Calculate the steady-state excess carrier concentration as a function of x. 6

   (b) Define ambipolar transport. Find ambipolar transport equation. Hence show that diffuse and drift behavior of excess majority carrier is determined by excess minority carrier. 7

   (c) Briefly describe different types of generation and recombination processes. Use appropriate diagram if necessary. 7

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Contents

1(a) Three Classes of Solids **11**
1 Theory . . . 11
2 Answer . . . 12
3 Keyword Explanation . . . 13
4 Notes and Remarks . . . 13

1(b) Atoms in a Face-Centered Cubic Lattice **13**
1 Theory . . . 13
2 Answer . . . 14
3 Keyword Explanation . . . 15
4 Notes and Remarks . . . 15

1(c) Average Thermal Speed of Electrons and Holes **15**
1 Theory . . . 15
2 Answer . . . 17
3 Keyword Explanation . . . 18
4 Notes and Remarks . . . 18

1(d) Schottky Barrier and Ohmic Contact **19**
1 Theory . . . 19
2 Answer . . . 20
3 Keyword Explanation . . . 21
4 Notes and Remarks . . . 22
5 Final Concise Answer . . . 22

2(a) Carrier Concentrations in a Compensated Semiconductor **23**
1 Theory . . . 23
2 Answer . . . 25
3 Keyword Explanation . . . 26
4 Notes and Remarks . . . 26

2(b) Hall Effect Measurement on a Silicon Sample **27**
1 Theory . . . 27
2 Answer . . . 28
3 Keyword Explanation . . . 29
4 Notes and Remarks . . . 29

2(c) Carrier Concentration Versus Temperature **30**
1 Theory . . . 30
2 Answer . . . 31
3 Keyword Explanation . . . 32
4 Notes and Remarks . . . 32

2(d) Energy Band Diagram of a Compensated Semiconductor **32**
1 Theory . . . 32
2 Answer . . . 33
3 Keyword Explanation . . . 33
4 Notes and Remarks . . . 34
5 Final Concise Answer . . . 34

3(a) Density of States, Fermi Function and Carrier Distribution in n-Type Material **35**

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1 Theory . . . 35
2 Answer . . . 36
3 Keyword Explanation . . . 37
4 Notes and Remarks . . . 37

3(b) How the Barrier Potential Keeps the p and n Regions in Equilibrium 38
1 Theory . . . 38
2 Answer . . . 39
3 Keyword Explanation . . . 40
4 Notes and Remarks . . . 40

3(c) Dielectric Relaxation Time 40
1 Theory . . . 40
2 Answer . . . 41
3 Keyword Explanation . . . 42
4 Notes and Remarks . . . 43

3(d) Donor Concentration and Mobility of Compensated n-Type Silicon 43
1 Theory . . . 43
2 Answer . . . 44
3 Keyword Explanation . . . 45
4 Notes and Remarks . . . 45
5 Final Concise Answer . . . 45

4(a) Effect of Temperature and Doping on Mobility 47
1 Theory . . . 47
2 Answer . . . 48
3 Keyword Explanation . . . 49
4 Notes and Remarks . . . 49

4(b) Ambipolar Transport 50
1 Theory . . . 50
2 Answer . . . 51
3 Keyword Explanation . . . 53
4 Notes and Remarks . . . 53

4(c) Built-In Potential, Depletion Width and Peak Field at 300 K and 80 K 54
1 Theory . . . 54
2 Answer . . . 55
3 Keyword Explanation . . . 56
4 Notes and Remarks . . . 57

4(d) Time and Space Dependence of Excess Carriers 57
1 Theory . . . 57
2 Answer . . . 58
3 Keyword Explanation . . . 59
4 Notes and Remarks . . . 60
5 Final Concise Answer . . . 60

5(a) Steady-State Excess Carriers Generated at the Origin 61
1 Theory . . . 61
2 Answer . . . 62

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3 Keyword Explanation . . . 63
4 Notes and Remarks . . . 63

5(b) Ideal PN Junction Current **64**
1 Theory . . . 64
2 Answer . . . 64
3 Keyword Explanation . . . 67
4 Notes and Remarks . . . 67

5(c) Quasi-Fermi Energy Levels **68**
1 Theory . . . 68
2 Answer . . . 69
3 Keyword Explanation . . . 70
4 Notes and Remarks . . . 70

5(d) Built-In Potential, Depletion Width and Peak Field Under Reverse Bias **71**
1 Theory . . . 71
2 Answer . . . 71
3 Keyword Explanation . . . 73
4 Notes and Remarks . . . 73
5 Final Concise Answer . . . 73

6(a) Inversion Charge and Threshold Voltage of a Nanoscale MOSFET **74**
1 Theory . . . 74
2 Answer . . . 76
3 Keyword Explanation . . . 77
4 Notes and Remarks . . . 77

6(b) Operating Mode of an NPN Transistor **78**
1 Theory . . . 78
2 Answer . . . 79
3 Keyword Explanation . . . 79
4 Notes and Remarks . . . 80

6(c) Advantages of the MOSFET **80**
1 Theory . . . 80
2 Answer . . . 80
3 Keyword Explanation . . . 81
4 Notes and Remarks . . . 81

6(d) Base Doping and Emitter Injection Efficiency **81**
1 Theory . . . 81
2 Answer . . . 82
3 Keyword Explanation . . . 83
4 Notes and Remarks . . . 84
5 Final Concise Answer . . . 84

7(a) Electron and Hole Concentrations from Charge Neutrality **85**
1 Theory . . . 85
2 Answer . . . 86
3 Keyword Explanation . . . 88
4 Notes and Remarks . . . 88

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7(b) Variation of the Fermi Level in an Extrinsic Semiconductor **88**
1 Theory . . . 88
2 Answer . . . 89
3 Keyword Explanation . . . 90
4 Notes and Remarks . . . 90

7(c) Minimum Band Gap of a Designed n-Type Semiconductor **91**
1 Theory . . . 91
2 Answer . . . 92
3 Keyword Explanation . . . 92
4 Notes and Remarks . . . 93

7(d) Occupation Probability at the Conduction Band Edge and the Carrier Concentrations **93**
1 Theory . . . 93
2 Answer . . . 94
3 Keyword Explanation . . . 95
4 Notes and Remarks . . . 95
5 Final Concise Answer . . . 95

8(a) Steady-State Excess Carriers in an Infinite p-Type Crystal **96**
1 Theory . . . 96
2 Answer . . . 97
3 Keyword Explanation . . . 98
4 Notes and Remarks . . . 98

8(b) The Ambipolar Transport Equation **99**
1 Theory . . . 99
2 Answer . . . 100
3 Keyword Explanation . . . 102
4 Notes and Remarks . . . 102

8(c) Generation and Recombination Processes **103**
1 Theory . . . 103
2 Answer . . . 104
3 Keyword Explanation . . . 105
4 Notes and Remarks . . . 105
5 Final Concise Answer . . . 106

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1. (a) Describe three classifications of solids-amorphous, polycrystalline, and single crystal. Marks: 6

   (b) The lattice constant of a face-centered cubic lattice is 0.425 nm. Determine the Marks: 6

   i. effective number of atoms per unit cell
   ii. volume density of atoms

   (c) If the free electrons in a semiconductor have effective mass $m_c = 0.1m_o$, where m_o is the mass of the electron in vacuum, what is their average speed at $T = 300K$? If the free holes in the same material have effective mass $m_v = 1.5m_o$, what is their average speed at the same temperature? Assume a Maxwell-Boltzmann distribution of the electron energies. Marks: 7

   (d) Explain Schottky barrier. How it differs from ohmic contact? Explain. Marks: 6

1.(a) Describe three classifications of solids-amorphous, polycrystalline, and single crystal. Marks: 6

1. Theory

1. What does "order" in a solid mean? (Core Idea)

A solid is a large number of atoms held close together by bonds. What makes one solid different from another, for electronics, is **how regularly those atoms are arranged**. Two words describe this:

- **Long-range order:** the same pattern of atoms repeats again and again over the whole piece, millions of atoms in every direction. If you know where a few atoms are, you can predict where every other atom is.
- **Short-range order:** each atom has roughly the right number of neighbours at roughly the right distance (its bonds are correct), but the pattern does not repeat. A few atoms away, the arrangement is unpredictable.

A region in which the pattern repeats perfectly is called a **grain** (or a crystallite). The surface where two grains of different orientation meet is a **grain boundary**. Using these words, Neamen (Section 1.1) sorts every solid into three classes: **amorphous**, **polycrystalline** and **single crystal**.

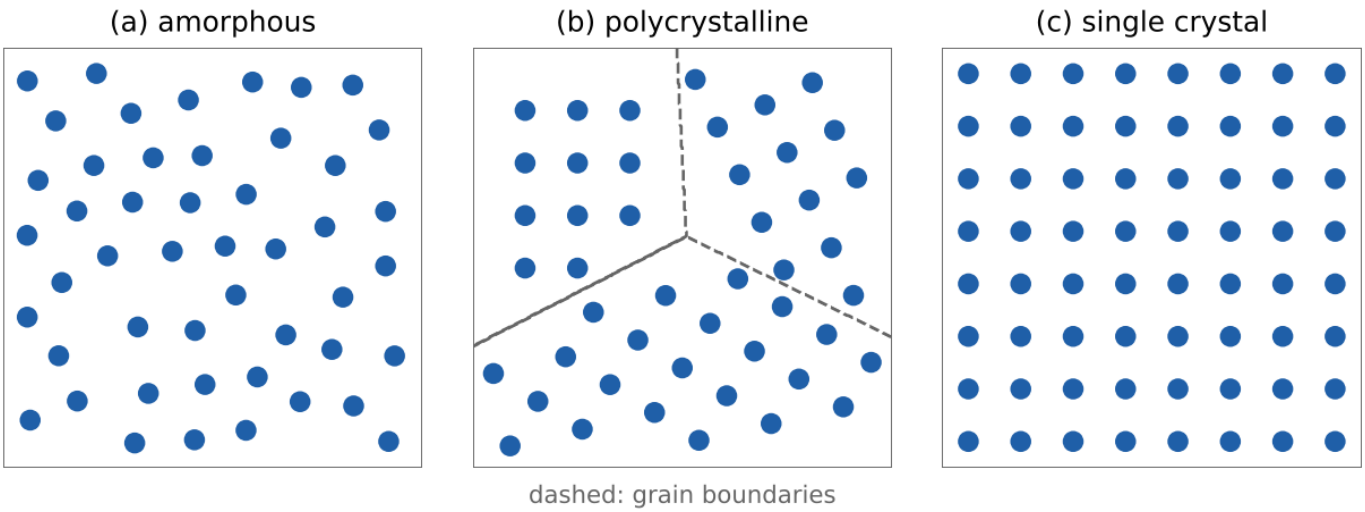


Figure 1: Two-dimensional pictures of the three classes of solids: amorphous, polycrystalline and single crystal

Reading the figure. Each dot is an atom. In panel (a) every atom has neighbours at about the same distance, but there is no repeating pattern: this is short-range order only. In panel (b) the atoms form small regions (grains) that are each perfectly regular, but the grains are tilted with respect to each other; the dashed grey lines are the grain boundaries, where the pattern breaks. In panel (c) one regular pattern runs through the whole piece.

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2. Why the arrangement matters for electrons

An electron moving through a perfectly periodic arrangement of atoms is not scattered by the atoms themselves; it is scattered only by defects, impurities and lattice vibrations. Every break in the pattern (a grain boundary, a disordered region) is a place where electrons are scattered or trapped. So:

- more disorder → more scattering → lower mobility, and more trap states in the band gap;
- perfect order → the best electrical properties.

This is why transistors and integrated circuits are made in **single-crystal** silicon wafers, while cheaper amorphous and polycrystalline silicon are used where top performance is not needed (thin-film transistors in displays, solar cells, MOSFET gate electrodes).

3. How each class is made

- **Single crystal:** grown slowly from a melt starting from a small seed crystal (the Czochralski method, Neamen Section 1.7), so every new atom joins the same pattern.
- **Polycrystalline:** deposited or cooled so that many seeds start growing at once, each with its own orientation, until the grains meet.
- **Amorphous:** deposited or cooled so fast (or at such a low temperature) that the atoms have no time to find their regular places.

2. Answer

Solids are classified by the **size of the region over which the atoms are arranged in a regular, repeating pattern** (Neamen, Section 1.1).

1. Amorphous solids

- The atoms have **order only within a few atomic distances** (short-range order): each atom has about the right neighbours, but there is no repeating pattern.
- There are no grains and no grain boundaries; the structure is random on a large scale (panel (a) of the figure above).
- Examples: glass, amorphous silicon (a-Si:H) used in thin-film solar cells and display transistors.
- Electrical effect: many defects and states in the band gap, so the carrier mobility is very low (around 1 $cm^2/V\ s$ in a-Si).

2. Polycrystalline solids

- The material has a **high degree of order over many atomic dimensions**, but only inside small regions called **grains** (crystallites).
- Neighbouring grains have different orientations and meet at **grain boundaries** (panel (b)).
- Example: polysilicon, used as the gate electrode of MOSFETs and for thin-film resistors.
- Electrical effect: grain boundaries scatter and trap carriers, so the properties are poorer than in a single crystal but better than in an amorphous solid.

3. Single-crystal solids

- The atoms have a **high degree of order throughout the entire volume** (long-range order): one geometric pattern, the **lattice**, repeats through the whole piece (panel (c)).
- There are no grain boundaries.
- Example: the silicon wafer on which integrated circuits are built.
- Electrical effect: the best electrical properties, because there are no grain boundaries to degrade them.

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Summary:

Class	Order	Grain boundaries	Example	Electrical quality
Amorphous	a few atoms only	none (no grains)	glass, a-Si	poorest
Polycrystalline	inside each grain	many	polysilicon gates	medium
Single crystal	whole volume	none	Si wafer	best

Table 1: Summary

3. Keyword Explanation

1. Lattice

The periodic arrangement of points that describes where the atoms of a single crystal sit; one pattern repeated in three dimensions.

2. Grain

A region of a polycrystalline solid inside which the atoms form one perfect lattice.

3. Grain boundary

The surface where two grains of different orientation meet; the lattice pattern breaks there, so carriers are scattered and trapped.

4. Long-range and short-range order

Long-range: the pattern repeats over the whole sample. Short-range: only the nearest neighbours are arranged regularly.

4. Notes and Remarks

- Neamen's Figure 1.1 shows the same three schematics; this answer follows its Section 1.1.
- A "single crystal" in practice still contains point defects (vacancies, interstitials) and dislocations, but no grain boundaries.
- Hydrogen is added to amorphous silicon (a-Si:H) to tie up broken bonds and reduce the states in the gap; that is what makes it usable in solar cells (Neamen, Section 14.2.5).

1.(b) The lattice constant of a face-centered cubic lattice is 0.425 nm. Determine the i. effective number of atoms per unit cell ii. volume density of atoms Marks: 6

1. Theory

1. Unit cell and lattice constant (Core Idea)

A single crystal is built by repeating one small block of atoms in every direction, like bricks in a wall. The smallest convenient block is the **unit cell**. For the cubic lattices the unit cell is a cube, and the length of its edge is the **lattice constant** a.

Three simple cubic lattices exist (Neamen, Figure 1.5):

- **simple cubic (sc):** an atom at each of the 8 corners;
- **body-centered cubic (bcc):** the 8 corners plus one atom at the centre of the cube;
- **face-centered cubic (fcc):** the 8 corners plus one atom at the centre of each of the 6 faces.

2. Why we count fractions of atoms

The unit cells touch each other, so an atom on the surface of a cube is **shared** with the neighbouring cubes:

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- A **corner** atom is shared by the 8 cubes that meet at that corner. Only 1/8 of it belongs to one cube.
- A **face** atom is shared by the 2 cubes on either side of that face. Only 1/2 of it belongs to one cube.
- A **body** (centre) atom belongs to one cube only.

Adding these fractions gives the **effective number of atoms per unit cell**: how many whole atoms one cell really contains when the crystal is built from it.

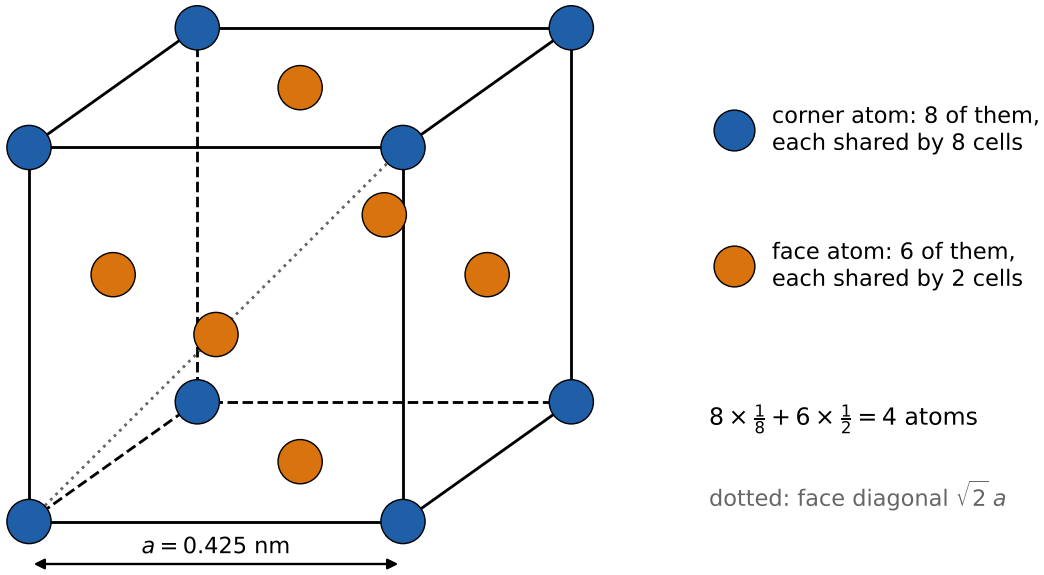


Figure 2: Face-centered cubic unit cell: 8 corner atoms shared by 8 cells and 6 face atoms shared by 2 cells

Reading the figure. The cube has edge a. The 8 blue corner atoms each sit where 8 cubes meet, so each gives 1/8 to this cube. The 6 orange face atoms each sit on a face shared by 2 cubes, so each gives 1/2. The dotted line marks one face diagonal: along it the corner and face atoms touch, which is why the nearest-neighbour distance is $a/\sqrt{2}$.

3. Volume density of atoms

The **volume density** is the number of atoms per unit volume. Since the crystal is made of identical unit cells, the density of the whole crystal equals the density of one cell (Neamen, Example 1.1):

$$\text{Volume density} = \frac{\text{effective number of atoms per unit cell}}{\text{volume of the unit cell}} = \frac{n}{a^3}$$

Semiconductor work uses cm^{-3}, so a is written in cm ($1\ \text{nm} = 10^{-7}\ \text{cm}$).

2. Answer

Given: face-centered cubic lattice, $a = 0.425\ \text{nm} = 0.425 \times 10^{-7}\ \text{cm} = 4.25 \times 10^{-8}\ \text{cm}$ (4.25 Å).

Step 1: Effective number of atoms per unit cell

An fcc cell has 8 corner atoms, each shared by 8 cells, and 6 face atoms, each shared by 2 cells:

$$n = 8 \times \frac{1}{8} + 6 \times \frac{1}{2} = 1 + 3 = 4$$

$$\boxed{n = 4 \text{ atoms per unit cell}}$$

Step 2: Volume of the unit cell

$$a^3 = (4.25 \times 10^{-8}\ \text{cm})^3 = 7.677 \times 10^{-23}\ \text{cm}^3$$

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Step 3: Volume density of atoms

$$\text{Density} = \frac{n}{a^3} = \frac{4}{7.677 \times 10^{-23}\ \text{cm}^3} = 5.21 \times 10^{22}\ \text{cm}^{-3}$$

$$\boxed{\text{Volume density} = 5.21 \times 10^{22}\ \text{atoms/cm}^3}$$

This is the order of magnitude of every solid: about 5×10^{22} atoms in each cubic centimetre (silicon has 5.0×10^{22} cm^{-3}).

Final Answer:

- (i) 4 atoms per unit cell
- (ii) 5.21×10^{22} cm^{-3}

3. Keyword Explanation

1. Unit cell

The small block of the lattice that, repeated in every direction, builds the whole crystal.

2. Lattice constant

The edge length a of the cubic unit cell.

3. Effective number of atoms per unit cell

The number of whole atoms that belong to one cell after the shared corner and face atoms are divided among the cells that share them.

4. Notes and Remarks

- **Book check:** this is Neamen's Exercise Ex 1.1 (fcc, $a = 4.25$ Å); the book's answers are (a) 4 and (b) 5.21×10^{22} cm^{-3}, the same as here.
- For comparison: sc has $8 \times \frac{1}{8} = 1$ atom per cell, bcc has $1 + 1 = 2$ (Neamen, Example 1.1), and the diamond lattice of silicon has 8 (the fcc 4 plus 4 inside the cell).
- In an fcc lattice the corner and face atoms touch along the face diagonal, so the nearest-neighbour distance is $a/\sqrt{2} = 0.3005$ nm, and the fraction of the volume filled by touching hard spheres is $\pi/(3\sqrt{2}) = 0.74$, the largest possible.

1.(c) If the free electrons in a semiconductor have effective mass $m_c = 0.1m_o$, where m_o is the mass of the electron in vacuum, what is their average speed at $T = 300K$? If the free holes in the same material have effective mass $m_v = 1.5m_o$, what is their average speed at the same temperature? Assume a Maxwell-Boltzmann distribution of the electron energies. Marks: 7

1. Theory

1. Free carriers move all the time (Core Idea)

Even with no applied voltage, the free electrons in the conduction band and the holes in the valence band are not at rest. They move in random directions, colliding with the vibrating lattice and with impurities. This **random thermal motion** is the same motion that gas molecules have in air. Its average energy is set by the temperature alone.

In a crystal, an electron near the bottom of the conduction band responds to forces as if it had a mass m^*, its **effective mass**, instead of the free-electron mass $m_0 = 9.11 \times 10^{-31}$ kg. The effective mass includes the effect of the periodic crystal potential (Neamen, Section 3.2). So in every formula of motion, m_0 is replaced by m^*: here $m_c = 0.1m_0$ for electrons and $m_v = 1.5m_0$ for holes.

2. The Maxwell-Boltzmann distribution of speeds

When the carriers are non-degenerate (few of them compared with the available states), the chance of finding a carrier with energy E falls as $e^{-E/kT}$: this is the Maxwell-Boltzmann distribution (the

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Boltzmann approximation of the Fermi-Dirac function, Neamen Section 3.5). For a particle of mass m with kinetic energy $E = \frac{1}{2}mv^2$ moving freely in three dimensions, the fraction of particles with speed between v and $v + dv$ is

$$f(v)\,dv = 4\pi \left(\frac{m}{2\pi kT}\right)^{3/2} v^2 \exp\left(-\frac{mv^2}{2kT}\right) dv$$

where

- $4\pi v^2\,dv$ counts the velocity states on a thin spherical shell of radius v (the number of directions grows as v^2)
- $\exp(-mv^2/2kT)$ is the Boltzmann factor (fast carriers are rare)
- the front factor makes $\int_0^\infty f(v)\,dv = 1$

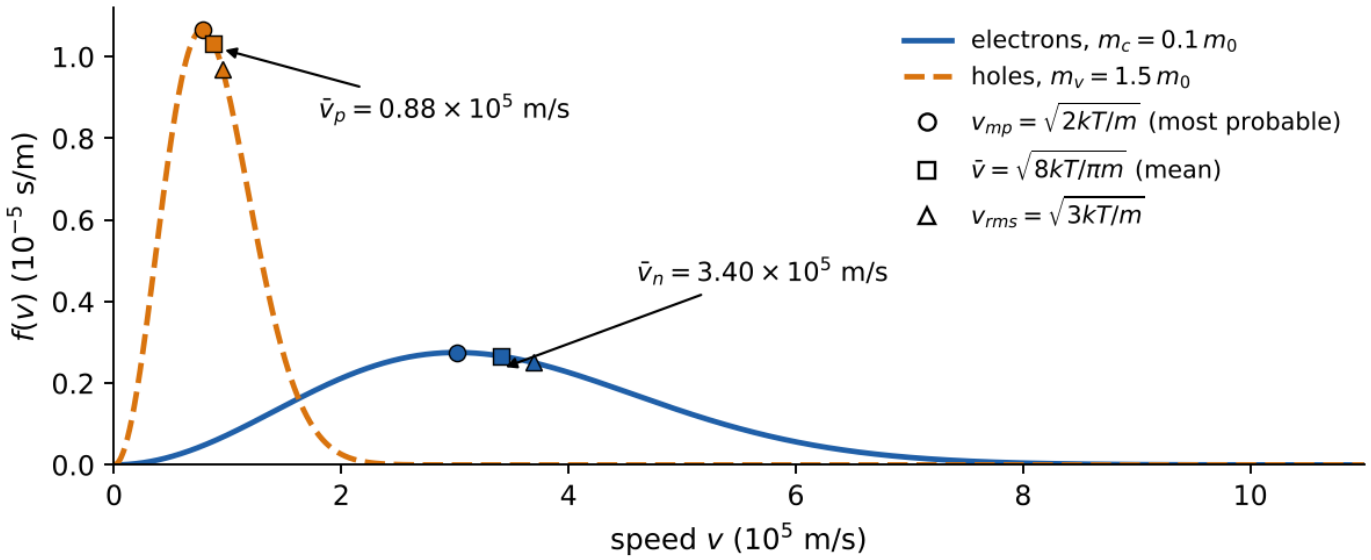


Figure 3: Maxwell-Boltzmann speed distributions of electrons (0.1 m0) and holes (1.5 m0) at 300 K

Reading the figure. The solid blue curve is the speed distribution of the light electrons ($m_c = 0.1m_0$), the dashed orange curve that of the heavy holes ($m_v = 1.5m_0$), both at 300 K. Each curve starts at zero (few velocity states at low speed), rises to a peak, then falls (the Boltzmann factor). The three marks on each curve are the most probable speed v_{mp}, the mean speed $\bar{v}$ and the rms speed v_{rms}, always in that order. The light electrons are spread to much higher speeds: their speeds are $\sqrt{15} = 3.87$ times those of the holes.

3. Three "average" speeds

From $f(v)$, three speeds are commonly quoted:

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Speed	Formula	Meaning
Most probable	$v_{mp} = \sqrt{2kT/m}$	the peak of $f(v)$
Mean (average)	$\bar{v} = \sqrt{8kT/(\pi m)}$	the arithmetic average of the speed
Root-mean-square	$v_{rms} = \sqrt{3kT/m}$	from the average kinetic energy $\frac{1}{2}m\overline{v^2} = \frac{3}{2}kT$

Table 2: Summary

The question asks for the **average speed** with a Maxwell-Boltzmann distribution, which is the mean speed $\bar{v}$. The rms speed, often called the **thermal velocity**, is also given in the Answer, since many textbooks use it as "the average thermal speed".

2. Answer

Given: $T = 300$ K, $k = 1.38 \times 10^{-23}$ J/K, $m_0 = 9.11 \times 10^{-31}$ kg; electrons $m_c = 0.1m_0 = 9.11 \times 10^{-32}$ kg; holes $m_v = 1.5m_0 = 1.3665 \times 10^{-30}$ kg.
Assumption: the carriers are non-degenerate and move freely between collisions with effective masses m_c and m_v, so their speeds follow the Maxwell-Boltzmann distribution.
The Answer derives the mean speed from the distribution. Step 2 is a general integral and is marked (Basic).

Step 1: The speed distribution

With the Boltzmann factor $e^{-E/kT}$, $E = \frac{1}{2}mv^2$, and $4\pi v^2\,dv$ velocity states in a shell of speeds v to $v + dv$:

$$f(v) = 4\pi \left(\frac{m}{2\pi kT}\right)^{3/2} v^2 \exp\left(-\frac{mv^2}{2kT}\right)$$

Step 2 (Basic): The integral $\int_0^\infty v^3 e^{-\alpha v^2} dv$

Put $u = v^2$, $du = 2v\,dv$, so $v^3\,dv = \frac{1}{2}u\,du$:

$$\int_0^\infty v^3 e^{-\alpha v^2} dv = \frac{1}{2}\int_0^\infty u e^{-\alpha u} du = \frac{1}{2}\cdot\frac{1}{\alpha^2} = \frac{1}{2\alpha^2}$$

using $\int_0^\infty u e^{-\alpha u} du = 1/\alpha^2$ (integration by parts).

Step 3: The mean speed

The mean speed is $\bar{v} = \int_0^\infty v f(v)\,dv$. With $\alpha = m/(2kT)$:

$$\bar{v} = 4\pi \left(\frac{m}{2\pi kT}\right)^{3/2} \cdot \frac{1}{2\alpha^2} = 4\pi \left(\frac{m}{2\pi kT}\right)^{3/2} \cdot \frac{1}{2}\left(\frac{2kT}{m}\right)^2$$

Collect the powers of m/kT: $(m/kT)^{3/2}\cdot(kT/m)^2 = (kT/m)^{1/2}$, and the numbers give $4\pi\cdot\frac{1}{(2\pi)^{3/2}}\cdot\frac{1}{2}\cdot 4 = \frac{8\pi}{(2\pi)^{3/2}} = \sqrt{\frac{8}{\pi}}$. So

$$\boxed{\bar{v} = \sqrt{\frac{8kT}{\pi m^*}}}$$

Step 4: Electrons, $m_c = 0.1m_0$

$$\frac{8kT}{\pi} = \frac{8(1.38 \times 10^{-23})(300)}{\pi} = 1.054 \times 10^{-20}\ \text{J}$$

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$$\bar{v}_n = \sqrt{\frac{1.054 \times 10^{-20}}{9.11 \times 10^{-32}}} = 3.40 \times 10^5 \text{ m/s}$$

$$\boxed{\bar{v}_n = 3.40 \times 10^5 \text{ m/s} = 3.40 \times 10^7 \text{ cm/s}}$$

Step 5: Holes, $m_v = 1.5m_0$

$$\bar{v}_p = \sqrt{\frac{1.054 \times 10^{-20}}{1.3665 \times 10^{-30}}} = 8.78 \times 10^4 \text{ m/s}$$

$$\boxed{\bar{v}_p = 8.78 \times 10^4 \text{ m/s} = 8.78 \times 10^6 \text{ cm/s}}$$

The ratio is $\bar{v}_n/\bar{v}_p = \sqrt{1.5/0.1} = \sqrt{15} = 3.87$: the lighter carriers move faster at the same temperature, because both have the same average kinetic energy $\frac{3}{2}kT = 0.0388$ eV.

Step 6: The rms (thermal) speed, for comparison

From $\frac{1}{2}m^*\overline{v^2} = \frac{3}{2}kT$, $v_{rms} = \sqrt{3kT/m^*}$:

- electrons: $v_{rms} = \sqrt{3(1.38 \times 10^{-23})(300)/9.11 \times 10^{-32}} = 3.69 \times 10^5$ m/s
- holes: $v_{rms} = \sqrt{3(1.38 \times 10^{-23})(300)/1.3665 \times 10^{-30}} = 9.53 \times 10^4$ m/s

Final Answer:

- Electrons ($0.1m_0$): average (mean) speed 3.40×10^5 m/s (3.40×10^7 cm/s); rms 3.69×10^5 m/s
- Holes ($1.5m_0$): average (mean) speed 8.78×10^4 m/s (8.78×10^6 cm/s); rms 9.53×10^4 m/s

3. Keyword Explanation

1. Effective mass

The mass a carrier appears to have when it is accelerated inside the crystal; it replaces m_0 in Newton's law and in the kinetic energy.

2. Maxwell-Boltzmann distribution

The distribution of speeds of non-degenerate particles in equilibrium; the probability of an energy E falls as $e^{-E/kT}$.

3. Thermal velocity

The random speed carriers have because of temperature alone, usually taken as $v_{rms} = \sqrt{3kT/m^*}$, about 10^7 cm/s in silicon at 300 K.

4. Notes and Remarks

- **Which "average"?** The mean speed $\sqrt{8kT/\pi m}$ is the true average of the speed. Some books (and some answer keys) take $v = \sqrt{3kT/m}$ from $\frac{1}{2}mv^2 = \frac{3}{2}kT$ and call it the average speed; that gives 3.69×10^5 m/s and 9.53×10^4 m/s, about 8.5 % larger. Either is accepted if the formula is stated.
- **Compared with drift:** Neamen (Section 5.1.3) notes that the mean thermal velocity in silicon is about 10^7 cm/s, while a typical drift velocity is 10^5 cm/s, only 1 % of it. Drift is a small bias on top of a fast random motion.
- The random motion gives zero net current: as many carriers cross any plane in one direction as in the other.

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1.(d) Explain Schottky barrier. How it differs from ohmic contact? Explain. Marks: 6

1. Theory

1. Work function and electron affinity (Core Idea)

Every metal-semiconductor junction is understood by comparing energies measured from the **vacuum level**, the energy an electron has when it is just free of the solid and at rest.

- **Work function** $e\phi$: the energy needed to take an electron from the Fermi level to the vacuum level. A metal has $e\phi_m$, a semiconductor $e\phi_s$.
- **Electron affinity** $e\chi$: the energy from the bottom of the conduction band E_c of the semiconductor to the vacuum level. It does not depend on doping (for Si, $\chi = 4.01$ V).

For an n-type semiconductor, $e\phi_s = e\chi + (E_c - E_F)$.

2. What happens when a metal touches a semiconductor

In thermal equilibrium the Fermi level must be **flat** through the whole system (Neamen, Section 7.2). If the two Fermi levels differ before contact, electrons flow from the side with the higher E_F to the side with the lower E_F until they line up. The charge that moves leaves fixed charge behind and bends the semiconductor's bands near the surface. The direction of the flow decides the type of contact:

- **Metal-n-type with $\phi_m > \phi_s$:** the semiconductor's E_F is higher, so electrons leave the semiconductor for the metal. They leave behind positive donor ions: a **depletion region** forms in the semiconductor, the bands bend *up*, and a barrier stands in the electrons' way. This is a **Schottky barrier** (rectifying contact).
- **Metal-n-type with $\phi_m < \phi_s$:** electrons flow from the metal into the semiconductor and pile up at its surface (an accumulation layer). The bands bend *down*, there is no barrier for electrons, and current flows easily in both directions. This is an **ohmic contact** (Neamen, Section 9.2.1).

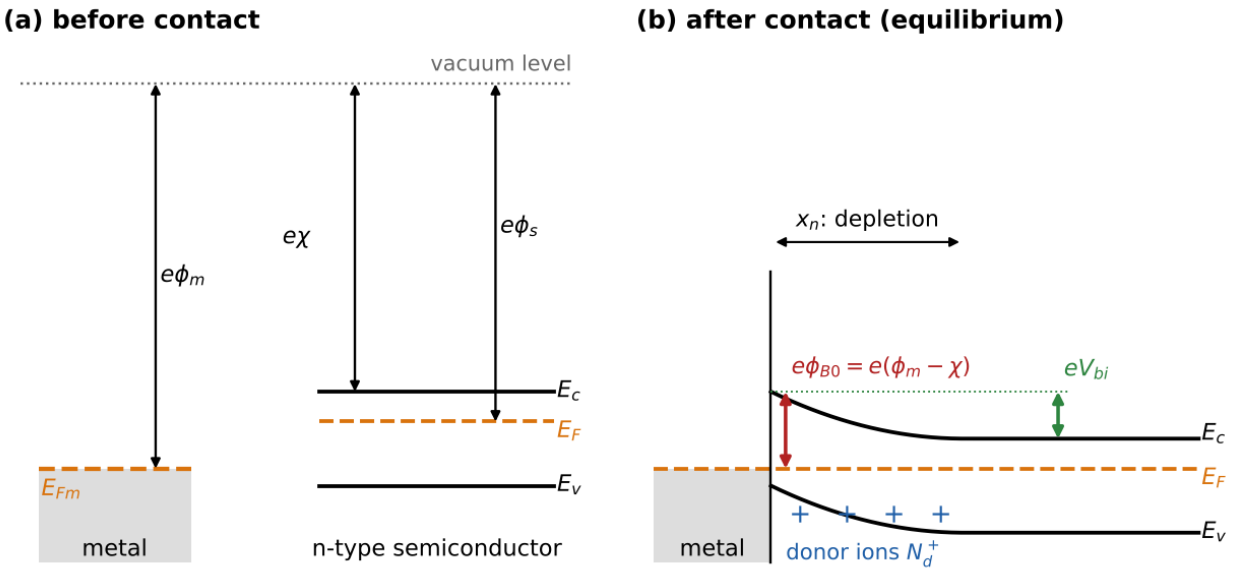


Figure 4: Schottky barrier: metal and n-type semiconductor before and after contact

Reading the figure. Panel (a): before contact, the metal's Fermi level E_{Fm} (dashed orange) lies $e\phi_m$ below the vacuum level, lower than the semiconductor's E_F because $\phi_m > \phi_s$. Panel (b): after contact the Fermi levels line up; electrons have left the semiconductor, uncovering positive donor ions (+) in the space charge region of width x_n, and the bands bend up. An electron in the metal sees the barrier $e\phi_{B0} = e(\phi_m - \chi)$ (red arrow); an electron in the semiconductor sees the smaller built-in barrier $eV_{bi} = e\phi_{B0} - e\phi_n$ (green arrow).

3. The barrier heights

From the figure (Neamen, Eqs. 9.1 and 9.2):

$$\phi_{B0} = \phi_m - \chi, \qquad V_{bi} = \phi_{B0} - \phi_n$$

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where $e\phi_n = E_c - E_F$ in the neutral semiconductor. The depletion region is exactly like the n side of a one-sided p^+n junction, so its width is (Neamen, Eq. 9.7)

$$W = x_n = \left[\frac{2\epsilon_s(V_{bi} + V_R)}{eN_d}\right]^{1/2}$$

4. Why the Schottky contact rectifies

- **Forward bias** (metal positive): the semiconductor-to-metal barrier eV_{bi} is lowered to $e(V_{bi} - V_a)$; electrons from the semiconductor flow over it into the metal by **thermionic emission**, and the current grows exponentially: $J = J_{sT}[\exp(eV_a/kT) - 1]$, with $J_{sT} = A^*T^2 \exp(-e\phi_{Bn}/kT)$ (Neamen, Section 9.1.6).
- **Reverse bias:** the semiconductor barrier rises; the metal-side barrier ϕ_{B0} stays the same, so only the small current J_{sT} flows.
- The current is carried by **majority carriers** (electrons) only. There is no minority carrier storage, so a Schottky diode switches much faster than a pn diode and has a lower turn-on voltage (about 0.3 V instead of 0.7 V).

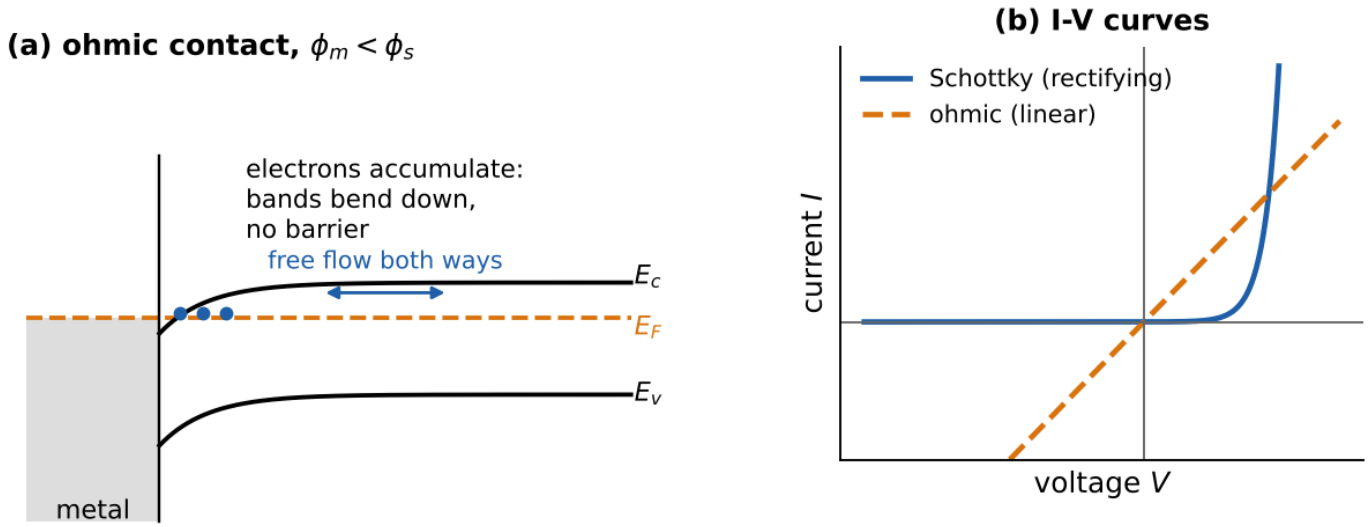


Figure 5: Ohmic contact band diagram and the I-V curves of rectifying and ohmic contacts

Reading the figure. Panel (a): with $\phi_m < \phi_s$ the electrons have moved into the semiconductor, the bands bend down at the surface, and the conduction band edge sits at or below E_F there: no barrier stops electrons in either direction. Panel (b): the solid blue curve is the Schottky (rectifying) contact, exponential in forward bias and a tiny saturation current in reverse; the dashed orange line is an ohmic contact, a straight line through the origin with a small resistance.

5. The practical ohmic contact: tunneling

Metals with a work function small enough for the rule above are rare, and surface states often pin the barrier. So in practice an ohmic contact is made by **heavily doping** the semiconductor under the metal ($N_d \gtrsim 10^{19}$ cm^{-3}). The barrier is still there, but its depletion width $W \propto 1/\sqrt{N_d}$ becomes only a few nanometres, so electrons **tunnel** straight through it in both directions (Neamen, Section 9.2.2). The quality of such a contact is given by the **specific contact resistance** R_c ($\Omega\,$cm^2), which falls steeply as N_d rises.

2. Answer

Schottky barrier. A Schottky barrier is the rectifying potential barrier formed at the junction of a metal and a lightly or moderately doped semiconductor when, for an n-type semiconductor, the metal work function is larger than the semiconductor work function ($\phi_m > \phi_s$) (Neamen, Section 9.1).

- On contact, electrons flow from the semiconductor (higher E_F) into the metal until the Fermi levels are equal.

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- Positive donor ions are left uncovered: a depletion region of width x_n forms in the semiconductor, the bands bend up, and the field points from the semiconductor towards the metal (see the Schottky figure above).
- Electrons in the metal face the **Schottky barrier height** $\phi_{B0} = \phi_m - \chi$; electrons in the semiconductor face the **built-in potential** $V_{bi} = \phi_{B0} - \phi_n$.
- Forward bias (metal +) lowers V_{bi} and electrons pour over the barrier by thermionic emission; reverse bias raises it and only a small saturation current flows:

$$J = J_{sT}\left[\exp\left(\frac{eV_a}{kT}\right) - 1\right], \qquad J_{sT} = A^*T^2 \exp\left(-\frac{e\phi_{Bn}}{kT}\right)$$

where A^* is the effective Richardson constant (120 A/cm^{2}K^2 for free electrons).

Ohmic contact. An ohmic contact is a metal-semiconductor contact with a **negligible resistance** compared with the bulk of the device, which passes current equally in both directions with a linear I-V characteristic (Neamen, Section 9.2). It is made in two ways:

1. **Ideal non-rectifying barrier:** for n-type, choose $\phi_m < \phi_s$. Electrons flow from the metal into the semiconductor, an accumulation layer forms, the bands bend down and there is no barrier to electron flow in either direction.
2. **Tunneling barrier** (the usual practice): dope the semiconductor very heavily under the metal. The depletion width becomes so thin (a few nm) that electrons tunnel through the barrier in both directions.

How they differ:

Point	Schottky barrier	Ohmic contact
Work functions (n-type)	$\phi_m > \phi_s$	$\phi_m < \phi_s$, or a heavily doped surface
Surface layer	depletion (bands bend up)	accumulation (bands bend down) or a very thin depletion layer
Barrier to electrons	yes, ϕ_{B0} and V_{bi}	none, or thin enough to tunnel through
Conduction	thermionic emission over the barrier	free flow or tunneling
I-V curve	rectifying, exponential	linear (a small resistance)
Use	Schottky diode, MESFET gate, fast switching	connecting every device to the outside circuit

Table 3: Summary

3. Keyword Explanation

1. Work function

The energy needed to remove an electron from the Fermi level to the vacuum level ($e\phi_m$ for a metal, $e\phi_s$ for a semiconductor).

2. Electron affinity

The energy from the conduction band edge to the vacuum level, $e\chi$; 4.01 eV for silicon.

3. Thermionic emission

Current carried by carriers that have enough thermal energy to go over a potential barrier.

4. Specific contact resistance

The resistance of a contact times its area, $R_c = (\partial J/\partial V)^{-1}$ at $V = 0$, in $\Omega\,\text{cm}^2$; the figure of merit of an ohmic contact.

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4. Notes and Remarks

- **Real barrier heights:** measured Schottky barriers depend much less on ϕ_m than $\phi_{B0} = \phi_m - \chi$ predicts, because interface states pin the Fermi level at the surface (Neamen, Figure 9.5 and Section 9.1.4). The Schottky effect (image-force lowering, Section 9.1.3) also lowers the barrier slightly with field.
- **Compared with a pn diode:** at the same current, a Schottky diode needs about 0.3 to 0.4 V less forward voltage, because J_{sT} is orders of magnitude larger than the pn saturation current; and it has no minority-carrier storage, so it switches in picoseconds rather than nanoseconds (Neamen, Section 9.1.7).
- For p-type semiconductors the rules reverse: a Schottky barrier needs $\phi_m < \phi_s$, an ohmic contact $\phi_m > \phi_s$.
- The 2018–19 paper asked the same comparison as 4(c) (2018–19 4(c)), extra reading only.

5. Final Concise Answer

(a) Amorphous solids have order only within a few atomic distances (short-range order, no grains); polycrystalline solids have perfect order inside small grains of different orientations separated by grain boundaries; single crystals have one regular lattice through the whole volume. Electrical quality rises from amorphous to single crystal, because disorder and grain boundaries scatter and trap carriers; devices are made in single-crystal wafers.

(b) fcc: $8 \times \frac{1}{8} + 6 \times \frac{1}{2} = 4$ atoms per unit cell; volume density $4/a^3 = 4/(4.25 \times 10^{-8}\ \text{cm})^3 = 5.21 \times 10^{22}$ cm^{-3} (Neamen Ex 1.1).

(c) Mean speed $\bar{v} = \sqrt{8kT/\pi m^*}$: electrons $(0.1m_0)$ 3.40×10^5 m/s, holes $(1.5m_0)$ 8.78×10^4 m/s, a ratio of $\sqrt{15}$; the rms (thermal) speeds $\sqrt{3kT/m^*}$ are 3.69×10^5 and 9.53×10^4 m/s.

(d) A Schottky barrier forms when a metal with $\phi_m > \phi_s$ meets n-type silicon: electrons leave the semiconductor, a depletion layer and an upward band bending appear, with barrier $\phi_{B0} = \phi_m - \chi$ from the metal and $V_{bi} = \phi_{B0} - \phi_n$ from the semiconductor; current flows by thermionic emission, $J = J_{sT}[e^{eV_a/kT} - 1]$, so the contact rectifies. An ohmic contact ($\phi_m < \phi_s$, or a heavily doped surface so electrons tunnel through a thin barrier) has no effective barrier, a linear I-V curve and a negligible resistance.

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2. (a) Starting from charge neutrality condition, Find thermal equilibrium carrier concentration at compensated semiconductor in terms of N_a, N_d and n_i Marks: 6

(b) Consider silicon at $T = 300$ K. A Hall effect device is fabricated with the following geometry: $d = 5 \times 10^{-3}$ cm, $W = 5 \times 10^{-2}$ cm and $L = 0.50$ cm. The electrical parameters measured are: $I_x = 0.50$ mA, $V_x = 1.25$ V, and $B_z = 650$ *gauss* $= 6.5 \times 10^{-2}$ *tesla*. The Hall field is $E_H = -16.5$ mV cm^{-1}. Determine: Marks: 8

i. the Hall voltage
ii. the conductivity type
iii. the majority carrier concentration and
iv. the majority carrier mobility

(c) Using proper graph, explain how carrier concentration varies with temperature. Marks: 5

(d) Sketch and briefly describe energy band diagram of compensated semiconductor. Marks: 6

2.(a) Starting from charge neutrality condition, Find thermal equilibrium carrier concentration at compensated semiconductor in terms of N_a, N_d and n_i Marks: 6

1. Theory

1. What is a compensated semiconductor? (Core Idea)

A semiconductor is **doped** by adding impurity atoms:

- **Donors** (P, As in Si) have one more valence electron than silicon. That extra electron is very loosely held and, at room temperature, escapes into the conduction band. The donor is left as a fixed **positive ion** N_d^+.
- **Acceptors** (B in Si) have one valence electron fewer. They grab an electron from the valence band, which leaves a free **hole**. The acceptor becomes a fixed **negative ion** N_a^-.

A **compensated semiconductor** contains *both* donors and acceptors in the same region (Neamen, Section 4.5). The extra electrons of the donors first drop into the empty acceptor states, so the two kinds of dopant partly cancel each other. Only the **difference** is left over to supply free carriers:

- $N_d > N_a$: compensated **n-type**, with about $N_d - N_a$ free electrons;
- $N_a > N_d$: compensated **p-type**, with about $N_a - N_d$ free holes;
- $N_d = N_a$: fully compensated, it behaves like intrinsic material.

Compensation happens all the time in device making: a p region is often made by diffusing acceptors into an n-type wafer, so that region contains both dopants.

2. Charge neutrality

A uniformly doped piece of semiconductor in **thermal equilibrium** (no light, no voltage, steady temperature) has **no net charge**: it was built from neutral atoms. Four charged species exist:

Species	Density	Charge
free electrons	n_0	$-e$ each
ionized acceptors	N_a^-	$-e$ each
free holes	p_0	$+e$ each
ionized donors	N_d^+	$+e$ each

Table 4: Summary

Total negative charge equals total positive charge, so (cancelling the common e)

$$n_0 + N_a^- = p_0 + N_d^+$$

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This is the **charge neutrality condition** (Neamen, Eq. 4.56).

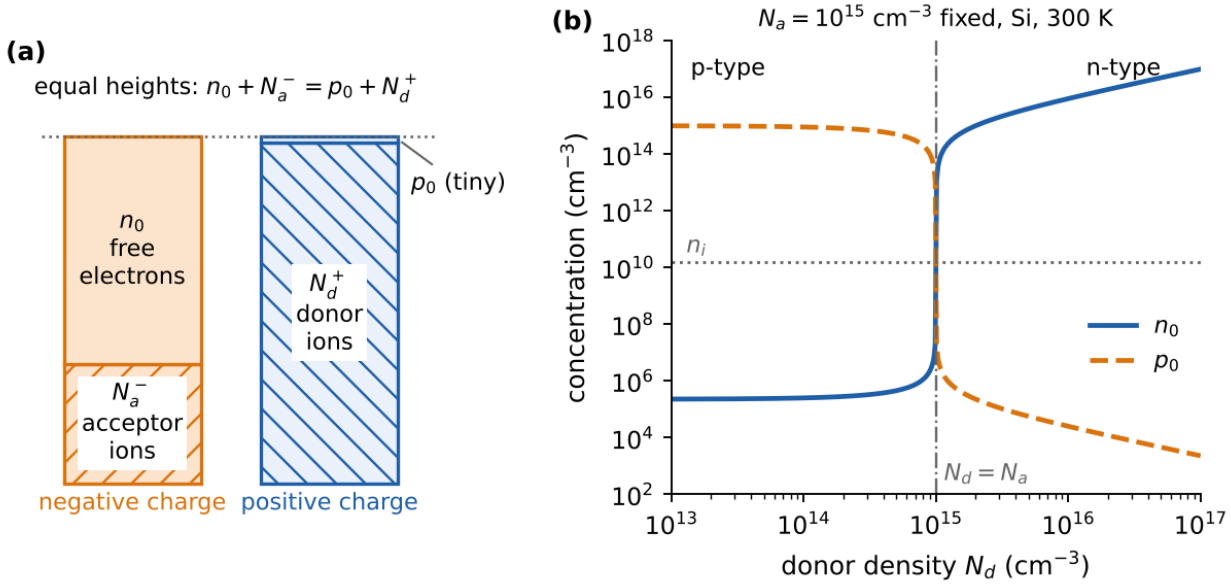


Figure 6: Charge balance in a compensated semiconductor, and n0 and p0 as the donor density is varied (reused from the 2020-21 paper)

Reading the figure. Panel (a) draws the neutrality condition as two stacks of equal height: on the left the negative charge (hatched acceptor ions plus free electrons n_0, orange outline), on the right the positive charge (hatched donor ions plus a tiny number of holes p_0, blue outline). Here $N_d > N_a$, so the material is n-type. Panel (b) is computed from the result of this question for Si with $N_a = 10^{15}$ cm^{-3} fixed and N_d varied: the solid blue curve is n_0, the dashed orange curve p_0, the dotted grey line n_i. Left of the dash-dot line ($N_d < N_a$) the material is p-type with $p_0 \approx N_a - N_d$; right of it, n-type with $n_0 \approx N_d - N_a$; at $N_d = N_a$ both equal n_i.

3. Two more facts we need

Complete ionization. In Si at room temperature practically every donor has given up its electron and every acceptor has taken one (the donor level is only about 0.045 eV below E_c, less than $2kT$). So $N_d^+ = N_d$ and $N_a^- = N_a$.

Mass-action law. In thermal equilibrium, for a non-degenerate semiconductor, the product of the electron and hole concentrations does not depend on the doping (Neamen, Eq. 4.43):

$$n_0 p_0 = n_i^2$$

where n_i is the intrinsic carrier concentration (1.5×10^{10} cm^{-3} for Si at 300 K). It comes from $n_0 = N_c e^{-(E_c - E_F)/kT}$ and $p_0 = N_v e^{-(E_F - E_v)/kT}$: in the product, E_F cancels and $n_0 p_0 = N_c N_v e^{-E_g/kT} = n_i^2$. Physically, more electrons give a hole more chances to recombine, so holes become rarer in exact proportion.

4. Why a quadratic, and which root?

After the two facts are used, the neutrality condition holds one unknown, n_0, but as both n_0 and $1/n_0$. Multiplying by n_0 turns it into a **quadratic**, which has two roots. One of them is always negative and is rejected, because a concentration cannot be negative.

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2. Answer

Assumptions: thermal equilibrium; complete ionization ($N_d^+ = N_d$, $N_a^- = N_a$); non-degenerate material, so $n_0 p_0 = n_i^2$ holds.

Step 4 is the general quadratic formula and is marked (Basic); a student who knows it can skip it.

Step 1: Charge neutrality

The negative charges (electrons and ionized acceptors) balance the positive charges (holes and ionized donors) (Neamen, Eq. 4.56):

$$n_0 + N_a^- = p_0 + N_d^+$$

With complete ionization (Neamen, Eq. 4.58):

$$n_0 + N_a = p_0 + N_d$$

Step 2: Remove the holes with the mass-action law

Put $p_0 = n_i^2/n_0$ (Neamen, Eq. 4.59a):

$$n_0 + N_a = \frac{n_i^2}{n_0} + N_d$$

Step 3: Make it a quadratic in n_0

Multiply every term by n_0:

$$n_0^2 + N_a n_0 = n_i^2 + N_d n_0$$

Bring everything to the left and group the n_0 terms (Neamen, Eq. 4.59b):

$$n_0^2 - (N_d - N_a)\, n_0 - n_i^2 = 0$$

Step 4 (Basic): The quadratic formula

For $ax^2 + bx + c = 0$, completing the square gives $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. Here $a = 1$, $b = -(N_d - N_a)$, $c = -n_i^2$:

$$n_0 = \frac{(N_d - N_a) \pm \sqrt{(N_d - N_a)^2 + 4n_i^2}}{2}$$

Take the 2 inside the square root ($\sqrt{X}/2 = \sqrt{X/4}$):

$$n_0 = \frac{N_d - N_a}{2} \pm \sqrt{\left(\frac{N_d - N_a}{2}\right)^2 + n_i^2}$$

Step 5: Choose the root

The square root is always larger than $|N_d - N_a|/2$, so the minus sign gives a negative n_0, which is impossible. Keep the plus sign:

$$\boxed{n_0 = \frac{N_d - N_a}{2} + \sqrt{\left(\frac{N_d - N_a}{2}\right)^2 + n_i^2}}$$

(Neamen, Eq. 4.60.)

Step 6: The hole concentration

Go back to Step 1 and remove the electrons instead, with $n_0 = n_i^2/p_0$ (Neamen, Eq. 4.61a):

$$\frac{n_i^2}{p_0} + N_a = p_0 + N_d$$

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Multiply by p_0 and collect terms:

$$n_i^2 + N_a p_0 = p_0^2 + N_d p_0 \quad \Rightarrow \quad p_0^2 - (N_a - N_d)\, p_0 - n_i^2 = 0$$

The quadratic formula with $a = 1$, $b = -(N_a - N_d)$, $c = -n_i^2$, and the positive root for the same reason as in Step 5, give

$$\boxed{p_0 = \frac{N_a - N_d}{2} + \sqrt{\left(\frac{N_a - N_d}{2}\right)^2 + n_i^2}}$$

(Neamen, Eq. 4.62.)

Steps to remember

Neutrality $n_0 + N_a = p_0 + N_d \to p_0 = n_i^2/n_0 \to$ multiply by n_0: quadratic $\to$ quadratic formula $\to$ keep the positive root $\to$ for p_0 put $n_0 = n_i^2/p_0$ instead.

Interpretation:

- Compensated n-type, $N_d - N_a \gg n_i$: $n_0 \approx N_d - N_a$ and $p_0 = n_i^2/(N_d - N_a)$.
- Compensated p-type, $N_a - N_d \gg n_i$: $p_0 \approx N_a - N_d$ and $n_0 = n_i^2/(N_a - N_d)$.
- Fully compensated, $N_d = N_a$: $n_0 = p_0 = n_i$.

Example (Si, 300 K)

$N_d = 10^{16}$, $N_a = 3 \times 10^{15}$ cm^{-3} gives $n_0 = 7.0 \times 10^{15}$ cm^{-3} and $p_0 = n_i^2/n_0 = 3.2 \times 10^4$ cm^{-3}.

3. Keyword Explanation

1. Compensated semiconductor

A semiconductor with both donors and acceptors in the same region; the free carriers come from the net doping $|N_d - N_a|$.

2. Charge neutrality

In a uniform semiconductor in equilibrium, the total positive charge (holes and ionized donors) equals the total negative charge (electrons and ionized acceptors).

3. Complete ionization

Every donor has given up its electron and every acceptor has accepted one; true in Si at room temperature for non-degenerate doping.

4. Mass-action law

$n_0 p_0 = n_i^2$ in thermal equilibrium for a non-degenerate semiconductor, whatever the doping.

4. Notes and Remarks

- **The equation sheet's form:** the paper's "Some useful equations" prints $n_0 = \frac{N_d - N_a}{2} + \sqrt{\frac{(N_d - N_a)^2}{2} + n_i^2}$. The square inside the root should be $\left(\frac{N_d - N_a}{2}\right)^2$, as Step 4 shows (the 2 is squared too); the result above is Neamen's Eq. 4.60.
- **When the formula fails:** at very low temperature the dopants are not all ionized (freeze-out), so $N_d^+ < N_d$; in degenerate material $n_0 p_0 = n_i^2$ no longer holds.
- **Near intrinsic:** $N_d = 5 \times 10^{10}$, $N_a = 3 \times 10^{10}$ cm^{-3} gives $n_0 = 2.80 \times 10^{10}$ and $p_0 = 8.03 \times 10^9$ cm^{-3}; here $n_0 \neq N_d - N_a$, so the full formula is needed.
- **Also asked in:** term test 1 of this session, 7(a), and the 2020-21 paper, 3(a) (extra reading).

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2.(b) Consider silicon at $T = 300$ K. A Hall effect device is fabricated with the following geometry: $d = 5 \times 10^{-3}$ cm, $W = 5 \times 10^{-2}$ cm and $L = 0.50$ cm. The electrical parameters measured are: $I_x = 0.50$ mA, $V_x = 1.25$ V, and $B_z = 650$ *gauss* $= 6.5 \times 10^{-2}$ *tesla*. The Hall field is $E_H = -16.5$ mV cm^{-1}. Determine: i. the Hall voltage ii. the conductivity type iii. the majority carrier concentration and iv. the majority carrier mobility Marks: 8

1. Theory

1. The Lorentz force (Core Idea)

A charge q moving with velocity $\mathbf{v}$ in a magnetic field $\mathbf{B}$ feels a force at right angles to both (Neamen, Eq. 5.48):

$$\mathbf{F} = q\,\mathbf{v} \times \mathbf{B}$$

In the **Hall effect**, a current I_x flows along a semiconductor bar in the $+x$ direction and a magnetic field B_z points along $+z$. The moving carriers are pushed sideways (along y) and pile up on one face of the bar. Their charge sets up a sideways electric field, the **Hall field** E_H, and a voltage across the bar, the **Hall voltage** V_H (Neamen, Section 5.4).

2. Why the sign tells the carrier type

- **Holes** move in $+x$ (with the current). $\mathbf{F} = (+e)(v_x\hat{x}) \times (B_z\hat{z}) = -ev_xB_z\,\hat{y}$: they are pushed to the $-y$ face ($y = 0$). Positive charge collects at $y = 0$, so the Hall field points in $+y$ and $V_H > 0$.
- **Electrons** move in $-x$ (against the current). $\mathbf{F} = (-e)(-v_x\hat{x}) \times (B_z\hat{z}) = -ev_xB_z\,\hat{y}$: they are pushed to the *same* face, $y = 0$. Now negative charge collects there, the Hall field points in $-y$ and $V_H < 0$.

So **a positive Hall voltage means p-type, a negative one means n-type.** This is the main use of the Hall effect: it measures the carrier type and concentration directly.

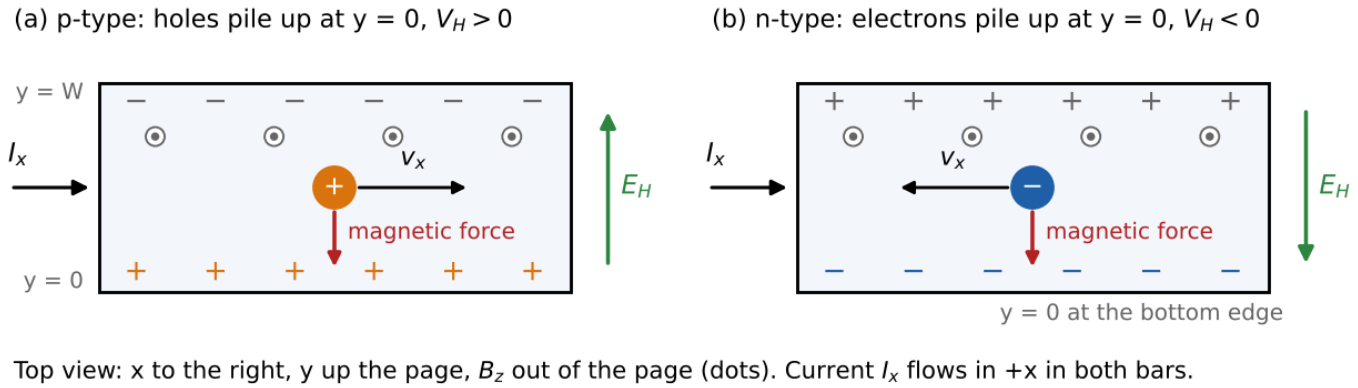


Figure 7: The Hall effect in p-type and n-type bars: carriers pile up at y = 0 (reused from the 2020-21 paper)

Reading the figure. Top view of the bar: x to the right, y up the page, B_z out of the page (dots). In panel (a) holes drift in $+x$, the magnetic force pushes them down to $y = 0$, positive charge collects there and the Hall field E_H (green) points up: $V_H > 0$. In panel (b) electrons drift in $-x$, the force again points down, negative charge collects at $y = 0$ and E_H points down: $V_H < 0$.

3. The force balance

The charge stops piling up when the electric force of the Hall field cancels the magnetic force (Neamen, Eq. 5.49):

$$qE_y = qv_xB_z \quad \Rightarrow \quad E_H = v_xB_z$$

The Hall voltage is the Hall field times the width W of the bar across which it acts (Neamen, Eq. 5.50):

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$$V_H = E_H W$$

4. Concentration and mobility from the measurement

For electrons the drift velocity follows from the current: $I_x = (-e)n(-v_x)(Wd)$, so $v_x = I_x/(enWd)$. Then $V_H = E_H W = -v_x B_z W$ for electrons (Hall field in $-y$), which gives (Neamen, Eqs. 5.55, 5.56)

$$V_H = -\frac{I_x B_z}{ned} \quad \Rightarrow \quad n = -\frac{I_x B_z}{edV_H}$$

where d is the thickness of the bar along z (the magnetic field direction).

The mobility comes from Ohm's law along the bar: $J_x = en\mu_n E_x$ with $J_x = I_x/(Wd)$ and $E_x = V_x/L$ (Neamen, Eq. 5.60):

$$\mu_n = \frac{I_x L}{enV_x W d}$$

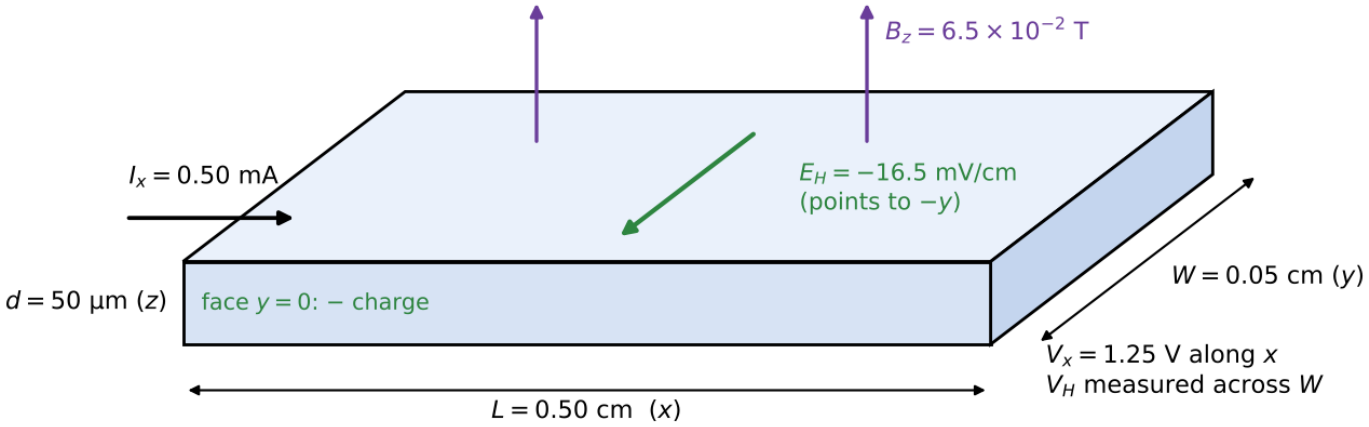


Figure 8: The Hall bar of the question with its dimensions and the measured values

Reading the figure. The bar is $L = 0.50$ cm long (along x, between the two ends where $V_x = 1.25$ V is applied), $W = 0.05$ cm wide (along y, across which the Hall voltage is measured) and $d = 50$ μm thick (along z, parallel to B_z). The current $I_x = 0.50$ mA flows in $+x$. The measured Hall field points in $-y$ (green arrow), so the face $y = 0$ is negative: the carriers are electrons.

2. Answer

Given: Si, $T = 300$ K; $d = 5 \times 10^{-3}$ cm $= 5 \times 10^{-5}$ m; $W = 5 \times 10^{-2}$ cm; $L = 0.50$ cm; $I_x = 0.50$ mA $= 5 \times 10^{-4}$ A; $V_x = 1.25$ V; $B_z = 6.5 \times 10^{-2}$ T; $E_H = -16.5$ mV/cm.

Relationships (Neamen, Section 5.4):

$$V_H = E_H W, \qquad n = -\frac{I_x B_z}{edV_H}, \qquad \mu_n = \frac{I_x L}{enV_x W d}$$

Step 1: Hall voltage

$$V_H = E_H W = (-16.5 \times 10^{-3} \text{ V/cm})(5 \times 10^{-2} \text{ cm}) = -8.25 \times 10^{-4} \text{ V}$$

$$\boxed{V_H = -0.825 \text{ mV}}$$

Step 2: Conductivity type

V_H (and E_H) is **negative**: negative charge has collected on the $y = 0$ face, so the majority carriers are electrons.

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$$\boxed{\text{n-type}}$$

Step 3: Majority carrier (electron) concentration

Use SI units in the formula (d in metres), then convert:

$$n = -\frac{I_x B_z}{e d V_H} = -\frac{(5\times10^{-4})(6.5\times10^{-2})}{(1.6\times10^{-19})(5\times10^{-5})(-8.25\times10^{-4})} = \frac{3.25\times10^{-5}}{6.6\times10^{-27}} = 4.92\times10^{21}\ \text{m}^{-3}$$

$$\boxed{n = 4.92\times10^{15}\ \text{cm}^{-3}}$$

Step 4: Majority carrier (electron) mobility

Now in cm units:

$$\mu_n = \frac{I_x L}{e n V_x W d} = \frac{(5\times10^{-4})(0.50)}{(1.6\times10^{-19})(4.92\times10^{15})(1.25)(5\times10^{-2})(5\times10^{-3})} = 1015\ \text{cm}^2/\text{V s}$$

$$\boxed{\mu_n = 1015\ \text{cm}^2/\text{V s}}$$

Check: $E_x = V_x/L = 2.5$ V/cm, so $v_x = \mu_n E_x = 2.54\times10^3$ cm/s $= 25.4$ m/s, and $E_H = v_x B_z = (25.4)(0.065) = 1.65$ V/m $= 16.5$ mV/cm, the measured value.

Final Answer:

- (i) $V_H = -0.825$ mV
- (ii) n-type
- (iii) $n = 4.92\times10^{15}$ cm^{-3}
- (iv) $\mu_n = 1015$ cm^2/V s

3. Keyword Explanation

1. Hall effect

The sideways voltage that appears across a current-carrying semiconductor in a perpendicular magnetic field.

2. Hall field

The transverse electric field E_H built up by the deflected carriers; in steady state it balances the magnetic force, $E_H = v_x B_z$.

3. Hall voltage

$V_H = E_H W$: positive for p-type, negative for n-type in the standard geometry.

4. Notes and Remarks

- **Book check:** this is Neamen's Problem 5.49 word for word. The book's answers (Appendix H) are (a) $V_H = -0.825$ mV, (b) n-type, (c) $n = 4.92\times10^{15}$ cm^{-3}, (d) $\mu_n = 1015$ cm^2/V s, the same as here.
- **Units:** in $n = -I_x B_z/(e d V_H)$, B_z in tesla needs d in metres, giving n in m^{-3}; mixing cm and T gives an answer a factor of 100 wrong.
- The mobility, 1015 cm^2/V s, is a little below the value Neamen's Figure 5.3 gives for silicon with 5×10^{15} cm^{-3} donors (about 1300 cm^2/V s); the problem's numbers are made up for practice. The conductivity is $\sigma = e n \mu_n = 0.80$ S/cm.
- The 2020-21 paper's 5(d) is another Hall effect problem (extra reading).

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2.(c) Using proper graph, explain how carrier concentration varies with temperature. Marks: 5

1. Theory

1. Two sources of free electrons (Core Idea)

In an n-type semiconductor there are two ways a free electron can appear in the conduction band:

1. **From a donor:** the donor's extra electron is bound by only $E_c - E_d \approx 0.045$ eV (P in Si). Very little energy frees it.
2. **From the valence band:** an electron is lifted across the whole band gap, $E_g = 1.12$ eV in Si. This makes an electron-hole pair, and its number is the intrinsic concentration n_i.

Temperature supplies the energy for both. Because the two energies are so different, they happen in different temperature ranges, which gives three regions.

2. The intrinsic concentration grows very fast with temperature

From $n_i^2 = N_c N_v \exp(-E_g/kT)$ with $N_c, N_v \propto T^{3/2}$ (Neamen, Eq. 4.23):

$$n_i = \sqrt{N_c N_v}\exp\left(-\frac{E_g}{2kT}\right) \propto T^{3/2}\exp\left(-\frac{E_g}{2kT}\right)$$

The exponential dominates. For Si, n_i is about 10^{10} cm^{-3} at 300 K but about 10^{14} cm^{-3} near 500 to 550 K: four decades for less than a factor of 2 in temperature.

3. The three regions

- **Freeze-out (low T, below about 100 K):** kT is too small to free even the donor electrons, so many electrons stay bound to their donors: $n_0 < N_d$. At $T \to 0$, $n_0 \to 0$ (Neamen, Section 4.4.2).
- **Extrinsic or saturation region (about 100 K to 500 K for $N_d = 10^{15}$):** all donors are ionized, but n_i is still much smaller than N_d, so $n_0 \approx N_d$, flat with temperature. Devices are designed to work here.
- **Intrinsic region (high T):** n_i becomes comparable with and then larger than N_d; electron-hole pairs created across the gap outnumber the donor electrons and $n_0 \approx n_i$, rising steeply.

The full curve follows the neutrality result $n_0 = \frac{N_d}{2} + \sqrt{\left(\frac{N_d}{2}\right)^2 + n_i^2}$ (complete ionization) joined to the freeze-out at low temperature.

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2. Answer

In an n-type semiconductor the electron concentration n_0 depends on temperature as shown below (Neamen, Figure 4.16).

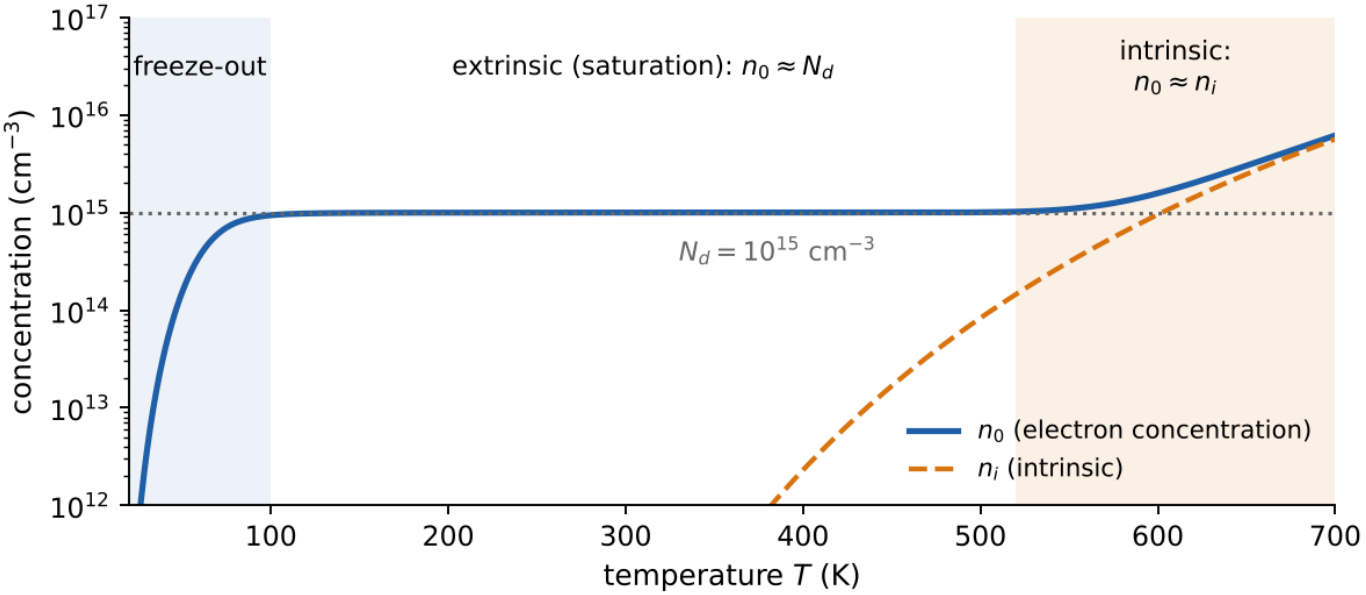


Figure 9: Electron concentration against temperature for n-type Si with Nd = 1e15 cm-3: freeze-out, extrinsic and intrinsic regions

Reading the figure. The solid blue curve is n_0 for silicon with $N_d = 10^{15}$ cm^{-3} (phosphorus, $E_c - E_d = 0.045$ eV), the dashed orange curve is n_i and the dotted grey line is N_d. On the left (shaded blue) the electrons freeze back onto the donors and n_0 falls; in the middle n_0 stays at N_d over a wide range; on the right (shaded orange) n_i overtakes N_d and n_0 follows n_i upward.

1. Freeze-out region (low temperature): the thermal energy kT is small compared with the donor ionization energy $E_c - E_d$, so most donor electrons stay bound: $n_0 \ll N_d$, and $n_0 \to 0$ as $T \to 0$ K. Here

$$\frac{n_d}{n_d + n_0} = \frac{1}{1 + \dfrac{N_c}{2N_d} \exp\left[-\dfrac{(E_c - E_d)}{kT}\right]}$$

(Neamen, Eq. 4.55) is the fraction of electrons still on the donors; it is near 1 at low T.

2. Extrinsic (saturation) region: once kT is a fair fraction of $E_c - E_d$, every donor is ionized (complete ionization), while $n_i \ll N_d$. So

$$n_0 \approx N_d, \qquad p_0 = \frac{n_i^2}{N_d}$$

n_0 is almost independent of temperature. For $N_d = 10^{15}$ cm^{-3} this covers roughly 100 K to 500 K, the working range of devices.

3. Intrinsic region (high temperature): $n_i \propto T^{3/2} e^{-E_g/2kT}$ rises steeply and becomes comparable with N_d; the full neutrality result

$$n_0 = \frac{N_d}{2} + \sqrt{\left(\frac{N_d}{2}\right)^2 + n_i^2}$$

then rises above N_d, and when $n_i \gg N_d$, $n_0 \approx p_0 \approx n_i$: the material behaves as intrinsic and loses its n-type character. For Si with $N_d = 10^{15}$ cm^{-3} this starts around 500 to 550 K.

Conclusion: at low temperature n_0 is limited by the ionization of donors, in the middle it equals the doping, and at high temperature it is set by thermal generation across the gap. The higher the doping, the higher the temperature at which intrinsic behaviour begins.

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3. Keyword Explanation

1. Freeze-out

The low-temperature state in which donor (or acceptor) atoms are not ionized, so the free carrier concentration is below the doping.

2. Extrinsic region

The temperature range in which all dopants are ionized and the majority carrier concentration equals the net doping.

3. Intrinsic temperature

The temperature at which n_i becomes comparable with the doping, so the material starts to behave as intrinsic.

4. Notes and Remarks

- The figure is drawn with Neamen's Si values (N_c, $N_v \propto T^{3/2}$, $E_g = 1.12$ eV held constant); real silicon's gap shrinks slightly with temperature, which moves the intrinsic region a little lower.
- Neamen's Figure 4.16 plots the same three regions against $1000/T$; the shape is the same mirrored.
- The ratio n_0/N_d against temperature, from freeze-out to intrinsic, is also drawn in the 2020-21 paper's 6(b) (extra reading).
- A p-type semiconductor behaves the same way with holes and acceptors.
- The same question is 1(a) of the 2018–19 paper (2018–19 1(a)), extra reading only.

2.(d) Sketch and briefly describe energy band diagram of compensated semiconductor. Marks: 6

1. Theory

1. Energy bands and dopant levels (Core Idea)

In a crystal the allowed electron energies form **bands**. In a semiconductor two matter: the **valence band** (top edge E_v), full at 0 K, and the **conduction band** (bottom edge E_c), empty at 0 K, separated by the band gap E_g (1.12 eV in Si).

Dopants add single energy levels **inside** the gap:

- a **donor level** E_d just below E_c (about 0.045 eV for P in Si). When the donor gives its electron to the conduction band it becomes N_d^+;
- an **acceptor level** E_a just above E_v (about 0.045 eV for B in Si). When it takes an electron from the valence band it becomes N_a^- and a hole appears.

The **Fermi level** E_F marks where a state is half likely to be filled; the closer it is to E_c, the more electrons there are (Neamen, Eq. 4.11). In an intrinsic semiconductor $E_F = E_{Fi}$, near the middle of the gap.

2. What compensation does to the electrons

When both N_d donors and N_a acceptors are present, the electrons of the donors have a lower-energy place to go than the conduction band: the **empty acceptor levels**. So, before anything else, $\min(N_d, N_a)$ donor electrons fall into acceptor states (Neamen, Figure 4.14). Then:

- if $N_d > N_a$: $N_d - N_a$ donor electrons are left over and go into the conduction band. The material is n-type and E_F lies above E_{Fi}, with $E_F - E_{Fi} = kT \ln[(N_d - N_a)/n_i]$;
- if $N_a > N_d$: $N_a - N_d$ acceptors stay empty and take electrons from the valence band, making holes. The material is p-type and E_F lies below E_{Fi};
- if $N_d = N_a$: every donor electron is used up by an acceptor; only thermally generated pairs remain, so $n_0 = p_0 = n_i$ and $E_F = E_{Fi}$.

Both dopants are fully **ionized** in every case: the donors are positive, the acceptors negative. Compensation does not remove the ions; it only cancels their free carriers.

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2. Answer

A compensated semiconductor contains both donor and acceptor impurities in the same region (Neamen, Section 4.5).

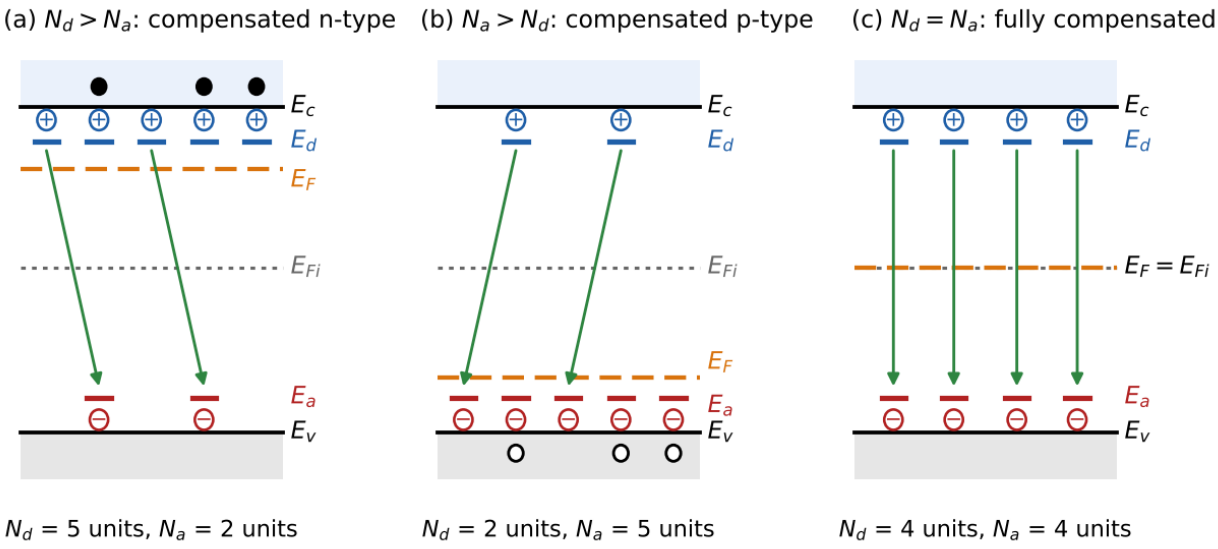


Figure 10: Energy band diagrams of a compensated semiconductor for Nd > Na, Na > Nd and Nd = Na (reused from the 2020-21 paper)

Reading the figure. In each panel the black lines are E_c and E_v, the short blue dashes the donor level E_d, the short red dashes the acceptor level E_a, the dotted grey line E_{Fi} and the dashed orange line E_F. Circled + signs are ionized donors, circled − signs ionized acceptors, filled dots free electrons and open circles holes. The green arrows show donor electrons dropping into acceptor states.

Description:

1. **Electrons move from donors to acceptors.** The donor level E_d lies near E_c and the acceptor level E_a near E_v. A donor electron can lower its energy a great deal by dropping into an empty acceptor state, so $\min(N_d, N_a)$ electrons do so (green arrows). Every donor becomes N_d^+ and every acceptor N_a^-.
2. **Case (a), $N_d > N_a$ (compensated n-type):** the $N_d - N_a$ remaining donor electrons go to the conduction band, $n_0 \approx N_d - N_a$, and E_F sits above E_{Fi}: $E_F - E_{Fi} = kT \ln[(N_d - N_a)/n_i]$.
3. **Case (b), $N_a > N_d$ (compensated p-type):** the $N_a - N_d$ acceptors still empty take electrons from the valence band, $p_0 \approx N_a - N_d$, and E_F sits below E_{Fi}: $E_{Fi} - E_F = kT \ln[(N_a - N_d)/n_i]$.
4. **Case (c), $N_d = N_a$ (fully compensated):** no free carriers come from the dopants; $n_0 = p_0 = n_i$ and $E_F = E_{Fi}$, as in intrinsic material, although the crystal is full of ions.
5. **Effect on transport:** the ions of both kinds scatter carriers, so the mobility depends on the **total** ionized impurity $N_I = N_d^+ + N_a^-$, not on the net doping. A compensated sample therefore has a lower mobility than an uncompensated one with the same carrier concentration (used in 3(d)).

3. Keyword Explanation

1. Donor level

The energy E_d of the extra electron of a donor atom, about 0.045 eV below E_c in Si.

2. Acceptor level

The energy E_a of the empty state of an acceptor atom, about 0.045 eV above E_v in Si.

3. Fully compensated

$N_d = N_a$: the dopants cancel, and the material has intrinsic carrier concentrations although it is heavily ionized.

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4. Notes and Remarks

- The carrier concentrations of each case come from the neutrality result derived in 2(a).
- The same diagram is the answer to 2(a) of the 2020-21 paper (2(a), extra reading).
- Compensation is how an n-type wafer is turned p-type: acceptors are diffused in at a concentration above N_d, and the region becomes compensated p-type.

5. Final Concise Answer

(a) Charge neutrality $n_0 + N_a = p_0 + N_d$ (complete ionization) with $p_0 = n_i^2/n_0$ gives $n_0^2 - (N_d - N_a)n_0 - n_i^2 = 0$, so $n_0 = \frac{N_d - N_a}{2} + \sqrt{\left(\frac{N_d - N_a}{2}\right)^2 + n_i^2}$ and $p_0 = \frac{N_a - N_d}{2} + \sqrt{\left(\frac{N_a - N_d}{2}\right)^2 + n_i^2}$ (Neamen Eqs. 4.60, 4.62); $n_0 \approx N_d - N_a$ when $N_d - N_a \gg n_i$.

(b) $V_H = E_H W = -0.825$ mV; negative, so n-type; $n = -I_x B_z/(edV_H) = 4.92 \times 10^{15}$ cm^{-3}; $\mu_n = I_x L/(enV_x W d) = 1015$ cm^2/V s (Neamen Problem 5.49).

(c) n_0 against T has three regions: freeze-out at low T ($n_0 < N_d$, donors not ionized), the extrinsic plateau ($n_0 = N_d$, about 100 to 500 K for 10^{15} cm^{-3}), and the intrinsic region at high T where $n_i \propto T^{3/2} e^{-E_g/2kT}$ overtakes N_d and $n_0 \to n_i$.

(d) Donor electrons drop into the acceptor levels, so all dopants are ionized; for $N_d > N_a$ the rest go to the conduction band ($n_0 \approx N_d - N_a$, E_F above E_{Fi}), for $N_a > N_d$ the empty acceptors make holes ($p_0 \approx N_a - N_d$, E_F below E_{Fi}), and for $N_d = N_a$ the material is intrinsic-like with $E_F = E_{Fi}$.

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3. (a) Sketch and explain density of states functions, Fermi distribution, and carrier concentration distribution curves of n-type material in a single graph. Marks: 8
   (b) Explain how barrier potential maintains equilibrium p, n regions. Marks: 4
   (c) Explain dielectric relaxation time. Find the equation for dielectric relaxation. Marks: 7
   (d) Consider a compensated n-type silicon at $T = 300$ K with a conductivity of ρ [*sic*: should be σ] $= 16\ (\Omega - \text{cm})^{-1}$ and an acceptor doping concentration of 10^{17} cm^{-3}. Determine the doner [*sic*: should be donor] concentration and the electron mobility. Marks: 6

3.(a) Sketch and explain density of states functions, Fermi distribution, and carrier concentration distribution curves of n-type material in a single graph. Marks: 8

1. Theory

1. Counting electrons needs two things (Core Idea)

To know how many electrons are in the conduction band, we need to know two things at every energy E:

1. **How many states exist** at that energy: the **density of states** $g(E)$, the number of allowed quantum states per unit energy per unit volume.
2. **How likely each state is to be filled**: the **Fermi-Dirac function** $f_F(E)$.

The number of electrons per unit energy is their product, and the total is the area under it (Neamen, Eqs. 4.1 and 4.3):

$$n(E) = g_c(E) f_F(E), \qquad n_0 = \int_{E_c}^{\infty} g_c(E) f_F(E)\, dE$$

For holes, a hole is an **empty** state, so the probability is $1 - f_F(E)$ (Neamen, Eq. 4.2):

$$p(E) = g_v(E)\,[1 - f_F(E)]$$

2. The density of states

Electrons near the band edges behave like free particles with an effective mass. Counting the quantum states of a particle in a box gives (Neamen, Eq. 3.72)

$$g_c(E) = \frac{4\pi(2m_n^*)^{3/2}}{h^3}\sqrt{E - E_c} \quad (E \geq E_c), \qquad g_v(E) = \frac{4\pi(2m_p^*)^{3/2}}{h^3}\sqrt{E_v - E} \quad (E \leq E_v)$$

Both are zero at the band edge and grow as the square root of the distance into the band; there are no states in the gap (dopant levels aside).

3. The Fermi-Dirac function

$$f_F(E) = \frac{1}{1 + \exp\left(\dfrac{E - E_F}{kT}\right)}$$

It is 1 well below E_F, exactly $\frac{1}{2}$ at E_F, and falls to 0 a few kT above E_F. Far above E_F (more than about $3kT$) it is close to the Boltzmann tail $e^{-(E-E_F)/kT}$.

4. What changes in n-type material

Doping does not change g_c or g_v; it moves E_F. In n-type material E_F lies **above** the middle of the gap, closer to E_c (Neamen, Figure 4.8). So:

- at E_c, f_F is much larger than in intrinsic material → many electrons;
- at E_v, $1 - f_F$ is much smaller → few holes.

The product $g_c f_F$ is zero at E_c (no states), rises as $\sqrt{E - E_c}$, and then falls as $e^{-(E-E_F)/kT}$: a peaked

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curve close to the band edge, with its maximum at $E_c + kT/2$. The same holds for holes, but with a much smaller area.

2. Answer

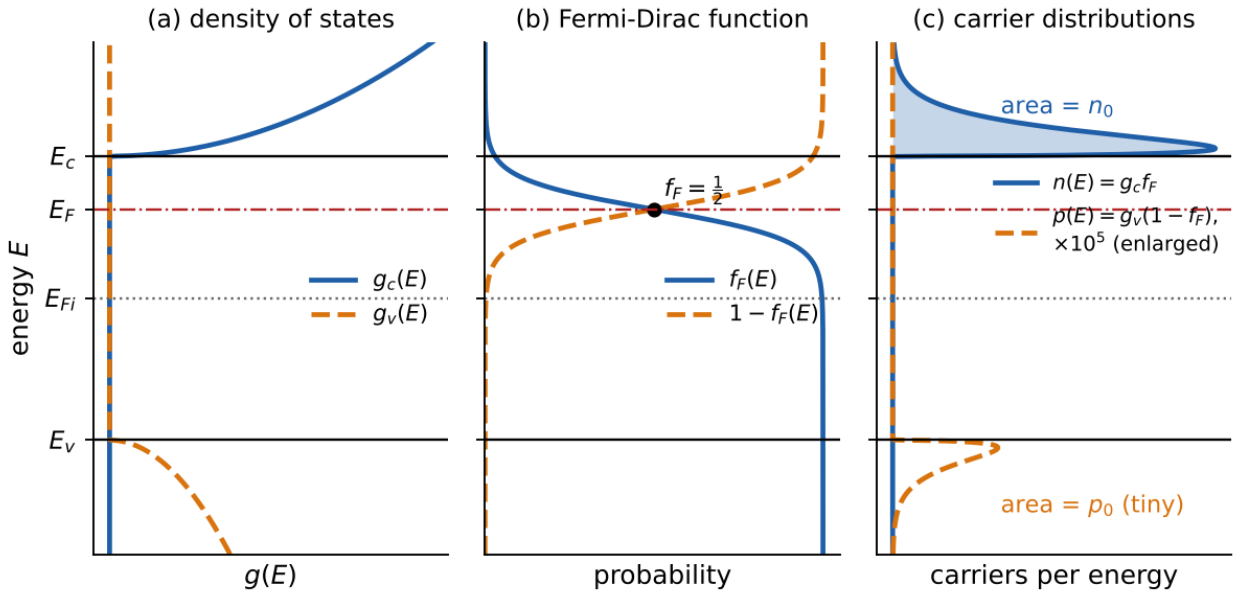


Figure 11: Density of states, Fermi-Dirac function and carrier distributions of n-type silicon on one energy axis

Reading the figure. Energy runs upward on all three panels, which share one axis with E_c, E_F, E_{Fi} and E_v marked. Left panel: the density of states, $g_c(E)$ (solid blue) above E_c and $g_v(E)$ (dashed orange) below E_v, both $\propto \sqrt{|E - E_{edge}|}$, zero inside the gap. Middle panel: the Fermi-Dirac function $f_F(E)$ (solid blue) and $1 - f_F(E)$ (dashed orange); $f_F = \frac{1}{2}$ at E_F, which lies well above E_{Fi} because the material is n-type. Right panel: the electron distribution $n(E) = g_c f_F$ (solid blue, filled) just above E_c, and the hole distribution $p(E) = g_v(1 - f_F)$ (dashed orange) just below E_v, which is drawn 10^5 times enlarged so that it can be seen at all: the areas are n_0 and p_0, and $n_0 \gg p_0$.

Explanation:

1. **Density of states** (Neamen, Eq. 3.72): $g_c(E) = \frac{4\pi(2m_n^*)^{3/2}}{h^3}\sqrt{E - E_c}$ for $E \geq E_c$ and $g_v(E) = \frac{4\pi(2m_p^*)^{3/2}}{h^3}\sqrt{E_v - E}$ for $E \leq E_v$. Zero in the band gap, it grows as the square root of the energy into each band. It is a property of the crystal and does not change with doping.
2. **Fermi-Dirac distribution:** $f_F(E) = 1/\{1 + \exp[(E - E_F)/kT]\}$ is the probability that a state at E is occupied. In n-type material the donors raise E_F above E_{Fi} towards E_c (for $N_d = 10^{16}$ cm^{-3}, $E_c - E_F = 0.21$ eV). So f_F is relatively large at E_c and $1 - f_F$ is very small at E_v. The function is symmetric about E_F.
3. **Carrier distributions** (Neamen, Eqs. 4.1, 4.2): $n(E) = g_c(E) f_F(E)$ and $p(E) = g_v(E)[1 - f_F(E)]$. Each starts at zero at its band edge (no states), peaks about $kT/2$ into the band and then dies away exponentially (the Boltzmann tail of f_F). Most carriers are within a few kT of the band edge.
4. **Concentrations:** the areas under the curves are the concentrations, $n_0 = \int g_c f_F \, dE = N_c \exp[-(E_c - E_F)/kT]$ and $p_0 = N_v \exp[-(E_F - E_v)/kT]$ (Neamen, Eqs. 4.11, 4.19). Because E_F is closer to E_c, the electron area is much larger than the hole area: $n_0 \gg p_0$, with $n_0 p_0 = n_i^2$.

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3. Keyword Explanation

1. Density of states

The number of allowed electron states per unit energy per unit volume, $g(E)$, in states/(eV cm^3) or J^{-1}m^{-3}.

2. Fermi-Dirac function

The probability $f_F(E)$ that a state at energy E is occupied by an electron in thermal equilibrium.

3. Effective density of states

$N_c = 2(2\pi m_n^* kT/h^2)^{3/2}$: the number of states that, all placed at E_c, would give the same n_0; 2.8×10^{19} cm^{-3} for Si at 300 K.

4. Notes and Remarks

- In p-type material the picture is mirrored: E_F is near E_v, the hole area is large and the electron area small (Neamen, Figure 4.9). In intrinsic material the two areas are equal.
- The peak of $n(E)$: with the Boltzmann form, $n(E) \propto \sqrt{x}\, e^{-x}$ with $x = (E - E_c)/kT$, whose maximum is at $x = \frac{1}{2}$, i.e. $E = E_c + kT/2$.
- The figure uses Neamen's Si values, $m_n^* = 1.08m_0$, $m_p^* = 0.56m_0$, $N_d = 10^{16}$ cm^{-3}, $T = 300$ K, with the energy spread of f_F widened for visibility; the shape, not the scale, is what the question asks for.

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3.(b) Explain how barrier potential maintains equilibrium p, n regions. Marks: 4

1. Theory

1. Two currents in a semiconductor (Core Idea)

Carriers move for two reasons:

- **Drift:** an electric field pushes holes along the field and electrons against it. $J_{drift} = e\mu_n nE + e\mu_p pE$.
- **Diffusion:** where the concentration changes with position, random thermal motion carries more carriers from the crowded side to the empty side. $J_{diff} = eD_n\, dn/dx - eD_p\, dp/dx$.

In thermal equilibrium no current can flow, so for each carrier the two must cancel exactly.

2. What happens when p and n are joined

On the p side there are many holes ($p_{p0} = N_a$) and very few electrons; on the n side, many electrons ($n_{n0} = N_d$) and very few holes. At the moment of contact, holes diffuse into the n side and electrons into the p side. Each carrier that leaves uncovers a fixed ion it was neutralizing: negative acceptor ions on the p side and positive donor ions on the n side. This **space charge region** produces an electric field pointing from n to p, and a potential difference, the **built-in potential barrier** V_{bi} (Neamen, Section 7.2).

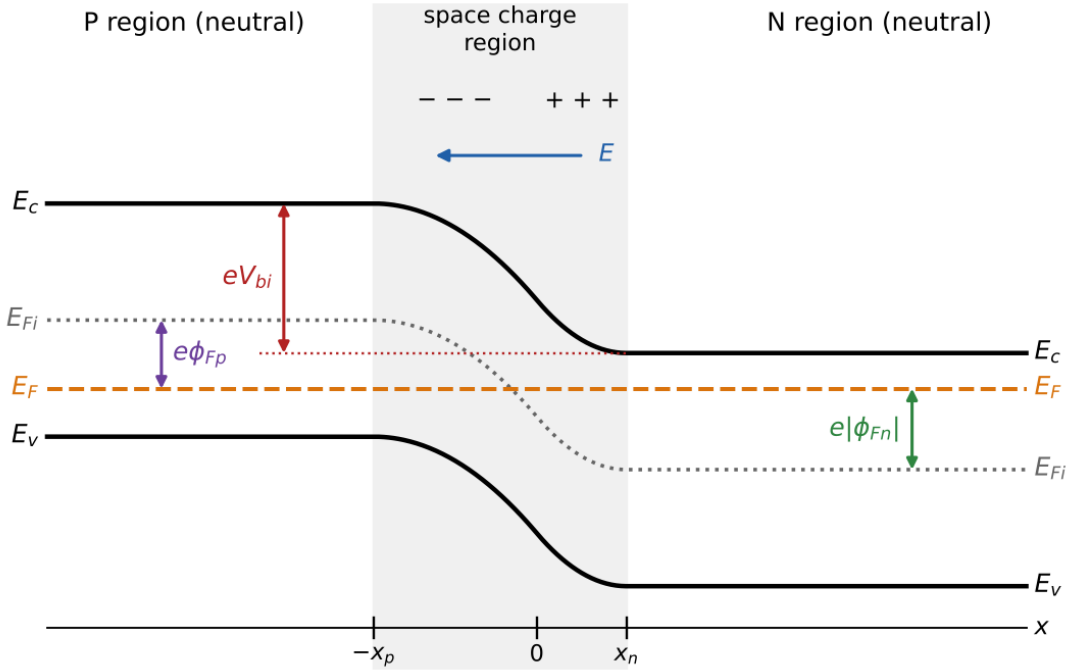


Figure 12: Equilibrium energy band diagram of a pn junction with the built-in barrier (reused from the 2020-21 paper)

Reading the figure. The dashed orange line is the Fermi level E_F, flat through the whole device in equilibrium. On the p side E_F is near E_v, on the n side near E_c, so the bands must bend inside the grey space charge region. The red arrow is the barrier eV_{bi}: an electron on the n side must climb this energy to reach the p side, and a hole on the p side must "climb down" the same step (holes gain energy downward). The blue arrow is the built-in field E, from the positive (n) side to the negative (p) side.

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2. Answer

When p and n regions are joined, holes diffuse from p to n and electrons from n to p, leaving uncovered acceptor ions (−) on the p side and donor ions (+) on the n side. These ions create an electric field from n to p and a potential barrier V_{bi} across the space charge region (see the figure above). The barrier keeps the two regions in equilibrium as follows:

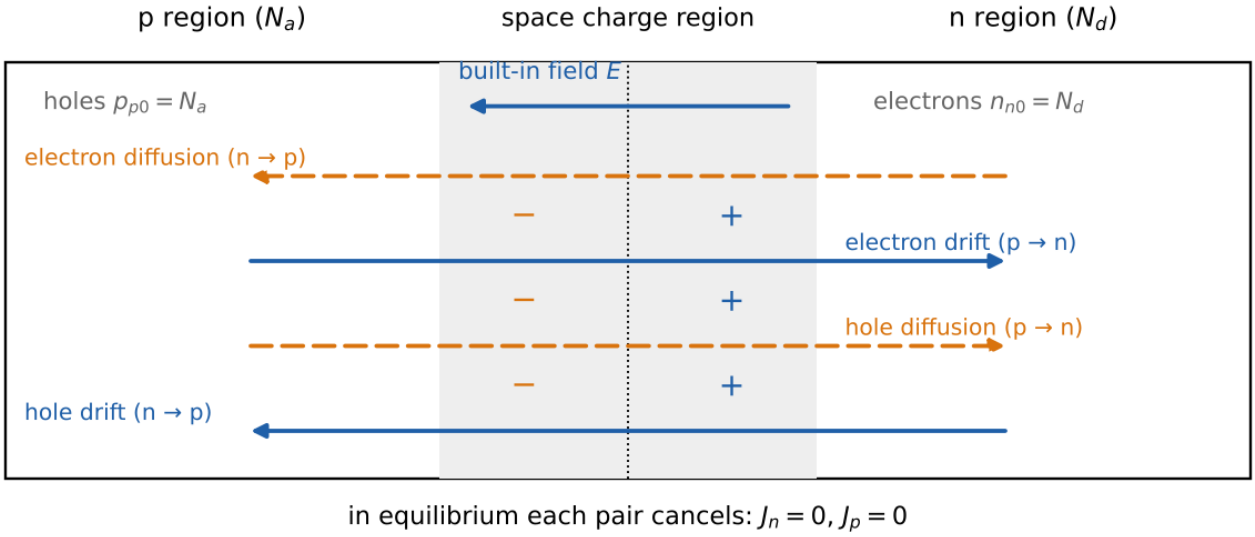


Figure 13: Diffusion and drift fluxes across the space charge region cancel in equilibrium

Reading the figure. Inside the space charge region the built-in field E (blue arrow, from n to p) acts. For electrons (top row) the diffusion flux from n to p (dashed orange arrow) is cancelled by the drift flux back from p to n (solid blue arrow); for holes (bottom row) the diffusion flux from p to n is cancelled by the drift flux from n to p. The ions shown are fixed and do not move.

1. **The field opposes diffusion.** The built-in field pushes electrons back to the n side and holes back to the p side, so a **drift current** flows opposite to the **diffusion current**. The barrier grows until, for each carrier type separately, drift exactly cancels diffusion:

$$J_n = e\mu_n nE + eD_n\frac{dn}{dx} = 0, \qquad J_p = e\mu_p pE - eD_p\frac{dp}{dx} = 0$$

2. **Only energetic carriers cross.** Seen as energies, an electron must have at least eV_{bi} of extra energy to climb from the n side into the p side. Only the Boltzmann fraction $e^{-eV_{bi}/kT}$ of the majority electrons can. That is exactly the right number to match the minority electrons on the p side (Neamen, Eq. 8.4):

$$n_{p0} = n_{n0}\exp\left(-\frac{eV_{bi}}{kT}\right)$$

The same holds for holes, $p_{n0} = p_{p0}\exp(-eV_{bi}/kT)$.

3. **The Fermi level is flat.** Zero net current means E_F is constant through the device; the bands bend by eV_{bi} to make the two sides' Fermi levels line up (Neamen, Eq. 7.10):

$$V_{bi} = \frac{kT}{e}\ln\left(\frac{N_a N_d}{n_i^2}\right)$$

4. **Result:** the p region stays p-type ($p_{p0} = N_a$) and the n region stays n-type ($n_{n0} = N_d$) right up to the edges of the space charge region; the large concentration differences are held in place by the barrier, with no net flow of either carrier.

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3. Keyword Explanation

1. Built-in potential barrier

The potential difference V_{bi} across the space charge region of a pn junction in equilibrium; it equals the difference of the two sides' Fermi levels before contact divided by e.

2. Space charge region

The region around the junction emptied of mobile carriers, holding the uncovered donor and acceptor ions.

3. Drift and diffusion currents

Current driven by an electric field, and current driven by a concentration gradient.

4. Notes and Remarks

- **What a bias does:** a forward voltage lowers the barrier to $V_{bi} - V_a$, so diffusion wins and current flows; a reverse voltage raises it, and only the tiny drift of minority carriers flows. This is the origin of rectification (5(b)).
- The barrier cannot be measured with a voltmeter across an unbiased diode: the contact potentials at the metal contacts cancel it.

3.(c) Explain dielectric relaxation time. Find the equation for dielectric relaxation. Marks: 7

1. Theory

1. A semiconductor does not keep a net charge (Core Idea)

Suppose some extra holes are suddenly injected into one small region of an n-type semiconductor, with no electrons to match them. That region now has a **net positive charge** $\rho = e\,\delta p$. Its electric field attracts electrons from the surrounding majority carriers. They flow in and neutralize the charge. The question is how fast.

The answer is the **dielectric relaxation time** τ_d: the time constant with which a net charge density in a conductor or semiconductor decays to zero by the flow of majority carriers (Neamen, Section 6.3.4). It is extremely short, about a picosecond, which is why we can assume **quasi-neutrality** ($\delta n = \delta p$) in the ambipolar transport equation.

2. The three laws we need

- **Ohm's law** (the current is drift of majority carriers): $\mathbf{J} = \sigma\mathbf{E}$ (Neamen, Eq. 6.72).
- **Poisson's equation** (Gauss's law in a dielectric): $\nabla \cdot \mathbf{E} = \rho/\epsilon$.
- **Continuity of charge** (charge cannot vanish; a decrease inside a volume is an outflow of current): $\nabla \cdot \mathbf{J} = -\partial\rho/\partial t$ (Neamen, Eq. 6.73).

Here σ is the conductivity and $\epsilon = \epsilon_r\epsilon_0$ the permittivity of the semiconductor. Generation and recombination are left out: they are far slower than the effect.

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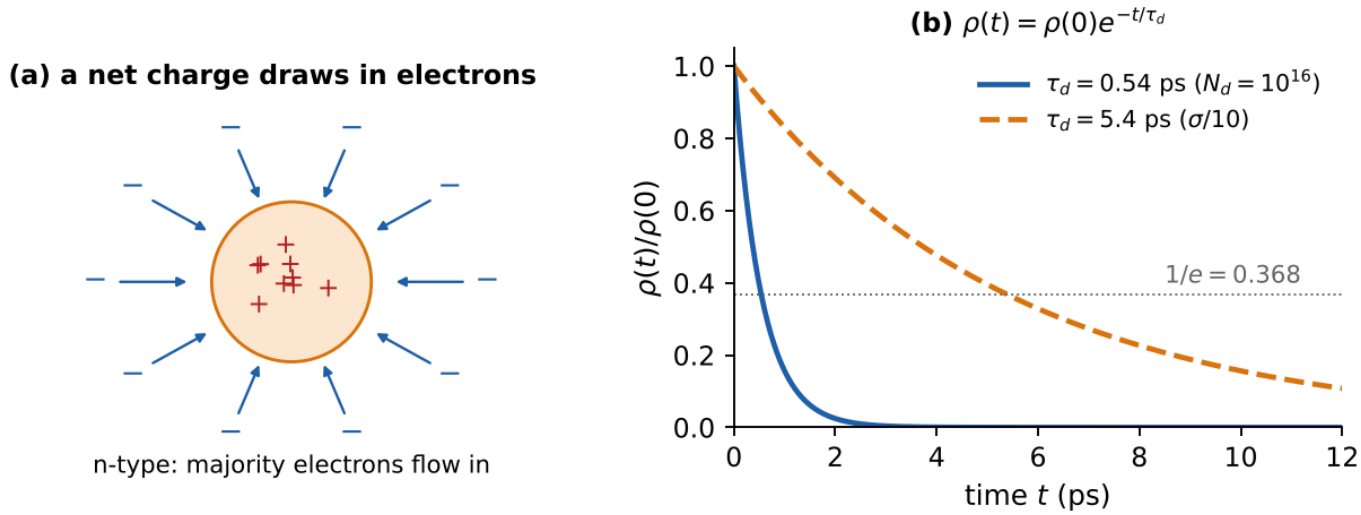


Figure 14: Decay of a net charge density with the dielectric relaxation time constant

Reading the figure. Panel (a): an injected hole cloud (+) in n-type material attracts the surrounding electrons (−) inwards, which neutralizes it. Panel (b): the net charge density $\rho(t)/\rho(0)$ falls exponentially, e^{-t/τ_d}; the solid blue curve is $N_d = 10^{16}$ cm^{-3} silicon ($\tau_d = 0.54$ ps), the dashed orange curve a ten times lower conductivity ($\tau_d = 5.4$ ps). After one τ_d, 36.8 % of the charge is left; after $5\tau_d$, less than 1 %.

2. Answer

Dielectric relaxation time. When a net charge density ρ appears inside a semiconductor (for example injected excess holes with no matching electrons), its field draws majority carriers in to neutralize it. The net charge decays exponentially, $\rho(t) = \rho(0)e^{-t/\tau_d}$, with the time constant $\tau_d = \epsilon/\sigma$, the **dielectric relaxation time**. For ordinary doping it is of the order of picoseconds, much shorter than carrier lifetimes, so excess electrons and holes always move together with $\delta n = \delta p$ (quasi-neutrality).

Derivation (Neamen, Section 6.3.4):

Step 5 solves a general first-order equation and is marked (Basic); a student who knows it can skip it.

Step 1: Start from the continuity of charge

Charge cannot be created or destroyed, so the rate at which charge density falls in a small volume equals the net current flowing out of it (Neamen, Eq. 6.73):

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho}{\partial t}$$

Step 2: The current is a drift current (Ohm's law)

The majority carriers move under the field of the net charge (Neamen, Eq. 6.72):

$$\mathbf{J} = \sigma \mathbf{E}$$

Take the divergence of both sides; σ is uniform, so it comes outside:

$$\nabla \cdot \mathbf{J} = \sigma \, \nabla \cdot \mathbf{E}$$

Step 3: Poisson's equation gives $\nabla \cdot \mathbf{E}$

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon}$$

so

$$\nabla \cdot \mathbf{J} = \frac{\sigma \rho}{\epsilon}$$

(Neamen, Eq. 6.74.)

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Step 4: Equate the two expressions for $\nabla \cdot \mathbf{J}$

From Steps 1 and 3:

$$\frac{\sigma\rho}{\epsilon} = -\frac{\partial\rho}{\partial t}$$

At a fixed point, ρ depends only on time, so the partial derivative is a total derivative (Neamen, Eqs. 6.75 and 6.76):

$$\frac{d\rho}{dt} + \frac{\sigma}{\epsilon}\rho = 0$$

Step 5 (Basic): Solve the first-order equation

Separate the variables: $\dfrac{d\rho}{\rho} = -\dfrac{\sigma}{\epsilon}dt$. Integrate from 0 to t:

$$\ln\frac{\rho(t)}{\rho(0)} = -\frac{\sigma}{\epsilon}t \quad \Rightarrow \quad \rho(t) = \rho(0)\exp\left(-\frac{t}{\tau_d}\right)$$

(Neamen, Eq. 6.77.)

Step 6: The time constant

$$\boxed{\tau_d = \frac{\epsilon}{\sigma}}$$

(Neamen, Eq. 6.78.) This is the **dielectric relaxation time constant**.

Steps to remember

Continuity $\nabla \cdot J = -\partial\rho/\partial t \rightarrow$ Ohm $J = \sigma E \rightarrow$ Poisson $\nabla \cdot E = \rho/\epsilon \rightarrow d\rho/dt + (\sigma/\epsilon)\rho = 0 \rightarrow \rho = \rho(0)e^{-t/\tau_d}$, $\tau_d = \epsilon/\sigma$.

Step 7: A number (Neamen, Example 6.6)

n-type Si with $N_d = 10^{16}$ cm^{-3}, $\mu_n \approx 1200$ cm^2/V s:

$$\sigma = e\mu_n N_d = (1.6\times10^{-19})(1200)(10^{16}) = 1.92\ (\Omega\text{-cm})^{-1}$$

$$\tau_d = \frac{\epsilon_r\epsilon_0}{\sigma} = \frac{(11.7)(8.85\times10^{-14})}{1.92} = 5.39\times10^{-13}\ \text{s} = 0.539\ \text{ps}$$

$$\boxed{\tau_d = 0.539\ \text{ps}}$$

This is about 2×10^5 times shorter than a typical minority carrier lifetime (10^{-7} s), so a net charge disappears long before any recombination happens: the excess electron and hole distributions are always neutral.

3. Keyword Explanation

1. Dielectric relaxation time

$\tau_d = \epsilon/\sigma$: the time constant with which a net charge in a conducting medium is neutralized by the flow of majority carriers.

2. Quasi-neutrality

The condition $\delta n \approx \delta p$ for excess carriers, justified because any charge imbalance relaxes in τ_d.

3. Continuity equation

The statement that charge is conserved: $\nabla \cdot \mathbf{J} + \partial\rho/\partial t = 0$.

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4. Notes and Remarks

- **Book check:** Step 7 is Neamen's Example 6.6; its answer is $\tau_d = 5.39 \times 10^{-13}$ s $= 0.539$ ps, as here.
- For the material of 3(d), $\sigma = 16$ S/cm gives $\tau_d = 1.035 \times 10^{-12}/16 = 6.5 \times 10^{-14}$ s: higher conductivity, faster relaxation.
- In a good metal τ_d is about 10^{-19} s; in an insulator it can be hours. The same formula explains why a charged insulator stays charged.
- The charge does not decay by recombination: the majority carriers simply move. Only later do the excess carriers (now neutral) recombine with the lifetime τ.

3.(d) Consider a compensated n-type silicon at $T = 300$ K with a conductivity of ρ [*sic*: should be σ] $= 16\ (\Omega - \text{cm})^{-1}$ and an acceptor doping concentration of 10^{17} cm^{-3}. Determine the doner [*sic*: should be donor] concentration and the electron mobility. Marks: 6

1. Theory

1. Conductivity (Core Idea)

The conductivity says how much current density a field produces, $J = \sigma E$. Both carriers contribute (Neamen, Eq. 5.19):

$$\sigma = e(\mu_n n + \mu_p p)$$

In an n-type sample $n \gg p$, so $\sigma \approx e\mu_n n$.

2. Compensation changes n and μ in different ways

In a compensated n-type sample (2(a)) the free electrons come from the **net** doping:

$$n_0 = N_d - N_a \quad (N_d - N_a \gg n_i)$$

But the electrons are scattered by *every* ionized impurity, positive or negative. So the mobility depends on the **total** ionized impurity concentration (Neamen, Section 5.1.2):

$$N_I = N_d^+ + N_a^-$$

and is read from Neamen's Figure 5.3 (mobility against impurity concentration). This is the key to the problem: σ depends on N_d twice, through $n_0 = N_d - N_a$ and through $\mu_n(N_d + N_a)$.

3. Why trial and error

Because μ_n is given only as a graph, we cannot solve $\sigma = e\mu_n(N_d + N_a)(N_d - N_a)$ in one line. We guess N_d, read μ_n at $N_I = N_d + N_a$, compute σ, and adjust the guess until $\sigma = 16$ (Neamen, Example 5.3).

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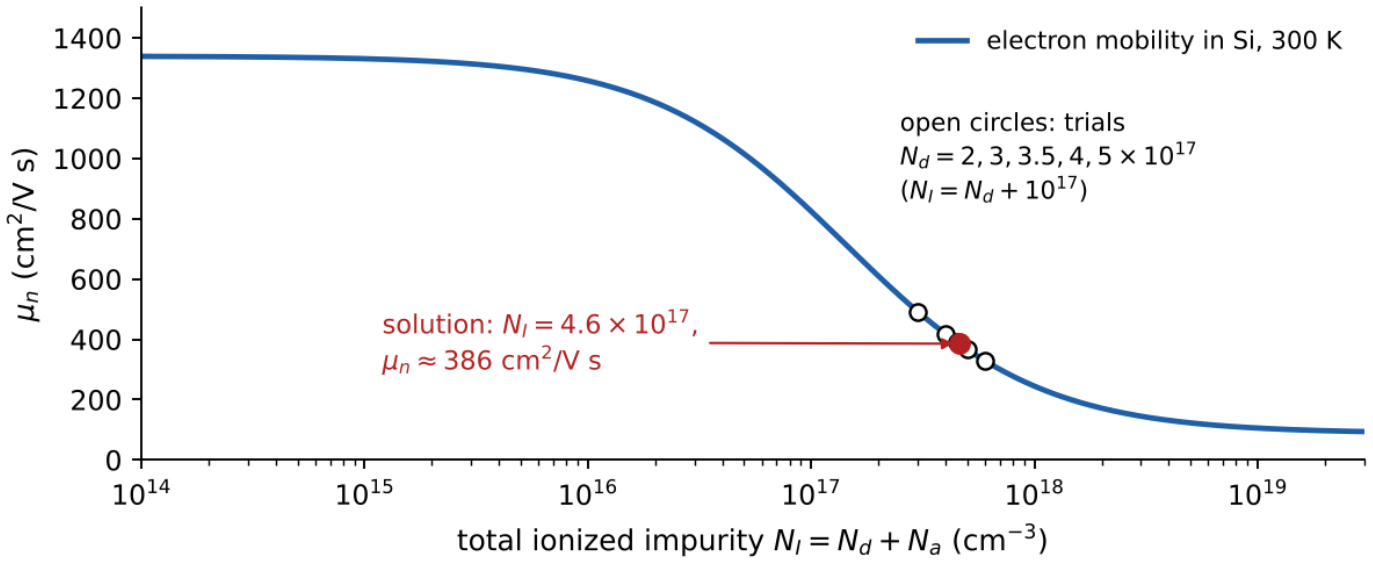


Figure 15: Electron mobility in silicon at 300 K against total ionized impurity concentration, with the trial points of 3(d)

Reading the figure. The solid blue curve is the electron mobility in Si at 300 K against the total ionized impurity concentration N_I; it follows Neamen's Figure 5.3 (an empirical fit, the Arora formula, gives 492 at 3×10^{17} and 329 at 6×10^{17} cm^{-3}, close to the book's readings of 510 and 325 there). The open circles are the trial values of N_d in the Answer. The filled red dot is the solution on this curve, $N_I = 4.59 \times 10^{17}$ cm^{-3} ($N_d = 3.59 \times 10^{17}$), $\mu_n = 386$ cm^2/V s; reading the book's graph gives about 400 cm^2/V s there.

2. Answer

Note on the paper

The question writes "a conductivity of $\rho = 16$ $(\Omega\text{-cm})^{-1}$". The unit $(\Omega\text{-cm})^{-1}$ and the word "conductivity" show that the quantity is the conductivity σ; the symbol ρ (resistivity) is a misprint. It is solved with $\sigma = 16$ $(\Omega\text{-cm})^{-1}$. "doner" means donor.

Given: compensated n-type Si, $T = 300$ K, $\sigma = 16$ $(\Omega\text{-cm})^{-1}$, $N_a = 10^{17}$ cm^{-3}.
Assumptions: complete ionization; $N_d - N_a \gg n_i$, so $n_0 = N_d - N_a$ and the hole term in σ is negligible; μ_n from Neamen's Figure 5.3 at $N_I = N_d + N_a$.

Step 1: The conductivity equation

$$\sigma \approx e\mu_n n_0 = e\mu_n(N_d - N_a)$$

$$16 = (1.6 \times 10^{-19})\,\mu_n\,(N_d - 10^{17}) \quad \Rightarrow \quad \mu_n(N_d - 10^{17}) = 1.0 \times 10^{20}\ (\text{V s cm})^{-1}$$

Step 2: Trial and error with the mobility curve

For each guess, $N_I = N_d + 10^{17}$ and μ_n is read from Figure 5.3 (here from the fit plotted above):

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Guess N_d (cm^{-3})	N_I (cm^{-3})	μ_n ($\mathrm{cm}^2/\mathrm{V\ s}$)	$\sigma = e\mu_n(N_d - N_a)$ (S/cm)
2×10^{17}	3×10^{17}	492	7.9 (too low)
3×10^{17}	4×10^{17}	418	13.4 (too low)
3.5×10^{17}	4.5×10^{17}	390	15.6 (close)
4×10^{17}	5×10^{17}	367	17.6 (too high)
5×10^{17}	6×10^{17}	329	21.1 (too high)

Table 5: Summary

The solution lies between 3.5×10^{17} and 4×10^{17} cm^{-3}, very close to 3.5×10^{17}. The fit gives exactly $N_d = 3.59 \times 10^{17}$ cm^{-3} with $\mu_n = 386$ $\mathrm{cm}^2/\mathrm{V\ s}$; reading Neamen's Figure 5.3 gives $\mu_n \approx 400$ $\mathrm{cm}^2/\mathrm{V\ s}$ at $N_I = 4.5 \times 10^{17}$, so that $\sigma = (1.6 \times 10^{-19})(400)(2.5 \times 10^{17}) = 16.0$ S/cm at $N_d = 3.5 \times 10^{17}$.

Step 3: Result

$$\boxed{N_d \approx 3.5 \times 10^{17}\ \mathrm{cm}^{-3}, \qquad \mu_n \approx 400\ \mathrm{cm}^2/\mathrm{V\ s}}$$

The free electron concentration is $n_0 = N_d - N_a = 2.5 \times 10^{17}$ cm^{-3}, while the electrons are scattered by $N_I = 4.5 \times 10^{17}$ cm^{-3} ions.

Final Answer: donor concentration $N_d \approx 3.5 \times 10^{17}$ cm^{-3}; electron mobility $\mu_n \approx 400$ $\mathrm{cm}^2/\mathrm{V\ s}$ (graph reading; the fitted curve gives 3.59×10^{17} cm^{-3} and 386 $\mathrm{cm}^2/\mathrm{V\ s}$).

3. Keyword Explanation

1. Total ionized impurity concentration

$N_I = N_d^+ + N_a^-$: every ion scatters carriers, whatever its sign.

2. Conductivity

$\sigma = e(\mu_n n + \mu_p p)$, the ratio of current density to field; its inverse is the resistivity ρ.

3. Ionized impurity scattering

The deflection of a carrier by the Coulomb field of a dopant ion; it lowers the mobility more as N_I rises.

4. Notes and Remarks

- **Book check:** this is Neamen's Example 5.3 word for word (with σ). The book's trials are $N_d = 2 \times 10^{17}$ ($\mu_n \approx 510$, $\sigma = 8.16$) and 5×10^{17} ($\mu_n \approx 325$, $\sigma = 20.8$), and its answer is $N_d \approx 3.5 \times 10^{17}$ cm^{-3}, $\mu_n \approx 400$ $\mathrm{cm}^2/\mathrm{V\ s}$, the same as here within graph-reading accuracy.
- **Why not just $N_d - N_a$?** If one took the mobility at the net doping 2.5×10^{17} cm^{-3} (about 540 $\mathrm{cm}^2/\mathrm{V\ s}$) the answer would be $N_d \approx 2.8 \times 10^{17}$ cm^{-3}, wrong because it ignores the scattering by the acceptor ions.
- Mobility is a strong function of doping in heavily doped material (Neamen's comment on this example); see 4(a).

5. Final Concise Answer

(a) On one energy axis: $g_c \propto \sqrt{E - E_c}$ and $g_v \propto \sqrt{E_v - E}$ (zero in the gap); $f_F(E) = 1/[1+e^{(E-E_F)/kT}]$ with E_F above E_{Fi}, near E_c for n-type; their products $n(E) = g_c f_F$ and $p(E) = g_v(1 - f_F)$ peak about $kT/2$ inside each band and their areas are $n_0 = N_c e^{-(E_c - E_F)/kT} \gg p_0 = N_v e^{-(E_F - E_v)/kT}$.

(b) Diffusion of holes (p → n) and electrons (n → p) uncovers ions that set up a field from n to p and a barrier $V_{bi} = (kT/e)\ln(N_a N_d/n_i^2)$; the field drives drift currents that exactly cancel the diffusion currents ($J_n = J_p = 0$), only the fraction $e^{-eV_{bi}/kT}$ of majority carriers can climb the barrier ($n_{p0} = n_{n0}e^{-eV_{bi}/kT}$), and E_F is flat, so each region keeps its equilibrium carrier concentrations.

(c) A net charge in a semiconductor is neutralized by majority carriers: continuity $\nabla \cdot J = -\partial\rho/\partial t$, Ohm $J = \sigma E$ and Poisson $\nabla \cdot E = \rho/\epsilon$ give $d\rho/dt + (\sigma/\epsilon)\rho = 0$, so $\rho(t) = \rho(0)e^{-t/\tau_d}$ with $\tau_d = \epsilon/\sigma$; for

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$N_d = 10^{16}$ cm^{-3} Si, $\tau_d = 0.539$ ps, which justifies quasi-neutrality.
(d) With σ (the paper's ρ is a misprint): $16 = e\mu_n(N_d - 10^{17})$ with μ_n read at $N_I = N_d + 10^{17}$; trial and error gives $N_d \approx 3.5 \times 10^{17}$ cm^{-3} and $\mu_n \approx 400$ cm^2/V s (Neamen Example 5.3).

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4. (a) With proper graphs, explain how mobility is affected by temperature and doping concentration. Marks: 6
   (b) Define ambipolar transport. Find ambipolar transport equation. Hence show that diffuse and drift behavior of excess majority carrier is determined by excess minority carrier. Marks: 7
   (c) For a pn junction with $N_a = 1.5 \times 10^{18}\text{cm}^{-3}$, $N_d = 1.5 \times 10^{16}\text{cm}^{-3}$ and $n_i = 2 \times 10^{10}\text{cm}^{-3}$ Marks: 6
      i. calculate the built-in potential barrier V_{bi}
      ii. calculate the depletion region with [*sic*: should be width] and E_{max} (Assume $\epsilon_s = 11.7\epsilon_o$) and
      iii. what will be the width at liquid nitrogen temperature (80K).
   (d) With proper graphs, explain time and spatial dependence of excess carrier concentration. Marks: 6

4.(a) With proper graphs, explain how mobility is affected by temperature and doping concentration. Marks: 6

1. Theory

1. What is mobility? (Core Idea)

In an electric field E a free carrier is accelerated, but it soon collides with something and loses the velocity it gained. On average it moves with a steady **drift velocity** proportional to the field (Neamen, Eqs. 5.4, 5.7):

$$v_d = \mu E$$

The constant μ is the **mobility** ($\text{cm}^2/\text{V s}$). Between collisions the carrier is accelerated freely for a mean time τ, so (Neamen, Eqs. 5.13, 5.14)

$$\mu = \frac{e\tau}{m^*}$$

A long time between collisions (little scattering) and a light effective mass give a high mobility. So **everything that changes the scattering changes the mobility**. Two scattering mechanisms dominate in silicon (Neamen, Section 5.1.2).

2. Lattice (phonon) scattering

The atoms of the crystal vibrate about their places; the higher the temperature, the larger the vibrations. These vibrations disturb the periodic potential and scatter carriers. More vibration means more scattering, so the lattice-limited mobility **falls** as temperature rises (Neamen, Eq. 5.15):

$$\mu_L \propto T^{-3/2}$$

3. Ionized impurity scattering

Every ionized dopant is a fixed charge whose Coulomb field deflects passing carriers. Two facts set its strength (Neamen, Eq. 5.16):

$$\mu_I \propto \frac{T^{3/2}}{N_I}$$

- **More ions** ($N_I = N_d^+ + N_a^-$ larger) → more scattering → lower μ_I.
- **Higher temperature** → faster carriers spend less time near each ion and are deflected less → **higher** μ_I.

4. Combining the two

The probabilities of the two kinds of collision in a time dt add, $dt/\tau = dt/\tau_I + dt/\tau_L$, so (Neamen, Eq. 5.18)

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$$\frac{1}{\mu} = \frac{1}{\mu_I} + \frac{1}{\mu_L}$$

The smaller of the two mobilities wins (Matthiessen's rule).

2. Answer

Mobility $\mu = v_d/E = e\tau/m^*$ is limited by two scattering mechanisms, combined by $1/\mu = 1/\mu_L + 1/\mu_I$ (Neamen, Eq. 5.18).

1. Effect of temperature

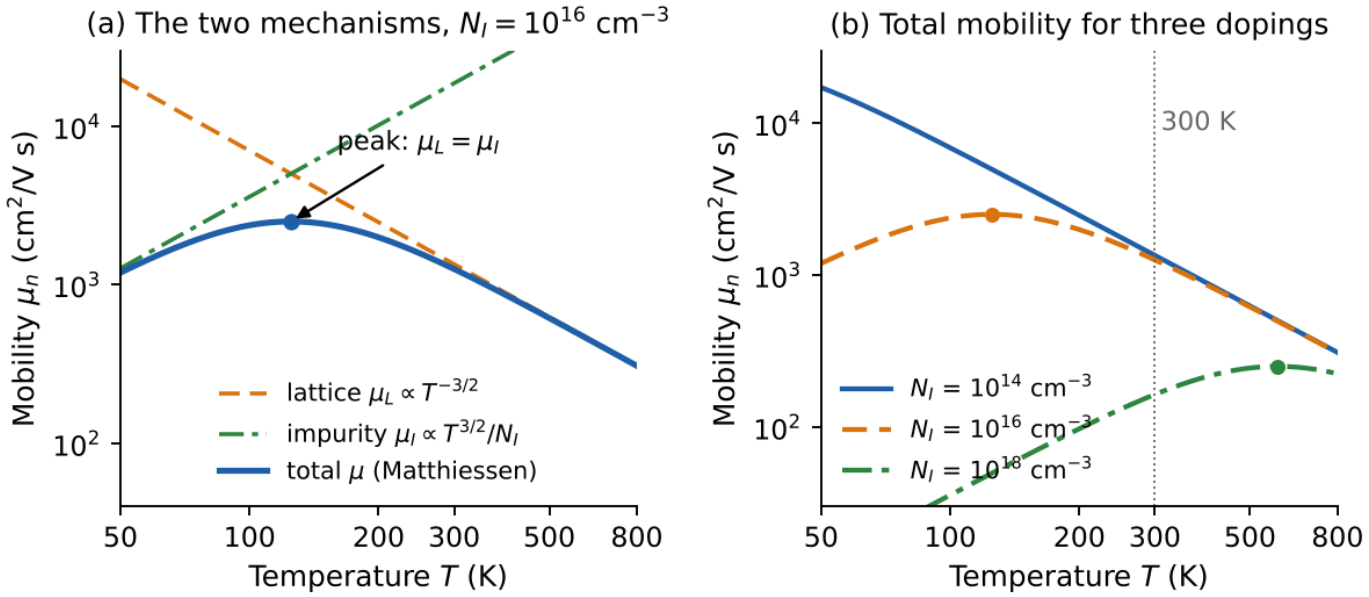


Figure 16: Electron mobility in silicon against temperature: the two scattering mechanisms and the total for three dopings (reused from the 2020-21 paper)

Reading the figure. Panel (a): for $N_I = 10^{16}$ cm^{-3}, the dashed orange line is the lattice-limited mobility ($\propto T^{-3/2}$), the dash-dot green line the impurity-limited mobility ($\propto T^{3/2}/N_I$), and the solid blue curve their combination, which follows the lower of the two and peaks where they cross. Panel (b): the total mobility for $N_I = 10^{14}$ (solid blue), 10^{16} (dashed orange) and 10^{18} cm^{-3} (dash-dot green).

- **Lattice scattering**, $\mu_L \propto T^{-3/2}$: as T rises the lattice vibrates more and scatters more, so mobility falls.
- **Ionized impurity scattering**, $\mu_I \propto T^{3/2}/N_I$: as T rises carriers move faster past the ions and are deflected less, so this mobility rises.
- **Total:** at low temperature impurity scattering dominates and μ rises with T; at high temperature lattice scattering dominates and μ falls with T. For lightly doped Si at room temperature, lattice scattering dominates and μ falls with T (Neamen, Figure 5.2).

2. Effect of doping concentration

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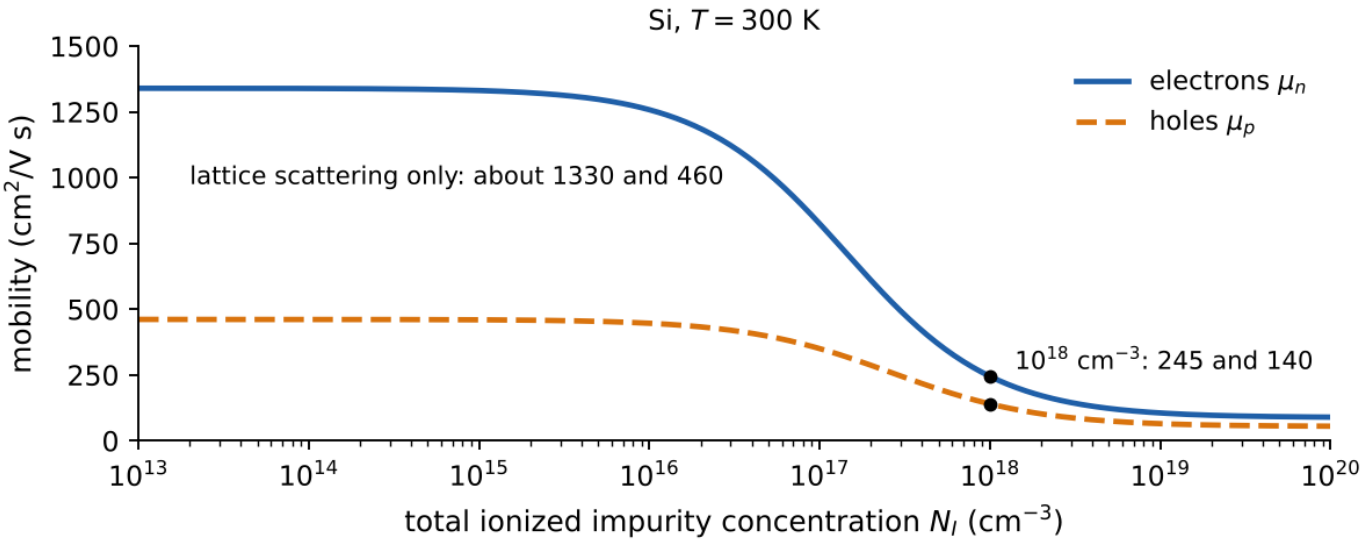


Figure 17: Electron and hole mobility in silicon at 300 K against the total ionized impurity concentration

Reading the figure. The solid blue curve is the electron mobility, the dashed orange curve the hole mobility, in Si at 300 K against N_I on a log axis. Both are flat below about 10^{15} cm^{-3} (about 1330 and 460 cm^2/V s, set by lattice scattering), then fall steeply between 10^{16} and 10^{19} cm^{-3} as impurity scattering takes over.

- At **low doping** ($N_I < 10^{15}$ cm^{-3}) $\mu_I \gg \mu_L$, so $\mu \approx \mu_L$: the mobility is set by the lattice and does not depend on doping ($\mu_n \approx 1330$, $\mu_p \approx 460$ cm^2/V s in Si; Neamen's Table 5.1 gives 1350 and 480 for pure Si).
- At **high doping** $\mu_I \propto 1/N_I$ becomes the smaller term and the mobility falls: at 10^{18} cm^{-3}, $\mu_n \approx 250$ and $\mu_p \approx 140$ cm^2/V s (Neamen, Figure 5.3).
- What matters is the **total** ionized impurity concentration $N_I = N_d^+ + N_a^-$: a compensated sample has a lower mobility than an uncompensated one with the same carrier concentration (3(d)).
- In panel (b) of the temperature figure, heavier doping lowers the curve and moves its peak to higher temperature.

3. Keyword Explanation

1. Mobility

The drift velocity per unit field, $\mu = v_d/E = e\tau/m^*$.

2. Lattice scattering

Scattering by thermal vibrations of the crystal atoms (phonons); stronger at higher temperature.

3. Matthiessen's rule

$1/\mu = 1/\mu_L + 1/\mu_I$: independent scattering mechanisms add their collision rates.

4. Notes and Remarks

- The doping curves are drawn with the Arora fit to measured silicon data, which reproduces Neamen's Figure 5.3 (its readings in Example 5.3, 510 and 325 cm^2/V s at 3×10^{17} and 6×10^{17} cm^{-3}, are within 4 %).
- Measured silicon follows T^{-n} with n about 2.4 for electrons and 2.2 for holes, steeper than the ideal 3/2, because other phonon modes also scatter (Neamen, Section 5.1.2).
- Electrons are faster than holes in Si because their conductivity effective mass is smaller ($0.26m_0$ against $0.39m_0$); the equation sheet's "$m^*/m_0 = 0.26$" is the electron value.

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- The 2020-21 paper's 4(a) covers the temperature part (extra reading).

4.(b) Define ambipolar transport. Find ambipolar transport equation. Hence show that diffuse and drift behavior of excess majority carrier is determined by excess minority carrier. Marks: 7

1. Theory

1. Excess carriers and why they move together (Core Idea)

When light (or injection) creates extra electron-hole pairs in a semiconductor, the extra electrons δn and extra holes δp are called **excess carriers**. Each kind would like to diffuse and drift on its own: electrons with D_n, μ_n, holes with D_p, μ_p, and in an applied field in opposite directions.

But as soon as the electron cloud and the hole cloud separate even slightly, a net charge appears, and with it an **internal electric field** E_{int} that pulls them back together (Poisson's equation, Neamen Eq. 6.32):

$$\frac{\partial E_{int}}{\partial x} = \frac{e(\delta p - \delta n)}{\epsilon_s}$$

A tiny imbalance makes a large field (the imbalance relaxes in the dielectric relaxation time, about a picosecond, 3(c)). So the excess electrons and holes **diffuse and drift together**, as one neutral cloud with $\delta n \approx \delta p$, with a single effective diffusion coefficient and a single effective mobility. This is called **ambipolar transport** (Neamen, Section 6.3).

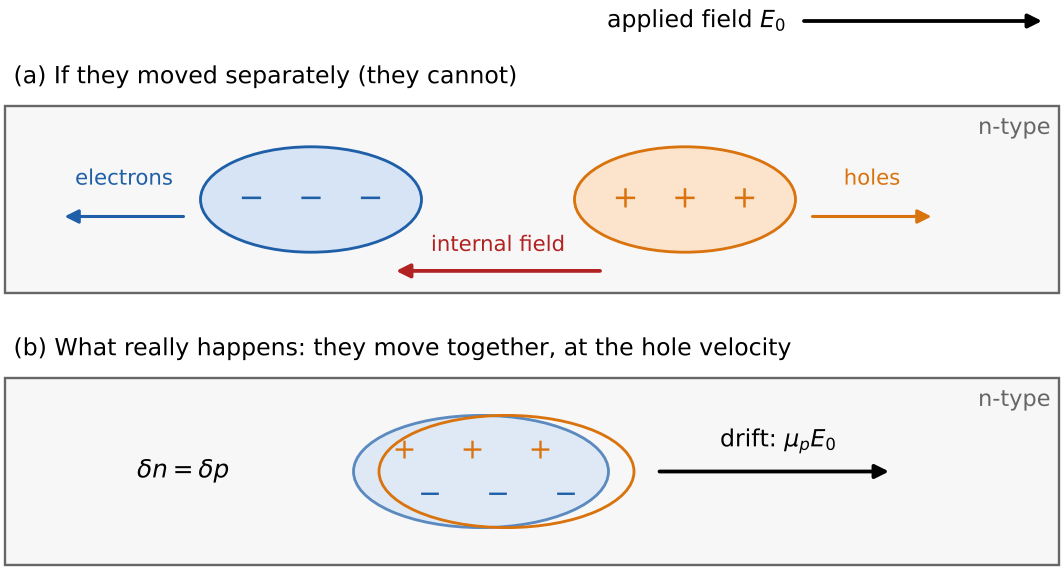


Figure 18: Excess electrons and holes are held together by their internal field and drift as one cloud (reused from the 2020-21 paper)

Reading the figure. Panel (a) shows what would happen if the clouds moved separately in an applied field E_0: electrons (blue, −) to the left, holes (orange, +) to the right, which would build a strong internal field (red arrow) opposing the separation. Panel (b) shows what really happens: the two clouds stay together ($\delta n = \delta p$) and drift as one, in n-type material at the hole (minority carrier) velocity $\mu_p E_0$.

2. The continuity equation (bookkeeping of carriers)

In a thin slice of width dx, the number of holes changes because of three things (Neamen, Section 6.2.1): holes flowing in minus holes flowing out (a flux F_p^+, holes per cm^2 per s), generation g_p, and recombination p/τ_{pt}. This bookkeeping gives the **continuity equation** (Neamen, Eq. 6.18):

$$\frac{\partial p}{\partial t} = -\frac{\partial F_p^+}{\partial x} + g_p - \frac{p}{\tau_{pt}}$$

The same for electrons, with flux F_n^-. The flux comes from drift and diffusion. Putting the current equations into the continuity equations gives one **time-dependent diffusion equation** for each carrier. Ambipolar transport combines the two.

2. Answer

Definition. **Ambipolar transport** is the motion of excess electrons and excess holes **together**, as one neutral distribution ($\delta n \approx \delta p$), with a single ambipolar diffusion coefficient D' and ambipolar mobility μ'. The internal electric field created by any small separation of the two charged clouds holds them together, so they diffuse and drift as one (Neamen, Section 6.3).

Derivation of the ambipolar transport equation (one dimension; Neamen, Sections 6.2 and 6.3). Steps 1 and 4 build general tools (the continuity bookkeeping and the product rule) and are marked (Basic); a student who knows them can skip them.

Step 1 (Basic): Continuity equation for holes

Take a slice of area A between x and $x + dx$. The hole flux (holes per unit area per second) entering at x is $F_p^+(x)$ and leaving at $x + dx$ is $F_p^+(x + dx) = F_p^+(x) + \frac{\partial F_p^+}{\partial x} dx$ (Taylor expansion, Neamen Eq. 6.15). The rate of change of the number of holes in the slice is inflow minus outflow, plus generation, minus recombination:

$$\frac{\partial p}{\partial t} A\, dx = \left[F_p^+(x) - F_p^+(x + dx)\right] A + g_p A\, dx - \frac{p}{\tau_{pt}} A\, dx$$

Divide by $A\, dx$ (Neamen, Eq. 6.18):

$$\frac{\partial p}{\partial t} = -\frac{\partial F_p^+}{\partial x} + g_p - \frac{p}{\tau_{pt}}$$

and for electrons, in the same way (Neamen, Eq. 6.19):

$$\frac{\partial n}{\partial t} = -\frac{\partial F_n^-}{\partial x} + g_n - \frac{n}{\tau_{nt}}$$

where g is the generation rate and τ_{pt}, τ_{nt} include thermal-equilibrium generation and recombination.

Step 2: The fluxes from drift and diffusion

The hole and electron current densities are drift plus diffusion (Neamen, Eqs. 6.20, 6.21):

$$J_p = e\mu_p pE - eD_p \frac{\partial p}{\partial x}, \qquad J_n = e\mu_n nE + eD_n \frac{\partial n}{\partial x}$$

A flux is a current divided by the charge of the carrier (Neamen, Eqs. 6.22, 6.23):

$$F_p^+ = \frac{J_p}{(+e)} = \mu_p pE - D_p \frac{\partial p}{\partial x}, \qquad F_n^- = \frac{J_n}{(-e)} = -\mu_n nE - D_n \frac{\partial n}{\partial x}$$

Step 3: Substitute into the continuity equations

$$\frac{\partial p}{\partial t} = -\mu_p \frac{\partial (pE)}{\partial x} + D_p \frac{\partial^2 p}{\partial x^2} + g_p - \frac{p}{\tau_{pt}}$$

$$\frac{\partial n}{\partial t} = +\mu_n \frac{\partial (nE)}{\partial x} + D_n \frac{\partial^2 n}{\partial x^2} + g_n - \frac{n}{\tau_{nt}}$$

(Neamen, Eqs. 6.24, 6.25.)

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Step 4 (Basic): Product rule on the drift term

$\frac{\partial(pE)}{\partial x} = E\frac{\partial p}{\partial x} + p\frac{\partial E}{\partial x}$ (Neamen, Eq. 6.26). So, rearranged:

$$D_p\frac{\partial^2 p}{\partial x^2} - \mu_p\left(E\frac{\partial p}{\partial x} + p\frac{\partial E}{\partial x}\right) + g_p - \frac{p}{\tau_{pt}} = \frac{\partial p}{\partial t}$$

$$D_n\frac{\partial^2 n}{\partial x^2} + \mu_n\left(E\frac{\partial n}{\partial x} + n\frac{\partial E}{\partial x}\right) + g_n - \frac{n}{\tau_{nt}} = \frac{\partial n}{\partial t}$$

These are the **time-dependent diffusion equations** (Neamen, Eqs. 6.27, 6.28; the first is on the paper's equation sheet).

Step 5: Write them for the excess carriers

Put $p = p_0 + \delta p$, $n = n_0 + \delta n$ with p_0, n_0 uniform in space and constant in time, so their derivatives vanish: $\partial^2 p/\partial x^2 = \partial^2(\delta p)/\partial x^2$, $\partial p/\partial x = \partial(\delta p)/\partial x$, $\partial p/\partial t = \partial(\delta p)/\partial t$. The same for electrons.

Step 6: Charge neutrality and equal rates

- The internal field holds the clouds together, so $\delta n = \delta p$ everywhere (the small charge difference is still kept in $\partial E/\partial x$, which is not zero).
- Electrons and holes are generated in pairs and recombine in pairs (Neamen, Eqs. 6.34, 6.35): $g_n = g_p \equiv g$ and $n/\tau_{nt} = p/\tau_{pt} \equiv R$.

Writing δn for both excess concentrations (Neamen, Eqs. 6.36, 6.37):

$$D_p\frac{\partial^2(\delta n)}{\partial x^2} - \mu_p\left(E\frac{\partial(\delta n)}{\partial x} + p\frac{\partial E}{\partial x}\right) + g - R = \frac{\partial(\delta n)}{\partial t} \qquad \text{(i)}$$

$$D_n\frac{\partial^2(\delta n)}{\partial x^2} + \mu_n\left(E\frac{\partial(\delta n)}{\partial x} + n\frac{\partial E}{\partial x}\right) + g - R = \frac{\partial(\delta n)}{\partial t} \qquad \text{(ii)}$$

Step 7: Eliminate $\partial E/\partial x$

Multiply (i) by $\mu_n n$ and (ii) by $\mu_p p$ and add. The field-gradient terms are $-\mu_n\mu_p np\,\partial E/\partial x$ and $+\mu_p\mu_n pn\,\partial E/\partial x$: they cancel. The rest gives (Neamen, Eq. 6.38)

$$(\mu_n n D_p + \mu_p p D_n)\frac{\partial^2(\delta n)}{\partial x^2} + \mu_n\mu_p(p-n)E\frac{\partial(\delta n)}{\partial x} + (\mu_n n + \mu_p p)(g - R) = (\mu_n n + \mu_p p)\frac{\partial(\delta n)}{\partial t}$$

Step 8: Divide by $(\mu_n n + \mu_p p)$: the ambipolar transport equation

$$\boxed{D'\frac{\partial^2(\delta n)}{\partial x^2} + \mu' E\frac{\partial(\delta n)}{\partial x} + g - R = \frac{\partial(\delta n)}{\partial t}}$$

(Neamen, Eq. 6.39), with the **ambipolar diffusion coefficient** and **ambipolar mobility** (Neamen, Eqs. 6.40, 6.41):

$$D' = \frac{\mu_n n D_p + \mu_p p D_n}{\mu_n n + \mu_p p}, \qquad \mu' = \frac{\mu_n\mu_p(p-n)}{\mu_n n + \mu_p p}$$

With the Einstein relation $\mu_n/D_n = \mu_p/D_p = e/kT$ (Neamen, Eq. 6.42), D' becomes (Neamen, Eq. 6.43)

$$D' = \frac{D_n D_p(n+p)}{D_n n + D_p p}$$

Hence: the excess majority carrier follows the minority carrier.

Step 9: p-type material at low injection

Low injection means $\delta n \ll p_0$, and in p-type $n_0 \ll p_0$. So $n = n_0 + \delta n \ll p = p_0 + \delta n \approx p_0$. In D' and μ' drop n next to p:

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$$D' = \frac{D_n D_p p}{D_p p} = D_n, \qquad \mu' = \frac{\mu_n \mu_p p}{\mu_p p} = \mu_n$$

(Neamen, Eqs. 6.45, 6.46.) The rates: $g - R = g' - \delta n/\tau_{n0}$ (Neamen, Eq. 6.51), with g' the excess generation rate and τ_{n0} the minority electron lifetime. The ambipolar equation becomes (Neamen, Eq. 6.55)

$$D_n \frac{\partial^2(\delta n)}{\partial x^2} + \mu_n E \frac{\partial(\delta n)}{\partial x} + g' - \frac{\delta n}{\tau_{n0}} = \frac{\partial(\delta n)}{\partial t}$$

Step 10: n-type material at low injection

Now $p \ll n \approx n_0$: $D' = D_p$ and $\mu' = -\mu_p$ (Neamen, Eqs. 6.47, 6.48), and (Neamen, Eq. 6.56)

$$D_p \frac{\partial^2(\delta p)}{\partial x^2} - \mu_p E \frac{\partial(\delta p)}{\partial x} + g' - \frac{\delta p}{\tau_{p0}} = \frac{\partial(\delta p)}{\partial t}$$

Steps to remember

Continuity for p and n → drift + diffusion fluxes → time-dependent diffusion equations → excess form with $\delta n = \delta p$, $g_n = g_p$, $R_n = R_p$ → multiply by $\mu_n n$ and $\mu_p p$, add: $\partial E/\partial x$ cancels → divide: D', μ' → low injection: D' = minority D, μ' = minority μ.

Conclusion. In an extrinsic semiconductor at low injection, D', μ' and the lifetime in the ambipolar equation are those of the **minority carrier** (electrons in p-type, holes in n-type), and μ' even has the minority carrier's sign (direction of drift). Since $\delta p = \delta n$, the excess **majority** carriers diffuse and drift with exactly the same distribution: their diffusion and drift behaviour is determined by the excess minority carriers. Physically, the majority carriers are so many that a tiny redistribution of them follows the minority cloud and neutralizes it, while the slow, rare minority carriers set the pace.

3. Keyword Explanation

1. Excess carriers

Electrons and holes above their thermal-equilibrium values, $\delta n = n - n_0$, $\delta p = p - p_0$.

2. Low-level injection

The excess carrier concentration is much smaller than the thermal-equilibrium majority carrier concentration.

3. Ambipolar diffusion coefficient and mobility

D' and μ': the single diffusion coefficient and mobility with which the neutral excess electron-hole distribution moves.

4. Continuity equation

The particle balance of a small volume: change = net inflow + generation − recombination.

4. Notes and Remarks

- **Intrinsic material:** with $n = p$, $\mu' = 0$ (the cloud does not drift at all) and $D' = 2D_n D_p/(D_n + D_p)$, between the two diffusion coefficients.
- The equation sheet of the paper carries the two time-dependent diffusion equations of Step 4 (in two forms) and the solution for a pulse; see 4(d).
- Also asked in term test 2 of this session, 8(b).

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4.(c) For a pn junction with $N_a = 1.5 \times 10^{18}\text{cm}^{-3}$, $N_d = 1.5 \times 10^{16}\text{cm}^{-3}$ and $n_i = 2 \times 10^{10}\text{cm}^{-3}$ i. calculate the built-in potential barrier V_{bi} ii. calculate the depletion region with [*sic*: should be width] and E_{max} (Assume $\epsilon_s = 11.7\epsilon_o$) and iii. what will be the width at liquid nitrogen temperature (80K). Marks: 6

1. Theory

1. The pn junction in equilibrium (Core Idea)

When a p region (N_a acceptors) meets an n region (N_d donors), holes diffuse into the n side and electrons into the p side. They leave behind fixed ions: $-eN_a$ on the p side over a width x_p, $+eN_d$ on the n side over x_n. This is the **space charge (depletion) region**. Its field opposes further diffusion, and the potential difference across it is the **built-in potential** V_{bi} (3(b)).

2. Built-in potential

The Fermi level is flat in equilibrium. On the p side E_F is $e\phi_{Fp} = kT\ln(N_a/n_i)$ below E_{Fi}; on the n side it is $e|\phi_{Fn}| = kT\ln(N_d/n_i)$ above E_{Fi}. The bands must shift by the sum (Neamen, Eqs. 7.1, 7.10):

$$V_{bi} = |\phi_{Fn}| + |\phi_{Fp}| = \frac{kT}{e}\ln\left(\frac{N_aN_d}{n_i^2}\right)$$

3. Depletion width and peak field

Integrating Poisson's equation $dE/dx = \rho/\epsilon_s$ over the two charge blocks gives a triangular field that is largest at the junction, and integrating again gives the potential (Neamen, Section 7.2.2). The results are (Neamen, Eqs. 7.17, 7.28, 7.31, 7.35):

$$N_ax_p = N_dx_n, \qquad W = x_n + x_p = \left\{\frac{2\epsilon_sV_{bi}}{e}\left[\frac{N_a+N_d}{N_aN_d}\right]\right\}^{1/2}$$

$$x_n = W\frac{N_a}{N_a+N_d}, \qquad |E_{max}| = \frac{eN_dx_n}{\epsilon_s} = \frac{2V_{bi}}{W}$$

The charge balance $N_ax_p = N_dx_n$ means the depletion region spreads mostly into the **lightly doped** side. Here $N_a = 100N_d$, so almost all of W lies on the n side (a one-sided p^+n junction).

4. What changes at low temperature

n_i depends strongly on temperature (Neamen, Eq. 4.23):

$$n_i^2 = N_cN_v\exp\left(-\frac{E_g}{kT}\right), \qquad N_c, N_v \propto T^{3/2} \quad \Rightarrow \quad n_i(T) = n_i(300)\left(\frac{T}{300}\right)^{3/2}\exp\left[-\frac{E_g}{2k}\left(\frac{1}{T}-\frac{1}{300}\right)\right]$$

At 80 K, n_i becomes astronomically small, so the logarithm in V_{bi} grows much faster than kT shrinks: V_{bi} **increases**, approaching E_g/e, and W **widens** as $\sqrt{V_{bi}}$.

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2. Answer

Given: $N_a = 1.5 \times 10^{18}$ cm^{-3}, $N_d = 1.5 \times 10^{16}$ cm^{-3}, $n_i = 2 \times 10^{10}$ cm^{-3} (at 300 K, as given), $\epsilon_s = 11.7\epsilon_0 = 11.7 \times 8.85 \times 10^{-14} = 1.0355 \times 10^{-12}$ F/cm, $e = 1.6 \times 10^{-19}$ C, $kT/e = 0.0259$ V at 300 K.

Note on the paper

"the depletion region with and E_{max}" is read as "the depletion region **width** and E_{max}". The temperature for parts (i) and (ii) is not printed; $T = 300$ K is taken, the temperature at which $n_i = 2 \times 10^{10}$ cm^{-3} is given.

Assumptions: abrupt junction, complete ionization at both temperatures, $E_g = 1.12$ eV and ϵ_s independent of temperature, no applied voltage.

Step 1: Built-in potential at 300 K

$$\frac{N_a N_d}{n_i^2} = \frac{(1.5 \times 10^{18})(1.5 \times 10^{16})}{(2 \times 10^{10})^2} = 5.625 \times 10^{13}, \qquad \ln\left(5.625 \times 10^{13}\right) = 31.661$$

$$V_{bi} = (0.0259)(31.661) = 0.820 \text{ V}$$

$$\boxed{V_{bi} = 0.820 \text{ V}}$$

Step 2: Depletion width at 300 K

$$\frac{N_a + N_d}{N_a N_d} = \frac{1.515 \times 10^{18}}{2.25 \times 10^{34}} = 6.733 \times 10^{-17} \text{ cm}^3, \qquad \frac{2\epsilon_s V_{bi}}{e} = \frac{2(1.0355 \times 10^{-12})(0.820)}{1.6 \times 10^{-19}} = 1.061 \times 10^{7} \text{ cm}^{-1}$$

$$W = \sqrt{(1.061 \times 10^{7})(6.733 \times 10^{-17})} = 2.673 \times 10^{-5} \text{ cm}$$

$$\boxed{W = 0.267 \ \mu\text{m}}$$

It splits as $x_n = W\, N_a/(N_a + N_d) = 0.2647$ μm and $x_p = 0.00265$ μm (2.65 nm): practically all on the lightly doped n side.

Step 3: Maximum field at 300 K

$$|E_{max}| = \frac{eN_d x_n}{\epsilon_s} = \frac{(1.6 \times 10^{-19})(1.5 \times 10^{16})(2.647 \times 10^{-5})}{1.0355 \times 10^{-12}} = 6.13 \times 10^{4} \text{ V/cm}$$

Check: $2V_{bi}/W = 2(0.820)/(2.673 \times 10^{-5}) = 6.13 \times 10^{4}$ V/cm.

$$\boxed{E_{max} = -6.13 \times 10^{4} \text{ V/cm}}$$

(the minus sign: the field points from n to p, in the $-x$ direction, Neamen Eq. 7.35).

Step 4: Intrinsic concentration at 80 K

$$\left(\frac{80}{300}\right)^{3/2} = 0.1377, \qquad \frac{E_g}{2k}\left(\frac{1}{80} - \frac{1}{300}\right) = \frac{1.12}{2(8.617 \times 10^{-5})}(0.009167) = 59.57$$

$$n_i(80 \text{ K}) = (2 \times 10^{10})(0.1377)\, e^{-59.57} = 3.70 \times 10^{-17} \text{ cm}^{-3}$$

(effectively no intrinsic carriers at all).

Step 5: Built-in potential at 80 K

$$\frac{kT}{e} = 0.0259 \times \frac{80}{300} = 0.006907 \text{ V}, \qquad \ln\frac{N_a N_d}{n_i^2} = \ln\frac{2.25 \times 10^{34}}{(3.70 \times 10^{-17})^2} = 154.77$$

$$V_{bi}(80 \text{ K}) = (0.006907)(154.77) = 1.069 \text{ V}$$

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Step 6: Depletion width at 80 K

$W \propto \sqrt{V_{bi}}$ with everything else the same:

$$W(80\text{ K}) = W(300\text{ K})\sqrt{\frac{1.069}{0.820}} = (0.2673)(1.1417) = 0.305\ \mu\text{m}$$

$$\boxed{W(80\text{ K}) = 0.305\ \mu\text{m}}$$

and the peak field rises to $|E_{max}| = 2(1.069)/(3.052 \times 10^{-5}) = 7.00 \times 10^4$ V/cm.

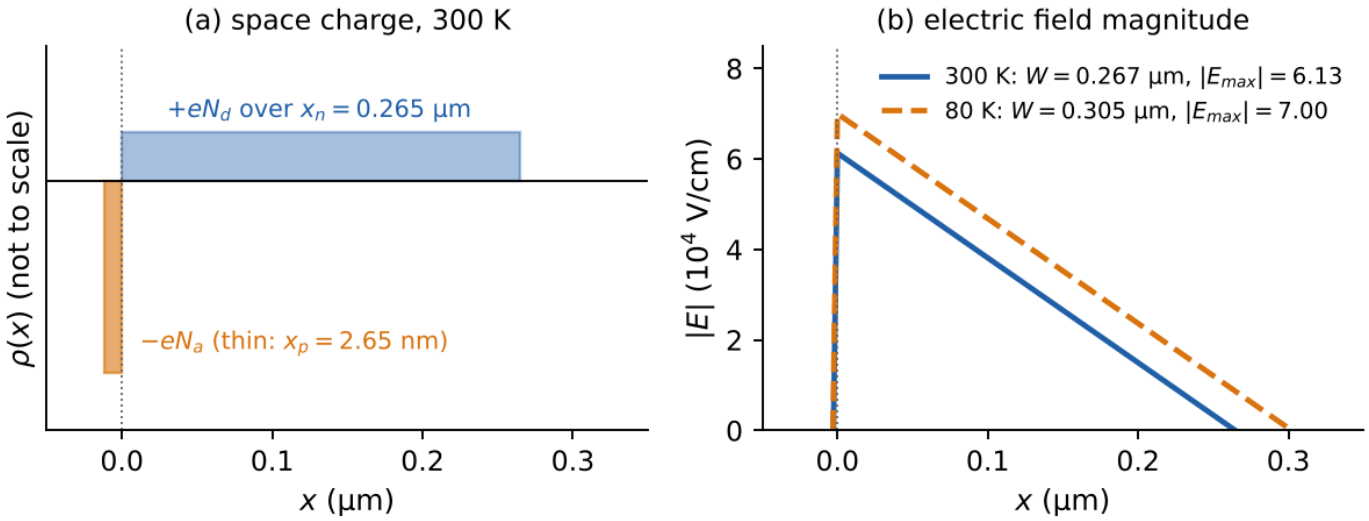


Figure 19: Charge density and electric field of the junction at 300 K and 80 K

Reading the figure. Panel (a): the space charge, $-eN_a$ in a very thin slice on the p side (2.65 nm, drawn wider so that it can be seen) and $+eN_d$ over the wide n side (heights not to scale: N_a is 100 times N_d). Panel (b): the field, a triangle peaked at the junction; the solid blue triangle is 300 K ($W = 0.267$ μm, $|E_{max}| = 6.13 \times 10^4$ V/cm), the dashed orange one 80 K ($W = 0.305$ μm, 7.00×10^4 V/cm). The area of each triangle is its V_{bi}.

Final Answer:

- (i) $V_{bi} = 0.820$ V (300 K)
- (ii) $W = 0.267$ μm ($x_n = 0.265$ μm, $x_p = 2.65$ nm), $E_{max} = -6.13 \times 10^4$ V/cm
- (iii) at 80 K: $n_i = 3.7{\times}10^{-17}$ cm^{-3}, $V_{bi} = 1.069$ V, $W = 0.305$ μm (14 % wider), $|E_{max}| = 7.00{\times}10^4$ V/cm

3. Keyword Explanation

1. Built-in potential

$V_{bi} = (kT/e)\ln(N_a N_d/n_i^2)$: the potential step across the space charge region in equilibrium.

2. One-sided junction

A junction with one side much more heavily doped; the depletion region lies almost entirely in the lightly doped side, $W \approx x_n$.

3. Liquid nitrogen temperature

About 77 K; the paper rounds it to 80 K.

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4. Notes and Remarks

- **Freeze-out at 80 K:** with phosphorus ($E_c - E_d = 0.045$ eV), only about 35 % of the 1.5×10^{16} cm^{-3} donors are ionized at 80 K (the ionization formula of 2(c)). The question assumes complete ionization, as above. Strictly, inside the depletion region the field sweeps out the electrons and every donor in it is ionized ($+eN_d$), so W is still set by N_d; what changes is V_{bi}, through the position of E_F in the neutral n region (closer to E_c), which raises V_{bi} slightly more.
- **Band gap at 80 K:** real Si has $E_g \approx 1.16$ eV at 80 K; using it would raise V_{bi} and W a little further. The answer keeps Neamen's 1.12 eV.
- **Check through the band edges:** $V_{bi}(80) = E_g/e - (kT/e)\ln(N_c N_v / N_a N_d)$ with Neamen's N_c, N_v scaled to 80 K gives 1.08 V, close to 1.069 V (the paper's $n_i = 2 \times 10^{10}$ differs from the n_i that Neamen's N_c, N_v give).
- The equation sheet prints $E_{max} = -\frac{e}{\epsilon_s}\left\{\frac{2\epsilon_s V_{bi}}{e}\frac{N_a N_d}{N_a+N_d}\right\}^{1/2}$; this is Neamen's Eq. 7.36 at $V_R = 0$, $E_{max} = -\{2eV_{bi}N_aN_d/[\epsilon_s(N_a + N_d)]\}^{1/2}$, written another way, and gives the same 6.13×10^4 V/cm.

4.(d) With proper graphs, explain time and spatial dependence of excess carrier concentration. Marks: 6

1. Theory

1. Excess carriers and their lifetime (Core Idea)

Light with $h\nu > E_g$, or injection from a junction, adds electron-hole pairs above the equilibrium values: $n = n_0 + \delta n$, $p = p_0 + \delta p$. As soon as the extra generation stops, the system returns to equilibrium by **recombination**. In low injection the recombination rate is proportional to the excess concentration (Neamen, Eqs. 6.13, 6.14):

$$R' = \frac{\delta n}{\tau_{n0}} \quad \text{(p-type)}, \qquad R' = \frac{\delta p}{\tau_{p0}} \quad \text{(n-type)}$$

where τ_{n0}, τ_{p0} are the **minority carrier lifetimes**: the average time an excess carrier survives.

2. The governing equation

All the cases below are solutions of the ambipolar transport equation for low injection (4(b)); for n-type (Neamen, Eq. 6.56):

$$D_p\frac{\partial^2(\delta p)}{\partial x^2} - \mu_p E\frac{\partial(\delta p)}{\partial x} + g' - \frac{\delta p}{\tau_{p0}} = \frac{\partial(\delta p)}{\partial t}$$

Each case keeps only the terms its conditions allow:

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Case	Kept terms	Dependence
Uniform, generation switched off	$-\delta p/\tau_{p0} = \partial\delta p/\partial t$	time only
Uniform, generation switched on	$g' - \delta p/\tau_{p0} = \partial\delta p/\partial t$	time only
Steady injection at a surface	$D_p\delta p'' - \delta p/\tau_{p0} = 0$	space only
A pulse, with or without a field	all but g'	time and space

Table 6: Summary

2. Answer

1. Time dependence (uniform, no field). With a uniform distribution there is no diffusion ($\partial^2/\partial x^2 = 0$) and no drift. After the generation is switched off at $t = 0$ (Neamen, Eqs. 6.57, 6.58):

$$\frac{d(\delta p)}{dt} = -\frac{\delta p}{\tau_{p0}} \quad \Rightarrow \quad \delta p(t) = \delta p(0)\, e^{-t/\tau_{p0}}$$

The excess decays exponentially with the minority carrier lifetime; $\delta n = \delta p$ decays with it. If instead a uniform generation g' is switched on at $t = 0$, $g' - \delta p/\tau_{p0} = d(\delta p)/dt$ gives (Neamen, Eq. 6.61)

$$\delta p(t) = g'\tau_{p0}\left(1 - e^{-t/\tau_{p0}}\right)$$

which rises to the steady value $g'\tau_{p0}$.

2. Spatial dependence (steady state, no field). Excess holes injected steadily at $x = 0$ into an n-type bar: no time change, no field, no generation for $x > 0$, so (Neamen, Eqs. 6.62 to 6.65)

$$D_p\frac{d^2(\delta p)}{dx^2} - \frac{\delta p}{\tau_{p0}} = 0 \quad \Rightarrow \quad \delta p(x) = \delta p(0)\, e^{-x/L_p}, \qquad L_p = \sqrt{D_p\tau_{p0}}$$

(the growing solution e^{+x/L_p} is dropped, since δp must stay finite as $x \to \infty$). The excess falls to $1/e$ in one **diffusion length** L_p, the average distance a minority carrier diffuses before it recombines.

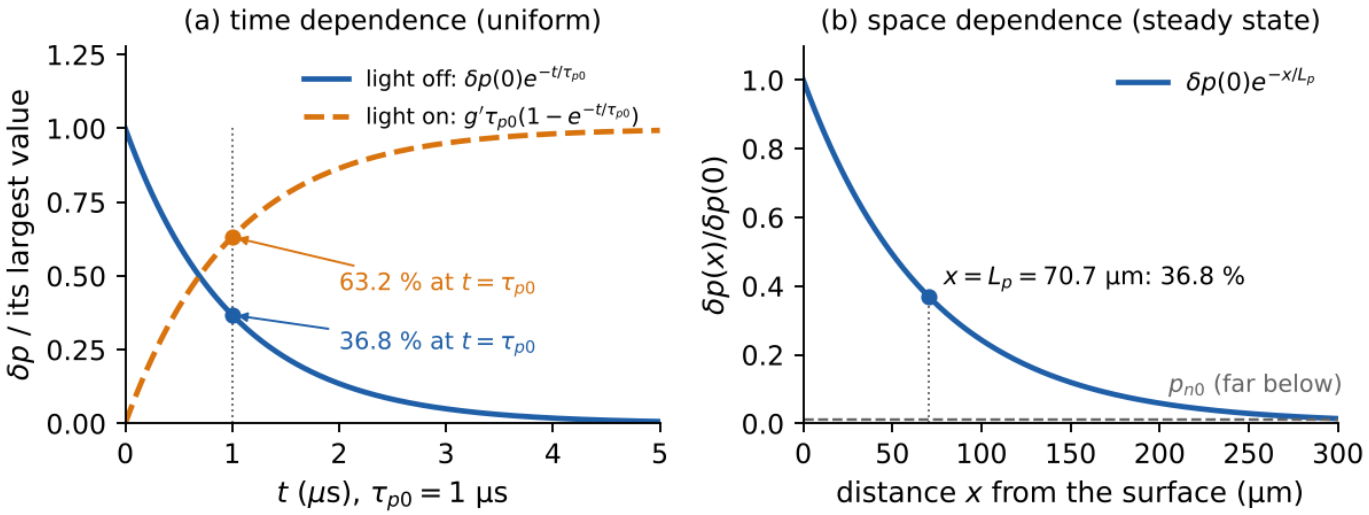


Figure 20: Time decay and build-up of a uniform excess, and the steady-state spatial decay from a surface

Reading the figure. Panel (a), time dependence with $\tau_{p0} = 1$ μs: the solid blue curve is the decay

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$\delta p(0)e^{-t/\tau_{p0}}$ after the light is switched off (36.8 % left at $t = \tau_{p0}$), the dashed orange curve the build-up $g'\tau_{p0}(1-e^{-t/\tau_{p0}})$ after it is switched on (63.2 % of the final value at $t = \tau_{p0}$). Panel (b), space dependence with $L_p = 70.7$ μm ($D_p = 10$ cm^2/s, $\tau_{p0} = 5$ μs): the solid blue curve is $\delta p(0)e^{-x/L_p}$, 36.8 % left at $x = L_p$; the dashed grey line is the equilibrium p_{n0}, on a linear scale far below.

3. Time and space together (a pulse). A short pulse of N_0 excess carriers per cm^2 generated at $x = 0$, $t = 0$ in n-type material with an applied field E_0 (Neamen, Eq. 6.70; the equation sheet's form):

$$\delta p(x,t) = \frac{e^{-t/\tau_{p0}}}{(4\pi D_p t)^{1/2}} \exp\left[-\frac{(x-\mu_p E_0 t)^2}{4D_p t}\right]$$

- the Gaussian **spreads** with time (its width grows as $\sqrt{4D_p t}$): diffusion;
- its area **shrinks** as $e^{-t/\tau_{p0}}$: recombination;
- its peak **moves** at $\mu_p E_0$ in the direction of the field: the whole neutral cloud drifts with the minority carrier mobility.

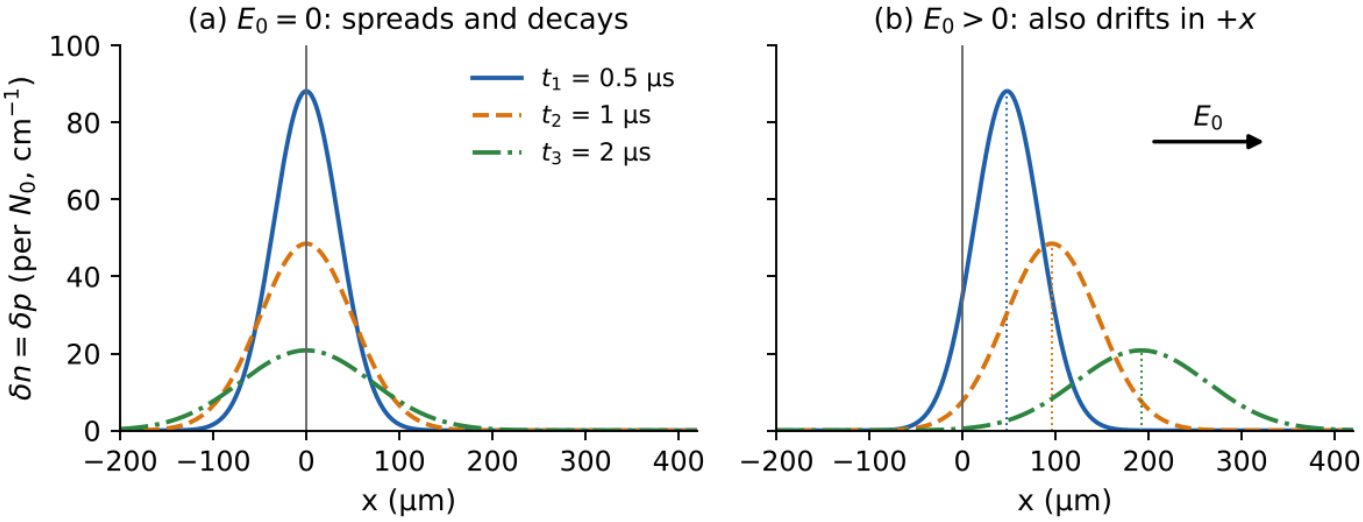


Figure 21: An excess carrier pulse at three times, without and with an applied field (reused from the 2020-21 paper)

Reading the figure. Panel (a), $E_0 = 0$: the pulse at $t_1 = 0.5$ μs (solid blue), $t_2 = 1$ μs (dashed orange) and $t_3 = 2$ μs (dash-dot green) stays centred at $x = 0$ but becomes lower and wider. Panel (b), $E_0 > 0$: the same pulses, with each peak moved to $x = \mu_p E_0 t$ (dotted lines).

3. Keyword Explanation

1. Minority carrier lifetime

τ_{n0} or τ_{p0}: the mean time an excess minority carrier exists before it recombines.

2. Diffusion length

$L = \sqrt{D\tau}$: the mean distance an excess minority carrier diffuses before it recombines.

3. Low-level injection

Excess concentration much smaller than the majority carrier concentration, so the recombination rate is linear in the excess.

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4. Notes and Remarks

- **Equation sheet:** the paper prints this pulse with $e^{-t/\rho_{p0}}$ in the numerator; ρ_{p0} is a misprint for the lifetime τ_{p0} (Neamen, Eq. 6.70).
- Neamen's TYU 6.5 uses $\tau_{p0} = 5$ μs, $D_p = 10$ cm^2/s, $\mu_p = 386$ cm^2/V s, $E_0 = 10$ V/cm; the peak moves 38.6 μm in 1 μs.
- The steady-state solution for injection in the middle of a crystal, decaying in both directions, is 5(a).
- The pulse was asked as a derivation in the 2020-21 paper's 5(a) (extra reading).

5. Final Concise Answer

(a) $\mu = e\tau/m^*$ is set by scattering: lattice scattering grows with temperature ($\mu_L \propto T^{-3/2}$), ionized impurity scattering grows with N_I and weakens with temperature ($\mu_I \propto T^{3/2}/N_I$), and $1/\mu = 1/\mu_L + 1/\mu_I$; so μ rises with T at low T (impurities dominate) and falls at high T (lattice dominates), and it is flat at low doping (about 1350 and 480 cm^2/V s for electrons and holes in pure Si) and falls steeply above 10^{16} cm^{-3} of total ionized impurity.

(b) Ambipolar transport is the joint drift and diffusion of excess electrons and holes, held together ($\delta n = \delta p$) by their internal field. From continuity with drift and diffusion fluxes, eliminating $\partial E/\partial x$: $D'\partial^2\delta n/\partial x^2 + \mu' E\,\partial\delta n/\partial x + g - R = \partial\delta n/\partial t$ with $D' = D_nD_p(n+p)/(D_nn + D_pp)$, $\mu' = \mu_n\mu_p(p-n)/(\mu_n n + \mu_p p)$; at low injection in p-type $D' = D_n$, $\mu' = \mu_n$ (in n-type D_p, $-\mu_p$), so the excess majority carriers move with the minority carrier's parameters.

(c) $V_{bi} = 0.0259\ln(5.625\times10^{13}) = 0.820$ V; $W = 0.267$ μm ($x_n = 0.265$ μm, $x_p = 2.65$ nm), $E_{max} = -6.13\times10^4$ V/cm; at 80 K, $n_i = 3.7\times10^{-17}$ cm^{-3}, $V_{bi} = 1.069$ V and $W = 0.305$ μm ($|E_{max}| = 7.0\times10^4$ V/cm), assuming complete ionization.

(d) Uniform excess: $\delta p(t) = \delta p(0)e^{-t/\tau_{p0}}$ after the generation stops, or $g'\tau_{p0}(1 - e^{-t/\tau_{p0}})$ after it starts; steady injection at a surface: $\delta p(x) = \delta p(0)e^{-x/L_p}$, $L_p = \sqrt{D_p\tau_{p0}}$; a pulse: $\delta p(x,t) = \frac{e^{-t/\tau_{p0}}}{\sqrt{4\pi D_p t}}e^{-(x-\mu_pE_0t)^2/4D_pt}$, which spreads, decays and drifts at μ_pE_0.

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5. (a) Consider a p-type semiconductor that is homogeneous and infinite in extent. Assume a zero applied electric field. For a one-dimensional crystal, assume that excess carriers are being generated at $x = 0$ only. The excess carriers being generated at $x = 0$ will being [*sic*: should be will begin] diffusing in both the $-x$ and $+x$ directions. Calculate the steady-state excess carrier concentration as a function of x. Marks: 7

(b) Find ideal pn junction current and sketch ideal electron and hole current components. Marks: 8

(c) Consider an n-type semiconductor at $T = 300$ K with carrier concentrations of $n_o = 10^{15}$ cm^{-3}, $n_i = 10^{10}$ cm^{-3} and $p_o = 10^5$ cm^{-3}. In non-equilibrium, assume that the excess carrier concentrations are $\delta n = \delta p = 10^{-13}$ [*sic*: should be 10^{13}] cm^{-3}. Calculate the quasi-Fermi energy levels. Marks: 4

(d) An abrupt silicon pn junction has dopant concentrations of $N_a = 2 \times 10^{16}$ cm^{-3} and $N_d = 2 \times 10^{15}$ cm^{-3} at $T = 300$ K. Calculate: Marks: 6

i. V_{bi}
ii. W at $V_R = 0$ and $V_R = 8$ V and
iii. the maximum electric field in the space charge region at $V_R = 0$ and $V_R = 8$ V

5.(a) Consider a p-type semiconductor that is homogeneous and infinite in extent. Assume a zero applied electric field. For a one-dimensional crystal, assume that excess carriers are being generated at $x = 0$ only. The excess carriers being generated at $x = 0$ will being [*sic*: should be will begin] diffusing in both the $-x$ and $+x$ directions. Calculate the steady-state excess carrier concentration as a function of x. Marks: 7

1. Theory

1. Excess carriers spread and die (Core Idea)

Suppose light falls on a thin slice of an infinitely long p-type bar at $x = 0$ and keeps creating electron-hole pairs there. The extra (**excess**) electrons δn and holes δp are crowded at $x = 0$, so they **diffuse** outward, to the left and to the right. While they diffuse they **recombine**: an excess minority electron survives on average a time τ_{n0}, its lifetime. After a while a **steady state** is reached: at every point, the carriers arriving by diffusion are exactly matched by those lost to recombination. The concentration then no longer changes with time, only with distance.

In p-type material the excess electrons are **minority carriers**. By ambipolar transport (4(b)) the excess holes follow them exactly ($\delta p = \delta n$), and the whole distribution moves with the minority carrier parameters D_n, μ_n, τ_{n0}.

2. The ambipolar transport equation for p-type material

At low injection in p-type material (Neamen, Eq. 6.55):

$$D_n \frac{\partial^2(\delta n)}{\partial x^2} + \mu_n E \frac{\partial(\delta n)}{\partial x} + g' - \frac{\delta n}{\tau_{n0}} = \frac{\partial(\delta n)}{\partial t}$$

The four terms are diffusion, drift, excess generation and recombination; the right side is the change in time.

3. The diffusion length

When the equation reduces to $D_n\,\delta n'' = \delta n/\tau_{n0}$, the combination

$$L_n = \sqrt{D_n \tau_{n0}}$$

has the unit of length (cm^2/s $\times$ s = cm^2, then the square root). It is the **minority carrier diffusion length**: the average distance an excess electron diffuses before it recombines. In silicon it is typically tens of micrometres.

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2. Answer

Given: p-type, homogeneous, infinite, one-dimensional; $E = 0$; excess carriers generated only at $x = 0$; steady state.

Step 3 solves a general second-order equation and is marked (Basic); a student who knows it can skip it.

Step 1: The governing equation

The excess electrons are minority carriers in p-type material, so the ambipolar transport equation at low injection uses the electron parameters (Neamen, Eq. 6.55):

$$D_n \frac{\partial^2(\delta n)}{\partial x^2} + \mu_n E \frac{\partial(\delta n)}{\partial x} + g' - \frac{\delta n}{\tau_{n0}} = \frac{\partial(\delta n)}{\partial t}$$

Step 2: Apply the conditions

- zero applied field: $E = 0$, so the drift term vanishes;
- generation only at $x = 0$: $g' = 0$ for $x \neq 0$;
- steady state: $\partial(\delta n)/\partial t = 0$.

What is left (Neamen, Eq. 6.62):

$$D_n \frac{d^2(\delta n)}{dx^2} - \frac{\delta n}{\tau_{n0}} = 0$$

Divide by D_n and put $L_n^2 = D_n \tau_{n0}$ (Neamen, Eq. 6.63):

$$\frac{d^2(\delta n)}{dx^2} - \frac{\delta n}{L_n^2} = 0$$

Step 3 (Basic): Solve the second-order equation

Try $\delta n = e^{sx}$: then $s^2 e^{sx} - e^{sx}/L_n^2 = 0$, so $s^2 = 1/L_n^2$ and $s = \pm 1/L_n$. The general solution is the sum of both (Neamen, Eq. 6.64):

$$\delta n(x) = A e^{-x/L_n} + B e^{+x/L_n}$$

Step 4: Boundary conditions for $x > 0$

Far away the excess must vanish: $\delta n \to 0$ as $x \to +\infty$. The term $B e^{+x/L_n}$ would grow without limit, so $B = 0$ for $x \geq 0$. At $x = 0$, $\delta n = \delta n(0)$, so $A = \delta n(0)$ (Neamen, Eq. 6.65a):

$$\delta n(x) = \delta n(0)\, e^{-x/L_n}, \qquad x \geq 0$$

Step 5: Boundary conditions for $x < 0$

Now $\delta n \to 0$ as $x \to -\infty$, where e^{-x/L_n} would grow, so $A = 0$ and $B = \delta n(0)$ (Neamen, Eq. 6.65b):

$$\delta n(x) = \delta n(0)\, e^{+x/L_n}, \qquad x \leq 0$$

Step 6: The complete result

Both halves join at $x = 0$ with the same value:

$$\boxed{\delta n(x) = \delta p(x) = \delta n(0) \exp\left(-\frac{|x|}{L_n}\right), \qquad L_n = \sqrt{D_n \tau_{n0}}}$$

The excess holes equal the excess electrons at every point (ambipolar transport, charge neutrality).

Steps to remember

Ambipolar equation with minority (electron) parameters → $E = 0$, $g' = 0$ for $x \neq 0$, steady state → $\delta n'' - \delta n/L_n^2 = 0$ → $A e^{-x/L_n} + B e^{x/L_n}$ → finite at $\pm\infty$ → $\delta n(0) e^{-|x|/L_n}$.

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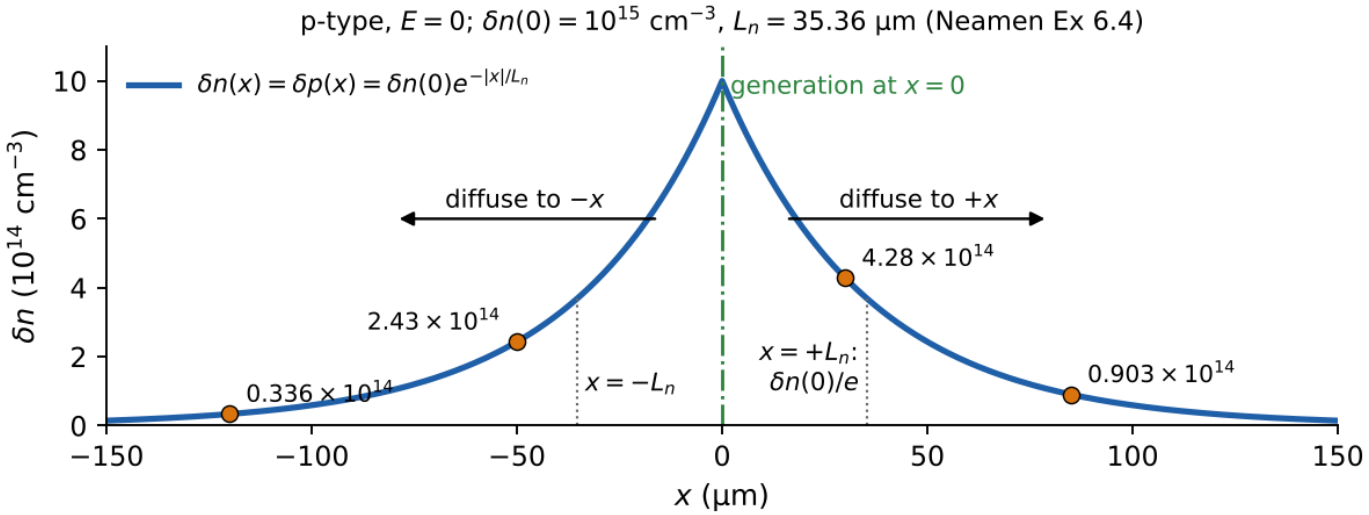


Figure 22: Steady-state excess carrier concentration with generation at x = 0 in an infinite p-type crystal

Reading the figure. The solid blue curve is $\delta n(x) = \delta p(x)$ for Neamen's Exercise Ex 6.4 values ($\delta n(0) = 10^{15}$ cm^{-3}, $D_n = 25$ cm^2/s, $\tau_{n0} = 5 \times 10^{-7}$ s, so $L_n = 35.36$ μm). It peaks at the generation plane $x = 0$ and decays symmetrically on both sides; at $x = \pm L_n$ it is $\delta n(0)/e$ (dotted lines). The dots mark the book's check points at +30, −50, +85 and −120 μm.

Example (Neamen, Ex 6.4)

with $L_n = \sqrt{(25)(5 \times 10^{-7})} = 3.536 \times 10^{-3}$ cm $= 35.36$ μm and $\delta n(0) = 10^{15}$ cm^{-3}: $\delta n = 4.28 \times 10^{14}$ at +30 μm, 2.43×10^{14} at −50 μm, 9.03×10^{13} at +85 μm and 3.36×10^{13} cm^{-3} at −120 μm.

Final Answer: $\delta n(x) = \delta p(x) = \delta n(0)e^{-|x|/L_n}$, i.e. $\delta n(0)e^{-x/L_n}$ for $x \geq 0$ and $\delta n(0)e^{+x/L_n}$ for $x \leq 0$, with $L_n = \sqrt{D_n \tau_{n0}}$; it falls to $1/e$ of its peak one diffusion length from the source.

3. Keyword Explanation

1. Steady state

A condition in which nothing changes with time: $\partial/\partial t = 0$.

2. Minority carrier diffusion length

$L_n = \sqrt{D_n \tau_{n0}}$: the mean distance an excess electron diffuses in p-type material before it recombines.

3. Excess carriers

Carriers above the equilibrium values: $\delta n = n - n_0$, $\delta p = p - p_0$.

4. Notes and Remarks

- **Book check:** the question is Neamen's Example 6.4 word for word ("will begin diffusing"; the paper's "will being" is a misprint). The book's answer is the same, $\delta n(x) = \delta n(0)e^{-x/L_n}$ for $x \geq 0$ and $\delta n(0)e^{+x/L_n}$ for $x \leq 0$, and its comment that the excess falls to $1/e$ at $x = L_n$. The example numbers above are its Exercise Ex 6.4, whose answers ($L_n = 35.36$ μm; 4.28×10^{14}, 2.43×10^{14}, 9.04×10^{13}, 3.36×10^{13} cm^{-3}) match.
- **The pen notes on the term-test scan** (beside the same question, 8(a)) write $E = 0$, $Ae^{-x/L_n} + Be^{x/L_n}$ and a boxed $L_n = \sqrt{D\tau}$: the same steps as here.
- The total electron concentration is $n = n_0 + \delta n$ and the hole concentration $p = p_0 + \delta n \approx p_0$ at low injection.
- Also asked in term test 2 of this session, 8(a). The 2020-21 paper's 3(d) solves the one-sided case ($x > 0$) with numbers (extra reading).

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5.(b) Find ideal pn junction current and sketch ideal electron and hole current components. Marks: 8

1. Theory

1. What a bias does to the barrier (Core Idea)

In equilibrium the built-in barrier V_{bi} holds the majority carriers back so that drift and diffusion cancel (3(b)). A **forward bias** V_a (p side positive) lowers the barrier to $V_{bi} - V_a$. Then many more majority electrons from the n side can climb into the p side, and holes from the p side into the n side: minority carriers are **injected** across the junction. Once in the neutral region, these injected minority carriers diffuse away from the junction and recombine with majority carriers. The diffusion current of these minority carriers *is* the diode current.

2. The ideal-diode assumptions (Neamen, Section 8.1.2)

1. **Abrupt depletion layer:** the space charge region has sharp edges at $-x_p$ and x_n; outside it the semiconductor is neutral.
2. **Boltzmann approximation** holds (non-degenerate material).
3. **Low injection:** injected minority concentrations stay much smaller than the majority concentrations.
4. The total current is the same at every x; the electron and hole currents are continuous through the depletion region (**no generation or recombination inside it**).

Assumption 4 is the key: the hole current at x_n is the same as the hole current at $-x_p$, so the total current is the hole diffusion current at x_n plus the electron diffusion current at $-x_p$.

3. Minority carriers diffuse and recombine

In the neutral n region there is no field, so the injected holes obey the steady-state ambipolar equation with hole (minority) parameters, exactly as in 5(a): they decay as e^{-x/L_p} with $L_p = \sqrt{D_p \tau_{p0}}$.

2. Answer

Derivation (Neamen, Section 8.1; forward bias V_a; $n_{n0} = N_d$, $p_{p0} = N_a$, $n_{p0} = n_i^2/N_a$, $p_{n0} = n_i^2/N_d$; the n region is $x > x_n$, the p region $x < -x_p$). The four ideal-diode assumptions above apply.

Step 4 solves a general second-order equation and is marked (Basic); a student who knows it can skip it.

Step 1: Equilibrium link across the barrier

In equilibrium $V_{bi} = (kT/e)\ln(N_a N_d/n_i^2)$, which can be rearranged with $n_{n0} = N_d$ and $n_{p0} = n_i^2/N_a$ as (Neamen, Eqs. 8.1 to 8.4)

$$\frac{n_i^2}{N_a N_d} = \exp\left(-\frac{eV_{bi}}{kT}\right) \quad \Rightarrow \quad n_{p0} = n_{n0}\exp\left(-\frac{eV_{bi}}{kT}\right)$$

Only the fraction $e^{-eV_{bi}/kT}$ of the majority electrons can climb the barrier.

Step 2: The law of the junction (boundary conditions)

With a forward bias the barrier becomes $V_{bi} - V_a$. With low injection, n_{n0} is unchanged, so the electron concentration at the p-side edge is (Neamen, Eqs. 8.5, 8.6)

$$n_p(-x_p) = n_{n0}\exp\left[-\frac{e(V_{bi}-V_a)}{kT}\right] = n_{p0}\exp\left(\frac{eV_a}{kT}\right)$$

and in the same way for holes at the n-side edge (Neamen, Eq. 8.7):

$$p_n(x_n) = p_{n0}\exp\left(\frac{eV_a}{kT}\right)$$

These are the boundary conditions. Far from the junction the excess goes to zero: $p_n(x \to +\infty) = p_{n0}$, $n_p(x \to -\infty) = n_{p0}$.

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Step 3: Minority carrier diffusion equations

In the neutral n region $E = 0$, $g' = 0$, steady state; the ambipolar equation for the minority holes becomes (Neamen, Eq. 8.9)

$$\frac{d^2(\delta p_n)}{dx^2} - \frac{\delta p_n}{L_p^2} = 0 \quad (x > x_n), \qquad L_p^2 = D_p \tau_{p0}$$

and for the minority electrons in the p region (Neamen, Eq. 8.10)

$$\frac{d^2(\delta n_p)}{dx^2} - \frac{\delta n_p}{L_n^2} = 0 \quad (x < -x_p), \qquad L_n^2 = D_n \tau_{n0}$$

Step 4 (Basic): General solutions

Each is $\frac{d^2y}{dx^2} = y/L^2$, solved by $e^{\pm x/L}$ (Neamen, Eqs. 8.12, 8.13):

$$\delta p_n(x) = Ae^{x/L_p} + Be^{-x/L_p}, \qquad \delta n_p(x) = Ce^{x/L_n} + De^{-x/L_n}$$

Step 5: Apply the boundary conditions

The excess must vanish deep in each region (long diode), so $A = 0$ (n side, $x \to +\infty$) and $D = 0$ (p side, $x \to -\infty$). At the edges, $\delta p_n(x_n) = p_n(x_n) - p_{n0} = p_{n0}[e^{eV_a/kT} - 1]$ and $\delta n_p(-x_p) = n_{p0}[e^{eV_a/kT} - 1]$. So (Neamen, Eqs. 8.14, 8.15)

$$\delta p_n(x) = p_{n0}\left[\exp\left(\frac{eV_a}{kT}\right) - 1\right]\exp\left(\frac{x_n - x}{L_p}\right), \qquad x \geq x_n$$

$$\delta n_p(x) = n_{p0}\left[\exp\left(\frac{eV_a}{kT}\right) - 1\right]\exp\left(\frac{x_p + x}{L_n}\right), \qquad x \leq -x_p$$

(the second is on the paper's equation sheet).

Step 6: Minority diffusion currents at the edges

There is no field in the neutral regions, so the minority current is pure diffusion. The hole current at x_n (Neamen, Eqs. 8.20 to 8.22):

$$J_p(x_n) = -eD_p \frac{d(\delta p_n)}{dx}\bigg|_{x=x_n} = -eD_p \cdot p_{n0}\left[e^{eV_a/kT} - 1\right] \cdot \left(-\frac{1}{L_p}\right) = \frac{eD_p p_{n0}}{L_p}\left[\exp\left(\frac{eV_a}{kT}\right) - 1\right]$$

The electron current at $-x_p$ (Neamen, Eqs. 8.23, 8.24):

$$J_n(-x_p) = eD_n \frac{d(\delta n_p)}{dx}\bigg|_{x=-x_p} = eD_n \cdot n_{p0}\left[e^{eV_a/kT} - 1\right] \cdot \frac{1}{L_n} = \frac{eD_n n_{p0}}{L_n}\left[\exp\left(\frac{eV_a}{kT}\right) - 1\right]$$

Both are positive ($+x$ direction, from p to n).

Step 7: Total current

No recombination in the depletion region (assumption 4) means $J_n(x_n) = J_n(-x_p)$, so the total current, the same everywhere, is (Neamen, Eq. 8.25)

$$J = J_p(x_n) + J_n(-x_p) = \left[\frac{eD_p p_{n0}}{L_p} + \frac{eD_n n_{p0}}{L_n}\right]\left[\exp\left(\frac{eV_a}{kT}\right) - 1\right]$$

$$\boxed{J = J_s\left[\exp\left(\frac{eV_a}{kT}\right) - 1\right], \qquad J_s = \frac{eD_p p_{n0}}{L_p} + \frac{eD_n n_{p0}}{L_n}}$$

(Neamen, Eqs. 8.26, 8.27): the **ideal diode (Shockley) equation**. In reverse bias (V_a negative, a few kT/e) the current saturates at $-J_s$.

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Steps to remember

$n_{p0} = n_{n0}e^{-eV_{bi}/kT}$ → barrier $V_{bi} - V_a$: $n_p(-x_p) = n_{p0}e^{eV_a/kT}$, $p_n(x_n) = p_{n0}e^{eV_a/kT}$ → diffusion equations with L_p, L_n → decaying exponentials → diffusion currents at the edges → add: $J = J_s(e^{eV_a/kT} - 1)$.

Step 8: The current components (the sketch)

The minority diffusion currents decay away from the junction as the injected carriers recombine (Neamen, Eqs. 8.28, 8.29):

$$J_p(x) = \frac{eD_p p_{n0}}{L_p}\left[e^{eV_a/kT} - 1\right]\exp\left(\frac{x_n - x}{L_p}\right), \qquad J_n(x) = \frac{eD_n n_{p0}}{L_n}\left[e^{eV_a/kT} - 1\right]\exp\left(\frac{x_p + x}{L_n}\right)$$

Since the total current J is constant, the **majority** carrier current makes up the rest: $J_n(x) = J - J_p(x)$ in the n region and $J_p(x) = J - J_n(x)$ in the p region. The majority carriers flowing towards the junction supply the carriers that are injected and those that recombine with the injected minority carriers.

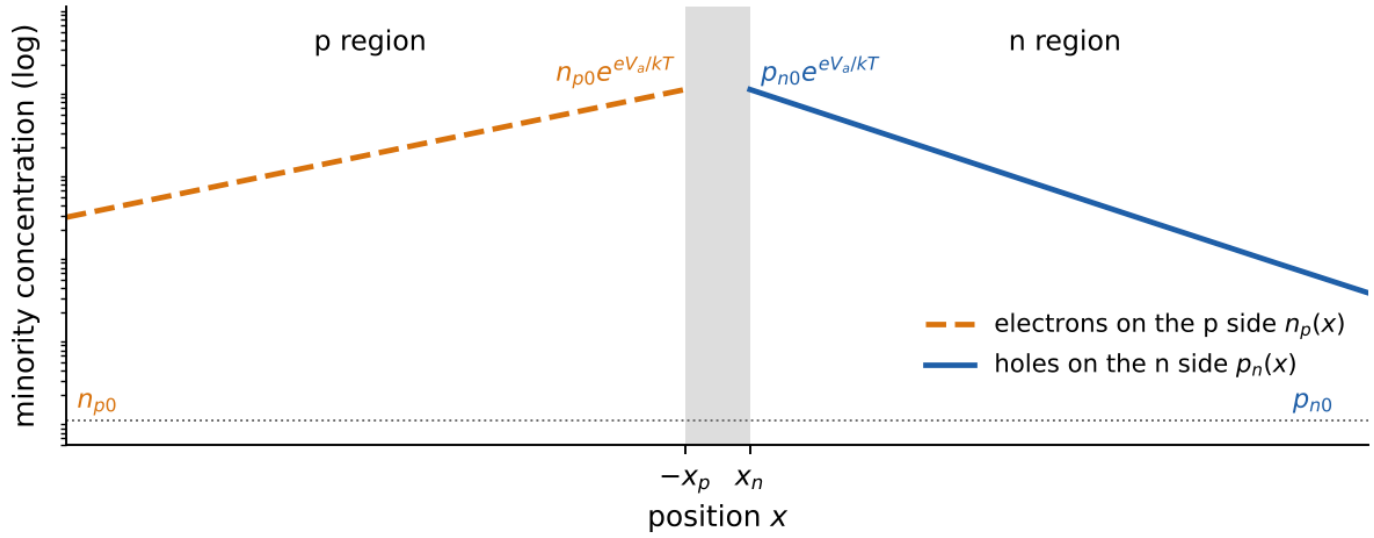


Figure 23: Minority carrier distributions on both sides of a forward-biased pn junction

Reading the figure. The space charge region runs from $-x_p$ to x_n (grey). On the p side (left) the dashed orange curve is the electron (minority) concentration, raised to $n_{p0}e^{eV_a/kT}$ at $-x_p$ and decaying to n_{p0} over a few L_n. On the n side (right) the solid blue curve is the hole (minority) concentration, raised to $p_{n0}e^{eV_a/kT}$ at x_n and decaying to p_{n0} over a few L_p (log scale; drawn with $N_a = N_d$, so $n_{p0} = p_{n0}$).

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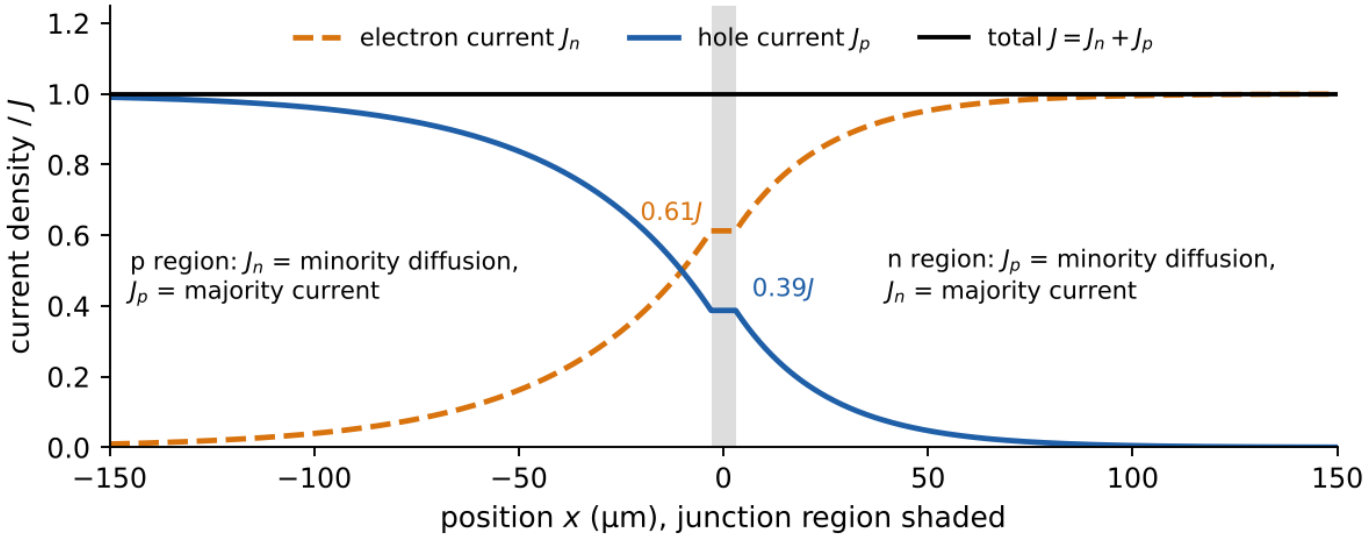


Figure 24: Ideal electron and hole current components through a forward-biased pn junction

Reading the figure. Current density against position, normalized to the total J. The dashed orange curve is the electron current J_n: on the n side (right) it is majority current, rising to J far from the junction; across the space charge region it is constant; on the p side it is the minority diffusion current, decaying away from $-x_p$. The solid blue curve is the hole current J_p, the mirror image. The flat black line is the total $J = J_n + J_p$, the same at every x. Drawn for $N_a = N_d = 10^{16}$ cm^{-3}, $D_n = 25$, $D_p = 10$ cm^2/s, $\tau_{n0} = \tau_{p0} = 0.5$ μs, so $J_n(-x_p) = 0.61J$ and $J_p(x_n) = 0.39J$.

Example

for that diode, $n_{p0} = p_{n0} = 2.25 \times 10^4$ cm^{-3}, $L_n = 35.4$ μm, $L_p = 22.4$ μm, and $J_s = 1.61 \times 10^{-11} + 2.55 \times 10^{-11} = 4.16 \times 10^{-11}$ A/cm^2; at $V_a = 0.6$ V, $J = 0.48$ A/cm^2.

3. Keyword Explanation

1. Law of the junction

$p_n(x_n) = p_{n0}e^{eV_a/kT}$, $n_p(-x_p) = n_{p0}e^{eV_a/kT}$: the minority carrier concentrations at the edges of the space charge region under bias.

2. Reverse saturation current density

$J_s = eD_p p_{n0}/L_p + eD_n n_{p0}/L_n$: the current in reverse bias, set by minority carriers generated within a diffusion length of the junction.

3. Minority carrier injection

The flow of carriers across a lowered barrier into the region where they are minority carriers.

4. Notes and Remarks

- **Which side dominates:** $p_{n0} = n_i^2/N_d$ and $n_{p0} = n_i^2/N_a$, so the lightly doped side's minority carriers carry most of the current. In a p^+n diode almost all of J is hole injection into the n side.
- **Short diode:** if a neutral region is shorter than a diffusion length, the exponentials become straight lines and L is replaced by the region's length (Neamen, Section 8.1.7).
- **Real diodes** also have a generation-recombination current in the space charge region, $\propto e^{eV_a/2kT}$, which dominates at low forward bias (Neamen, Section 8.2).
- The equation sheet prints $J = J_p(x_n) + J_n(-x_p) = \left[\frac{eD_p p_{n0}}{L_p} + \frac{eD_n n_{p0}}{L_n}\right][e^{eV_a/kT} - 1]$; it is the result of Step 7.
- The current components alone were asked as 4(a) of the 2018–19 paper (2018–19 4(a)), extra

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reading only.

5.(c) Consider an n-type semiconductor at $T = 300$ K with carrier concentrations of $n_o = 10^{15}$ cm^{-3}, $n_i = 10^{10}$ cm^{-3} and $p_o = 10^{5}$ cm^{-3}. In non-equilibrium, assume that the excess carrier concentrations are $\delta n = \delta p = 10^{-13}$ [*sic*: should be 10^{13}] cm^{-3}. Calculate the quasi-Fermi energy levels. Marks: 4

1. Theory

1. The Fermi level describes equilibrium only (Core Idea)

In thermal equilibrium both carrier concentrations follow from one Fermi level (Neamen, Eqs. 6.84a, 6.84b):

$$n_0 = n_i \exp\left(\frac{E_F - E_{Fi}}{kT}\right), \qquad p_0 = n_i \exp\left(\frac{E_{Fi} - E_F}{kT}\right)$$

and their product is n_i^2. When excess carriers are generated (by light, say), both n and p rise, so $np > n_i^2$. No single E_F can describe both any more.

2. One level for each carrier

The electrons among themselves come to equilibrium in about a picosecond, and so do the holes; only the exchange between the two (recombination) is slow. So each carrier type can be described by its own Fermi level, the **quasi-Fermi level** (Neamen, Eqs. 6.85a, 6.85b):

$$n_0 + \delta n = n_i \exp\left(\frac{E_{Fn} - E_{Fi}}{kT}\right), \qquad p_0 + \delta p = n_i \exp\left(\frac{E_{Fi} - E_{Fp}}{kT}\right)$$

where E_{Fn} is the quasi-Fermi level for electrons and E_{Fp} for holes. Their separation measures how far the material is from equilibrium: $np = n_i^2 \exp[(E_{Fn} - E_{Fp})/kT]$ (Neamen, Eq. 8.19).

3. What low injection does

In an n-type sample at low injection, $\delta n \ll n_0$: the electron concentration hardly changes, so E_{Fn} is practically at the old E_F. But δp can be enormous compared with the tiny p_0: the hole concentration changes by many orders of magnitude, so E_{Fp} moves far from E_F, towards E_v.

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2. Answer

Note on the paper

The paper prints $\delta n = \delta p = 10^{-13}$ cm^{-3}. An excess of 10^{-13} carriers per cm^3 is meaningless (less than one carrier in 10^{13} cm^3) and would leave the material in equilibrium. The question is Neamen's Example 6.7, which has $\delta n = \delta p = 10^{13}$ cm^{-3}; the exponent's minus sign is a misprint, and the question is solved with 10^{13} cm^{-3}.

Given: n-type, $T = 300$ K ($kT = 0.0259$ eV), $n_0 = 10^{15}$ cm^{-3}, $n_i = 10^{10}$ cm^{-3}, $p_0 = 10^5$ cm^{-3}, $\delta n = \delta p = 10^{13}$ cm^{-3}.

Step 1: Equilibrium Fermi level (for comparison)

$$E_F - E_{Fi} = kT \ln\left(\frac{n_0}{n_i}\right) = (0.0259)\ln\left(\frac{10^{15}}{10^{10}}\right) = (0.0259)(11.513) = 0.2982 \text{ eV}$$

Step 2: Quasi-Fermi level for electrons

$$n = n_0 + \delta n = 10^{15} + 10^{13} = 1.01 \times 10^{15} \text{ cm}^{-3}$$

$$E_{Fn} - E_{Fi} = kT \ln\left(\frac{n_0 + \delta n}{n_i}\right) = (0.0259)\ln\left(1.01 \times 10^5\right) = 0.2984 \text{ eV}$$

$$\boxed{E_{Fn} - E_{Fi} = 0.2984 \text{ eV}}$$

only 0.26 meV above the equilibrium E_F: the majority carriers barely notice the injection.

Step 3: Quasi-Fermi level for holes

$$p = p_0 + \delta p = 10^5 + 10^{13} \approx 1.0 \times 10^{13} \text{ cm}^{-3}$$

$$E_{Fi} - E_{Fp} = kT \ln\left(\frac{p_0 + \delta p}{n_i}\right) = (0.0259)\ln\left(10^3\right) = 0.179 \text{ eV}$$

$$\boxed{E_{Fi} - E_{Fp} = 0.179 \text{ eV}}$$

E_{Fp} lies 0.179 eV **below** E_{Fi}, whereas the equilibrium E_F was 0.298 eV above it: the minority hole concentration rose by a factor of 10^8.

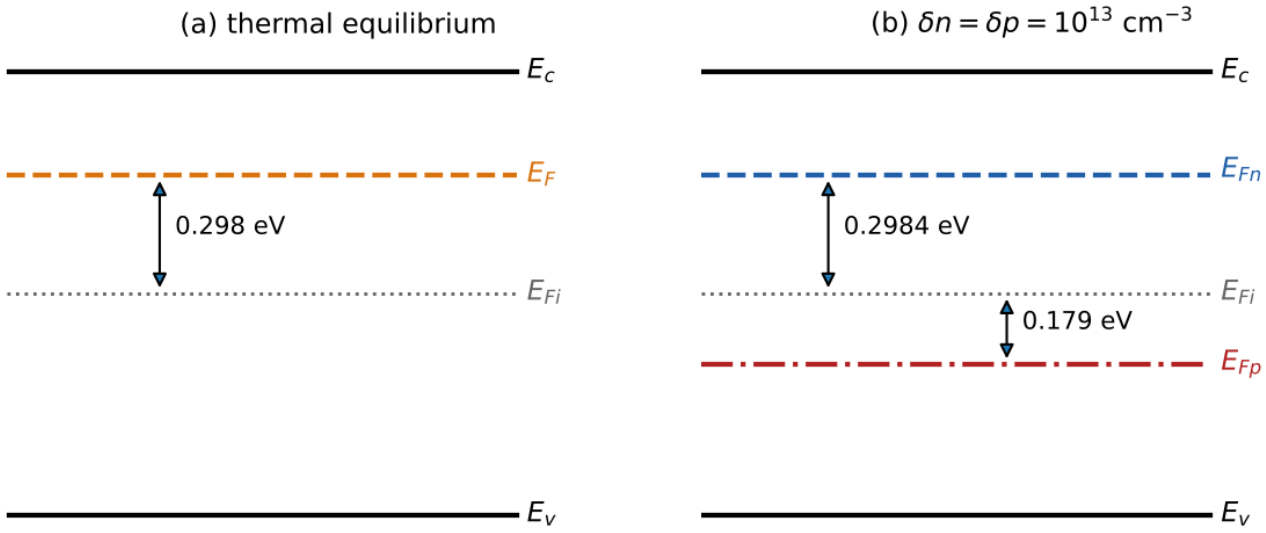


Figure 25: Band diagram in equilibrium and with excess carriers, showing the quasi-Fermi levels

Reading the figure. Panel (a), thermal equilibrium: the single Fermi level E_F (dashed orange) is

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0.298 eV above E_{Fi} (dotted grey). Panel (b), with $\delta n = \delta p = 10^{13}$ cm^{-3}: the electron quasi-Fermi level E_{Fn} (dashed blue) is practically where E_F was (0.2984 eV), while the hole quasi-Fermi level E_{Fp} (dash-dot red) has dropped to 0.179 eV below E_{Fi}. Their separation, 0.477 eV, measures the departure from equilibrium.

Final Answer: $E_{Fn} - E_{Fi} = 0.2984$ eV (above E_{Fi}); $E_{Fi} - E_{Fp} = 0.179$ eV (E_{Fp} below E_{Fi}); equilibrium $E_F - E_{Fi} = 0.2982$ eV; splitting $E_{Fn} - E_{Fp} = 0.477$ eV.

3. Keyword Explanation

1. Quasi-Fermi level

E_{Fn} or E_{Fp}: the level that, put in the equilibrium formula, gives the actual non-equilibrium electron or hole concentration.

2. Low-level injection

$\delta n = \delta p \ll n_0$ in n-type: the majority concentration is almost unchanged.

3. Non-equilibrium

A state with $np \neq n_i^2$, maintained by an external source such as light or bias.

4. Notes and Remarks

- **Book check:** Neamen's Example 6.7 gives $E_F - E_{Fi} = 0.2982$ eV, $E_{Fn} - E_{Fi} = 0.2984$ eV and $E_{Fi} - E_{Fp} = 0.179$ eV, the same as here.
- **With the printed 10^{-13}:** $p = 10^5 + 10^{-13} = 10^5$ cm^{-3} to any precision, so $E_{Fn} = E_{Fp} = E_F$ (0.2982 eV above E_{Fi}): nothing to calculate. This confirms the misprint.
- In a forward-biased pn junction the quasi-Fermi levels split by eV_a across the depletion region, $np = n_i^2 e^{eV_a/kT}$ (Neamen, Eq. 8.18); that is where the law of the junction in 5(b) comes from.

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5.(d) An abrupt silicon pn junction has dopant concentrations of $N_a = 2 \times 10^{16}$ cm^{-3} and $N_d = 2 \times 10^{15}$ cm^{-3} at $T = 300$ K. Calculate: i. V_{bi} ii. W at $V_R = 0$ and $V_R = 8$ V and iii. the maximum electric field in the space charge region at $V_R = 0$ and $V_R = 8$ V Marks: 6

1. Theory

1. The junction in equilibrium (Core Idea)

An abrupt pn junction has acceptors N_a on one side and donors N_d on the other, changing sharply at the metallurgical junction. Diffusion of carriers across it uncovers fixed ions, $-eN_a$ over x_p and $+eN_d$ over x_n, forming the **space charge region** of width $W = x_n + x_p$. Its field points from n to p, and the potential across it is the **built-in potential** (Neamen, Eq. 7.10):

$$V_{bi} = \frac{kT}{e} \ln\left(\frac{N_a N_d}{n_i^2}\right)$$

2. What a reverse bias does

A reverse voltage V_R (n side positive) adds to the barrier: the total potential across the space charge region becomes $V_{bi}+V_R$ (Neamen, Eq. 7.33). More charge must be uncovered to hold it, so the depletion region **widens** and the field **grows**. The equilibrium formulas still hold with V_{bi} replaced by $V_{bi} + V_R$ (Neamen, Eqs. 7.34, 7.36, 7.37):

$$W = \left\{\frac{2\epsilon_s(V_{bi} + V_R)}{e}\left[\frac{N_a + N_d}{N_a N_d}\right]\right\}^{1/2}, \qquad |E_{max}| = \frac{eN_d x_n}{\epsilon_s} = \frac{2(V_{bi} + V_R)}{W}$$

The field is a triangle peaked at the junction; its area equals the total potential $V_{bi} + V_R$, which is why $|E_{max}| = 2(V_{bi} + V_R)/W$.

3. Where the depletion lies

Charge balance, $N_a x_p = N_d x_n$ (Neamen, Eq. 7.17), means $x_n/x_p = N_a/N_d$. Here $N_a = 10N_d$, so 10/11 of W is on the n side:

$$x_n = W\frac{N_a}{N_a + N_d}, \qquad x_p = W\frac{N_d}{N_a + N_d}$$

2. Answer

Given: Si, $T = 300$ K; $N_a = 2 \times 10^{16}$ cm^{-3}, $N_d = 2 \times 10^{15}$ cm^{-3}; $n_i = 1.5 \times 10^{10}$ cm^{-3}, $kT/e = 0.0259$ V, $\epsilon_s = 11.7 \times 8.85 \times 10^{-14} = 1.0355 \times 10^{-12}$ F/cm (Neamen's values for Si).

Step 1: Built-in potential

$$\frac{N_a N_d}{n_i^2} = \frac{(2 \times 10^{16})(2 \times 10^{15})}{(1.5 \times 10^{10})^2} = 1.778 \times 10^{11}, \qquad \ln\left(1.778 \times 10^{11}\right) = 25.904$$

$$V_{bi} = (0.0259)(25.904) = 0.671 \text{ V}$$

$$\boxed{V_{bi} = 0.671 \text{ V}}$$

Step 2: Common factors

$$\frac{2\epsilon_s}{e} = \frac{2(1.0355 \times 10^{-12})}{1.6 \times 10^{-19}} = 1.294 \times 10^{7} \text{ (V cm)}^{-1}, \qquad \frac{N_a + N_d}{N_a N_d} = \frac{2.2 \times 10^{16}}{4 \times 10^{31}} = 5.5 \times 10^{-16} \text{ cm}^3$$

Their product is 7.119×10^{-9} cm^2/V, so $W = \sqrt{7.119 \times 10^{-9}\,(V_{bi} + V_R)}$ cm.

Step 3: Depletion width at $V_R = 0$

$$W = \sqrt{(7.119 \times 10^{-9})(0.671)} = 6.91 \times 10^{-5} \text{ cm}$$

$$\boxed{W(0 \text{ V}) = 0.691\ \mu\text{m}}$$

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with $x_n = 0.628$ μm and $x_p = 0.0628$ μm.

Step 4: Depletion width at $V_R = 8$ V

$$V_{bi} + V_R = 8.671 \text{ V}, \qquad W = \sqrt{(7.119 \times 10^{-9})(8.671)} = 2.48 \times 10^{-4} \text{ cm}$$

$$\boxed{W(8 \text{ V}) = 2.48 \ \mu\text{m}}$$

with $x_n = 2.26$ μm and $x_p = 0.226$ μm.

Step 5: Maximum field at $V_R = 0$

$$|E_{max}| = \frac{eN_d x_n}{\epsilon_s} = \frac{(1.6 \times 10^{-19})(2 \times 10^{15})(6.283 \times 10^{-5})}{1.0355 \times 10^{-12}} = 1.94 \times 10^4 \text{ V/cm}$$

Check: $2V_{bi}/W = 2(0.671)/(6.911 \times 10^{-5}) = 1.94 \times 10^4$ V/cm.

$$\boxed{E_{max}(0 \text{ V}) = -1.94 \times 10^4 \text{ V/cm}}$$

Step 6: Maximum field at $V_R = 8$ V

$$|E_{max}| = \frac{2(V_{bi} + V_R)}{W} = \frac{2(8.671)}{2.4845 \times 10^{-4}} = 6.98 \times 10^4 \text{ V/cm}$$

$$\boxed{E_{max}(8 \text{ V}) = -6.98 \times 10^4 \text{ V/cm}}$$

(The minus sign: the field points from n to p, in $-x$.)

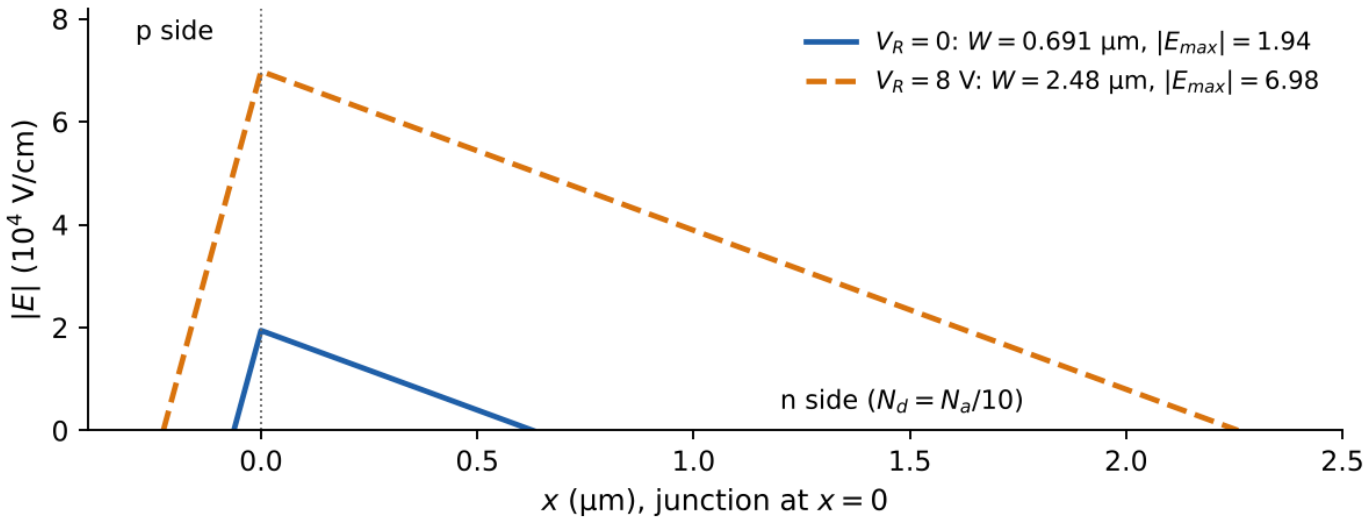


Figure 26: Electric field in the space charge region at VR = 0 and VR = 8 V

Reading the figure. The field magnitude against position, with the junction at $x = 0$. The solid blue triangle is $V_R = 0$ ($W = 0.691$ μm, peak 1.94×10^4 V/cm), the dashed orange triangle $V_R = 8$ V ($W = 2.48$ μm, peak 6.98×10^4 V/cm). Both extend ten times further into the lightly doped n side than into the p side. The area under each triangle is $V_{bi} + V_R$.

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Final Answer:

	$V_R = 0$	$V_R = 8$ **V**
$V_{bi} + V_R$	0.671 V	8.671 V
W	0.691 μm	2.48 μm
x_n, x_p	0.628, 0.0628 μm	2.26, 0.226 μm
E_{max}	-1.94×10^4 V/cm	-6.98×10^4 V/cm

Table 7: Summary

3. Keyword Explanation

1. Abrupt junction

A junction where the doping changes from N_a to N_d in a step at the metallurgical junction.

2. Reverse bias

A voltage making the n side positive with respect to the p side; it raises the barrier to $V_{bi} + V_R$.

3. Maximum electric field

The field at the metallurgical junction, the peak of the triangle, $|E_{max}| = 2(V_{bi} + V_R)/W$; it sets the breakdown voltage.

4. Notes and Remarks

- W grows as $\sqrt{V_{bi} + V_R}$: a 13-fold increase in total voltage makes W $\sqrt{12.92} = 3.6$ times wider, and E_{max} also 3.6 times larger.
- The junction capacitance per area is $C' = \epsilon_s/W$ (Neamen, Eq. 7.43): 1.50×10^{-8} F/cm^2 at 0 V and 4.17×10^{-9} F/cm^2 at 8 V; the equation sheet's C' formula gives the same.
- The peak field at 8 V is far below the breakdown field of Si at this doping (about 3×10^5 V/cm).
- The 2020-21 paper's 6(d) is the same calculation at zero bias with other doping (extra reading).

5. Final Concise Answer

(a) With $E = 0$, $g' = 0$ for $x \neq 0$ and steady state, the ambipolar equation for p-type becomes $D_n \delta n'' - \delta n/\tau_{n0} = 0$, so $\delta n = Ae^{-x/L_n} + Be^{x/L_n}$; finiteness at $\pm\infty$ gives $\delta n(x) = \delta p(x) = \delta n(0) e^{-|x|/L_n}$, $L_n = \sqrt{D_n \tau_{n0}}$ (Neamen Example 6.4).

(b) Law of the junction $p_n(x_n) = p_{n0} e^{eV_a/kT}$, $n_p(-x_p) = n_{p0} e^{eV_a/kT}$; minority diffusion in the neutral regions gives $\delta p_n = p_{n0}(e^{eV_a/kT} - 1) e^{(x_n - x)/L_p}$ and $\delta n_p = n_{p0}(e^{eV_a/kT} - 1) e^{(x_p + x)/L_n}$; their diffusion currents at the edges add to $J = \left[\frac{eD_p p_{n0}}{L_p} + \frac{eD_n n_{p0}}{L_n}\right](e^{eV_a/kT} - 1)$. The minority currents decay away from the junction and the majority currents rise so that $J_n + J_p = J$ everywhere.

(c) With $\delta n = \delta p = 10^{13}$ cm^{-3} (the printed 10^{-13} is a misprint): $E_F - E_{Fi} = 0.2982$ eV; $E_{Fn} - E_{Fi} = kT \ln[(n_0 + \delta n)/n_i] = 0.2984$ eV; $E_{Fi} - E_{Fp} = kT \ln[(p_0 + \delta p)/n_i] = 0.179$ eV (Neamen Example 6.7).

(d) $V_{bi} = 0.0259 \ln(1.778 \times 10^{11}) = 0.671$ V; $W = 0.691$ μm at $V_R = 0$ and 2.48 μm at 8 V; $E_{max} = -1.94 \times 10^4$ V/cm at 0 V and -6.98×10^4 V/cm at 8 V.

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6. (a) The body of a nanoscale N-channel MOSFET is doped at $N_A = 5^{18}$ [*sic*: should be 5×10^{18}] cm^{-3}. Marks: 12

   i. Determine the maximum inversion-layer charge per unit area (Q_I) that can be reached so that the electric field in the oxide remains below $E_{ox} = 0.6$ V/nm. Neglect any interface-trap and gate-oxide charge.
   ii. If the channel length and the channel width are $L = 50$ nm and $W = 500$ nm, respectively, determine how many electrons are creating this inversion layer.
   iii. Calculate the threshold voltage of a MOSFET having gate-oxide thickness $t_{ox} = 1$ nm and flat-band voltage $V_{FB} = 1.0$ V.
   iv. What gate voltage is needed to form the maximum inversion-layer charge?

   (b) Determine the mode of operation of an NPN BJT with $\beta \approx 450$, if it is known that: Marks: 6

   i. $V_{BE} = 0.7V$, $V_{CE} = 5.2V$
   ii. $V_{BE} = 0.7V$, $V_{CE} = 0.2V$
   iii. $V_{BE} = 0.8V$, $V_{BC} = 0.3V$
   iv. $V_{BE} = -0.8V$, $V_{BC} = 0.7V$
   v. $I_C = 455mA$, $I_B = 1mA$
   vi. $I_C = 455mA$, $I_E = 502mA$

   (c) Mention some advantages of using MOSFET. Marks: 3

   (d) Does the doping level in the base of a BJT affect the emitter efficiency γ_E? If so, why? Marks: 4

6.(a) The body of a nanoscale N-channel MOSFET is doped at $N_A = 5^{18}$ [*sic*: should be 5×10^{18}] cm^{-3}. i. Determine the maximum inversion-layer charge per unit area (Q_I) that can be reached so that the electric field in the oxide remains below $E_{ox} = 0.6$ V/nm. Neglect any interface-trap and gate-oxide charge. ii. If the channel length and the channel width are $L = 50$ nm and $W = 500$ nm, respectively, determine how many electrons are creating this inversion layer. iii. Calculate the threshold voltage of a MOSFET having gate-oxide thickness $t_{ox} = 1$ nm and flat-band voltage $V_{FB} = 1.0$ V. iv. What gate voltage is needed to form the maximum inversion-layer charge? Marks: 12

1. Theory

1. The MOS capacitor (Core Idea)

An n-channel MOSFET is built on a **p-type** body. On top of the body is a thin insulating layer, the **gate oxide** (SiO_2, thickness t_{ox}), and on top of that the **gate** electrode. Gate, oxide and body form a capacitor with capacitance per unit area (Neamen, Section 10.1)

$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}}, \qquad \epsilon_{ox} = 3.9\epsilon_0$$

A positive gate voltage puts positive charge on the gate. To balance it, negative charge must gather in the silicon under the oxide. It comes in two forms:

1. **Depletion charge** Q'_{SD}: holes are pushed away from the surface, leaving the fixed negative acceptor ions uncovered over a depth x_d; $|Q'_{SD}| = eN_a x_d$.
2. **Inversion charge** Q'_I: once the surface potential is large enough, free **electrons** gather in a thin layer right at the surface. This electron layer is the **channel** of the n-channel MOSFET.

2. Threshold and the maximum depletion width

The surface is said to be **inverted** when the electron concentration at the surface equals the hole concentration in the bulk, which happens when the bands at the surface have bent by $2\phi_{fp}$ (Neamen, Section 10.1.3):

$$\phi_{fp} = V_t \ln\left(\frac{N_a}{n_i}\right), \qquad \phi_s = 2\phi_{fp} \text{ at threshold}$$

Beyond threshold, any extra gate charge is balanced by extra **electrons** (the electron concentration rises exponentially with ϕ_s), so the depletion region **stops growing**. Its maximum width and charge are (Neamen, Eqs. 10.6, 10.27)

$$x_{dT} = \left(\frac{4\epsilon_s \phi_{fp}}{eN_a}\right)^{1/2}, \qquad |Q'_{SD}(max)| = eN_a x_{dT}$$

3. Gauss's law across the oxide

The oxide contains no charge (interface-trap and oxide charges are neglected). By Gauss's law the electric displacement in the oxide equals the total charge per unit area below it, in the silicon:

$$\epsilon_{ox} E_{ox} = |Q'_{SD}| + |Q'_I|$$

So the **oxide field** limits how much charge the gate can induce. Too large a field breaks the thin oxide down or makes it leak, which is why a nanoscale MOSFET with $t_{ox} = 1$ nm is limited to $E_{ox} \approx 0.6$ V/nm.

4. Threshold voltage and gate voltage

The gate voltage is shared between the oxide and the silicon (Neamen, Eq. 10.21 with ϕ_{ms} and the oxide charge collected in V_{FB}):

$$V_G = V_{FB} + \phi_s + V_{ox}, \qquad V_{ox} = E_{ox} t_{ox} = \frac{|Q'_{SD}| + |Q'_I|}{C_{ox}}$$

At threshold, $\phi_s = 2\phi_{fp}$ and $Q'_I = 0$ (Neamen, Eq. 10.31c):

$$V_T = V_{FB} + 2\phi_{fp} + \frac{|Q'_{SD}(max)|}{C_{ox}}$$

Above threshold, ϕ_s stays at about $2\phi_{fp}$ and each extra volt on the gate adds C_{ox} of inversion charge: $|Q'_I| = C_{ox}(V_G - V_T)$.

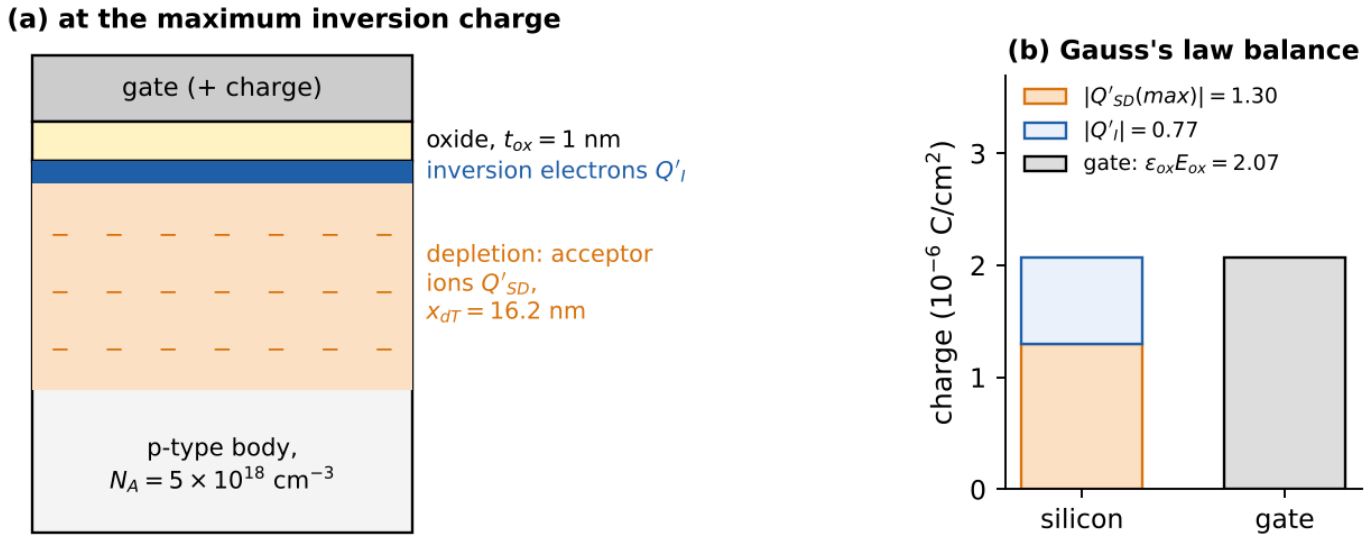


Figure 27: MOS structure at the maximum inversion charge: charges, oxide field and the Gauss balance

Reading the figure. Left: the cross section, gate on top of a 1 nm oxide on the p-type body; under the oxide a thin sheet of inversion electrons Q'_I (blue) above the depletion region of uncovered acceptors

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Q'_{SD} (orange, 16.2 nm deep). Right: the charge per area balanced by the gate charge, and the oxide field $E_{ox} = (|Q'_{SD}| + |Q'_I|)/\epsilon_{ox} = 0.6$ V/nm. Of the 2.07×10^{-6} C/cm^2 the oxide field allows, 1.30×10^{-6} C/cm^2 is taken by the depletion charge and 0.77×10^{-6} C/cm^2 is left for the channel.

2. Answer

Note on the paper

"$N_A = 5^{18}$ cm^{-3}" means $N_A = 5 \times 10^{18}$ cm^{-3} ($5^{18} = 3.8 \times 10^{12}$ would be absurd for a nanoscale MOSFET body); the "×10" was lost in printing.

Given: p-type body $N_A = 5 \times 10^{18}$ cm^{-3}; $E_{ox,max} = 0.6$ V/nm $= 6 \times 10^6$ V/cm; $L = 50$ nm, $W = 500$ nm; $t_{ox} = 1$ nm $= 10^{-7}$ cm; $V_{FB} = 1.0$ V. Constants (Neamen): $n_i = 1.5 \times 10^{10}$ cm^{-3}, $V_t = 0.0259$ V, $\epsilon_s = 11.7\epsilon_0 = 1.0355 \times 10^{-12}$ F/cm, $\epsilon_{ox} = 3.9\epsilon_0 = 3.4515 \times 10^{-13}$ F/cm, $e = 1.6 \times 10^{-19}$ C.

Assumptions: no interface-trap or oxide charge (as stated); in strong inversion the depletion charge stays at its threshold value $|Q'_{SD}(max)|$ and the surface potential at $2\phi_{fp}$.

Step 1: Fermi potential of the body

$$\phi_{fp} = V_t \ln\left(\frac{N_A}{n_i}\right) = (0.0259)\ln\left(\frac{5 \times 10^{18}}{1.5 \times 10^{10}}\right) = (0.0259)(19.625) = 0.5083 \text{ V}$$

so $2\phi_{fp} = 1.0166$ V.

Step 2: Maximum depletion width and depletion charge

$$x_{dT} = \left(\frac{4\epsilon_s \phi_{fp}}{eN_A}\right)^{1/2} = \left[\frac{4(1.0355 \times 10^{-12})(0.5083)}{(1.6 \times 10^{-19})(5 \times 10^{18})}\right]^{1/2} = 1.622 \times 10^{-6} \text{ cm} = 16.2 \text{ nm}$$

$$|Q'_{SD}(max)| = eN_A x_{dT} = (1.6 \times 10^{-19})(5 \times 10^{18})(1.622 \times 10^{-6}) = 1.298 \times 10^{-6} \text{ C/cm}^2$$

Step 3 (part i): Maximum inversion charge from Gauss's law

The total charge the oxide field allows:

$$\epsilon_{ox} E_{ox} = (3.4515 \times 10^{-13})(6 \times 10^6) = 2.071 \times 10^{-6} \text{ C/cm}^2$$

This must equal $|Q'_{SD}(max)| + |Q'_I|$, so

$$|Q'_I|_{max} = \epsilon_{ox} E_{ox} - |Q'_{SD}(max)| = 2.071 \times 10^{-6} - 1.298 \times 10^{-6} = 7.73 \times 10^{-7} \text{ C/cm}^2$$

$$\boxed{Q_I = -7.73 \times 10^{-7} \text{ C/cm}^2 = -7.73 \times 10^{-3} \text{ C/m}^2}$$

(negative: it is electron charge). As an electron density, $7.73 \times 10^{-7}/1.6 \times 10^{-19} = 4.83 \times 10^{12}$ electrons/cm^2.

Step 4 (part ii): Number of electrons in the channel

Channel area: $A = LW = (50 \times 10^{-7} \text{ cm})(500 \times 10^{-7} \text{ cm}) = 2.5 \times 10^{-10}$ cm^2.

$$N = \frac{|Q'_I|A}{e} = \frac{(7.73 \times 10^{-7})(2.5 \times 10^{-10})}{1.6 \times 10^{-19}} = 1208$$

$$\boxed{N \approx 1.2 \times 10^3 \text{ electrons}}$$

Only about 1200 electrons form the whole channel of this transistor.

Step 5 (part iii): Threshold voltage

$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}} = \frac{3.4515 \times 10^{-13}}{10^{-7}} = 3.4515 \times 10^{-6} \text{ F/cm}^2$$

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$$\frac{|Q'_{SD}(max)|}{C_{ox}} = \frac{1.298 \times 10^{-6}}{3.4515 \times 10^{-6}} = 0.376 \text{ V}$$

$$V_T = V_{FB} + 2\phi_{fp} + \frac{|Q'_{SD}(max)|}{C_{ox}} = 1.0 + 1.0166 + 0.376 = 2.39 \text{ V}$$

$$\boxed{V_T = 2.39 \text{ V}}$$

Step 6 (part iv): Gate voltage for the maximum inversion charge

Above threshold the extra gate voltage appears across the oxide and induces $|Q'_I| = C_{ox}(V_G - V_T)$:

$$V_G = V_T + \frac{|Q'_I|_{max}}{C_{ox}} = 2.39 + \frac{7.73 \times 10^{-7}}{3.4515 \times 10^{-6}} = 2.39 + 0.224 = 2.62 \text{ V}$$

Check from the voltage division: $V_G = V_{FB} + 2\phi_{fp} + E_{ox}t_{ox} = 1.0 + 1.0166 + (0.6 \text{ V/nm})(1 \text{ nm}) = 2.62$ V.

$$\boxed{V_G = 2.62 \text{ V}}$$

Final Answer:

- (i) $|Q_I|_{max} = 7.73 \times 10^{-7}$ C/cm^2 (4.83×10^{12} electrons/cm^2)
- (ii) about 1208 electrons
- (iii) $V_T = 2.39$ V
- (iv) $V_G = 2.62$ V

3. Keyword Explanation

1. Inversion layer

The thin layer of minority carriers (electrons in a p-type body) at the surface once $\phi_s > 2\phi_{fp}$; the channel of the MOSFET.

2. Threshold voltage

The gate voltage at which the surface just inverts, $\phi_s = 2\phi_{fp}$: $V_T = V_{FB} + 2\phi_{fp} + |Q'_{SD}(max)|/C_{ox}$.

3. Flat-band voltage

The gate voltage at which the bands in the silicon are flat (no charge in the silicon); it collects the work-function difference and the oxide charges.

4. Oxide field

$E_{ox} = (|Q'_{SD}| + |Q'_I|)/\epsilon_{ox}$; it is limited by oxide breakdown and leakage.

4. Notes and Remarks

- **Source of the question:** this is not a Neamen problem; it follows the textbook treatment of a nanoscale MOSFET in which the oxide field is the limit (the form of Dimitrijev's *Principles of Semiconductor Devices*). It is solved here with Neamen's equations (Section 10.1) and constants. A different n_i (some books use 1.02×10^{10} cm^{-3}) changes $2\phi_{fp}$ by about 0.02 V and the results in the third digit.
- With a 1 nm oxide, $E_{ox}t_{ox} = 0.6$ V is all the oxide can take; that is why such a MOSFET works with a supply of about 1 V and why high-k dielectrics replaced SiO_2.
- The depletion charge is large here (N_A is very high to stop short-channel effects), which leaves only 37 % of the allowed charge for the channel.

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6.(b) Determine the mode of operation of an NPN BJT with $\beta \approx 450$, if it is known that: i. $V_{BE} = 0.7V$, $V_{CE} = 5.2V$ ii. $V_{BE} = 0.7V$, $V_{CE} = 0.2V$ iii. $V_{BE} = 0.8V$, $V_{BC} = 0.3V$ iv. $V_{BE} = -0.8V$, $V_{BC} = 0.7V$ v. $I_C = 455mA$, $I_B = 1mA$ vi. $I_C = 455mA$, $I_E = 502mA$ Marks: 6

1. Theory

1. Two junctions, four modes (Core Idea)

An NPN bipolar transistor is an n-type emitter, a thin p-type base and an n-type collector: two pn junctions back to back, the base-emitter (B-E) and the base-collector (B-C) junction. Each can be forward or reverse biased, so there are four **modes of operation** (Neamen, Section 12.1.3):

Mode	B-E junction	B-C junction	For an NPN
Forward active	forward	reverse	$V_{BE} > 0$, $V_{BC} < 0$
Saturation	forward	forward	$V_{BE} > 0$, $V_{BC} > 0$
Cutoff	reverse	reverse	$V_{BE} < 0$, $V_{BC} < 0$
Inverse active	reverse	forward	$V_{BE} < 0$, $V_{BC} > 0$

Table 8: Summary

For an NPN, a junction is forward biased when its p side (the base) is positive: $V_{BE} = V_B - V_E > 0$ for B-E and $V_{BC} = V_B - V_C > 0$ for B-C.

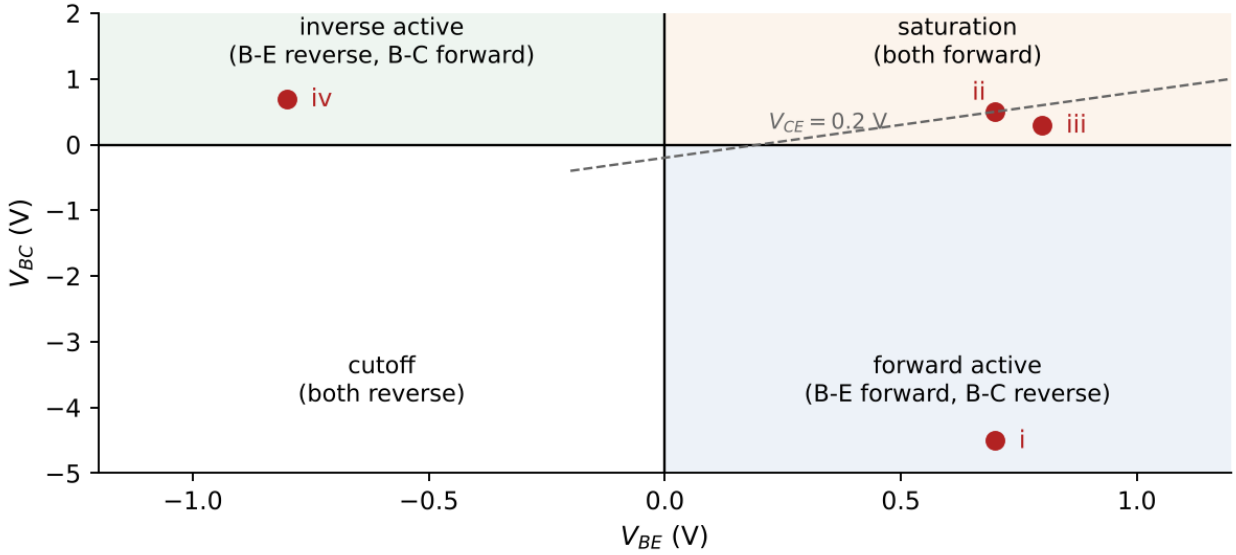


Figure 28: The four operating modes of an NPN transistor on the VBE-VBC plane, with the cases of the question

Reading the figure. The horizontal axis is V_{BE}, the vertical axis V_{BC}; each quadrant is one mode (forward active bottom right, saturation top right, inverse active top left, cutoff bottom left). The red dots are cases i to iv of the question at their (V_{BE}, V_{BC}); the dashed line $V_{BC} = V_{BE} - 0.2$ is $V_{CE} = 0.2$ V.

2. Voltage relations

The three terminal voltages are linked by $V_{CE} = V_{CB} + V_{BE} = V_{BE} - V_{BC}$. So $V_{BC} = V_{BE} - V_{CE}$: given V_{BE} and V_{CE}, the B-C bias follows.

3. Current relations

In **forward active** mode the collector current is set by the base current: $I_C = \beta I_B$, and $I_E = I_C + I_B$. In **saturation** the B-C junction also injects, which cuts the collector current: $I_C < \beta I_B$ (the transistor cannot deliver βI_B because the external circuit limits I_C). So comparing I_C/I_B with β tells the mode from currents alone.

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2. Answer

Given: NPN, $\beta \approx 450$. Rules: $V_{BC} = V_{BE} - V_{CE}$; forward active if B-E forward and B-C reverse; saturation if both forward; inverse active if B-E reverse and B-C forward; forward active if $I_C = \beta I_B$, saturation if $I_C < \beta I_B$ (with $I_B = I_E - I_C$).

Case i: $V_{BE} = 0.7$ V, $V_{CE} = 5.2$ V

$V_{BC} = 0.7 - 5.2 = -4.5$ V: B-E forward, B-C reverse.

Forward active

Case ii: $V_{BE} = 0.7$ V, $V_{CE} = 0.2$ V

$V_{BC} = 0.7 - 0.2 = +0.5$ V: both junctions forward biased.

Saturation

Case iii: $V_{BE} = 0.8$ V, $V_{BC} = 0.3$ V

Both V_{BE} and V_{BC} positive: both junctions forward biased ($V_{CE} = 0.8 - 0.3 = 0.5$ V).

Saturation (at the edge: soft saturation)

The B-C forward bias is small: its injection is $e^{-(0.8-0.3)/0.0259} = 4 \times 10^{-9}$ of the B-E injection, so the transistor still behaves almost as in the active mode.

Case iv: $V_{BE} = -0.8$ V, $V_{BC} = 0.7$ V

B-E reverse, B-C forward ($V_{CE} = -0.8 - 0.7 = -1.5$ V): the collector acts as the emitter.

Inverse (reverse) active

Case v: $I_C = 455$ mA, $I_B = 1$ mA

$I_C/I_B = 455 \approx \beta = 450$ (within about 1 %): the collector current is βI_B, as in the active mode.

Forward active

Case vi: $I_C = 455$ mA, $I_E = 502$ mA

$I_B = I_E - I_C = 502 - 455 = 47$ mA, so $I_C/I_B = 455/47 = 9.7 \ll \beta = 450$. In active mode the base current would give $\beta I_B = 21$ A; the collector carries far less, so the transistor is held in

Saturation

(Equivalently $I_C/I_E = 0.906$, far below $\alpha = \beta/(\beta+1) = 0.998$.)

Final Answer:

Case	Data	Mode
i	$V_{BC} = -4.5$ V	forward active
ii	$V_{BC} = +0.5$ V	saturation
iii	$V_{BE}, V_{BC} > 0$	saturation (soft)
iv	$V_{BE} < 0, V_{BC} > 0$	inverse active
v	$I_C/I_B = 455 \approx \beta$	forward active
vi	$I_C/I_B = 9.7 \ll \beta$	saturation

Table 9: Summary

3. Keyword Explanation

1. Forward active mode

B-E forward, B-C reverse: the transistor amplifies, $I_C = \beta I_B$.

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2. Saturation

Both junctions forward biased: V_{CE} is small (about 0.1 to 0.2 V) and $I_C < \beta I_B$; the transistor is a closed switch.

3. Inverse active mode

B-E reverse, B-C forward: emitter and collector swap roles, with a very small gain because the transistor is not built for it.

4. Notes and Remarks

- **Case iii by another rule:** some books call a transistor saturated only when the B-C forward bias is large enough to conduct (about 0.5 V and above). By that rule case iii, with only 0.3 V, would still count as forward active; by the junction-bias definition (Neamen), both forward means saturation. The answer gives the definition and notes that it is soft saturation.
- **Cutoff** (both reverse) does not occur among these cases.
- The minority carrier profiles of the four modes are drawn in the 2020-21 paper's 4(c) (extra reading).
- **Source:** like 6(a), this item is not a Neamen problem (it is in the style of Dimitrijev); it is answered with Neamen's definitions of the modes.

6.(c) Mention some advantages of using MOSFET. Marks: 3

1. Theory

1. How the MOSFET differs from the BJT (Core Idea)

A **MOSFET** controls the current in a channel by the **electric field** of a gate that is insulated from the channel by an oxide. A **BJT** controls its collector current by the **base current**, injecting minority carriers through forward-biased junctions. This one difference, an insulated, field-controlled input against a current-driven junction input, gives the MOSFET most of its advantages.

2. Majority carriers in the channel

The MOSFET current is carried by one type of carrier (electrons in an n-channel device) drifting along the channel. No minority carriers are stored in a base, so there is no storage time when the device switches off.

2. Answer

Advantages of the MOSFET (mainly compared with the BJT; Neamen, Chapters 10 and 11):

1. **Very high input impedance:** the gate is insulated by the oxide, so the DC gate current is practically zero ($\sim 10^{-15}$ A); the device is **voltage controlled** and draws almost no drive power.
2. **Low power, CMOS logic:** complementary n- and p-channel MOSFETs draw almost no current when the inputs are steady, so CMOS logic dissipates power mainly while switching.
3. **Small size and high density:** the device is simple (no isolation diffusions like a BJT's) and scales down well, which makes billions of transistors on one chip possible.
4. **Simple fabrication and lower cost** per transistor.
5. **Fast switching with no storage delay:** a majority carrier device has no minority charge stored to remove.
6. **Thermal stability:** the channel mobility falls as temperature rises, so the current falls; there is no thermal runaway, and MOSFETs can be paralleled easily.
7. **Good switch:** no offset voltage in the on state (a resistive channel), and the channel conducts in both directions.

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3. Keyword Explanation

1. Input impedance

The ratio of input voltage to input current; extremely high for an insulated gate.

2. CMOS

Complementary MOS: an n-channel and a p-channel MOSFET in series, so one is always off in a steady state.

3. Thermal runaway

In a BJT, heating raises the current, which raises the heating further; it does not happen in a MOSFET.

4. Notes and Remarks

- The disadvantages, for balance: the gate oxide is easily damaged by static charge; the transconductance per unit current is lower than a BJT's; and at very small sizes short-channel effects and oxide leakage appear (6(a)).
- For 3 marks, about four points with one line each are enough.

6.(d) Does the doping level in the base of a BJT affect the emitter efficiency γ_E? If so, why? Marks: 4

1. Theory

1. Two currents cross the base-emitter junction (Core Idea)

In an NPN transistor in forward active mode, the forward-biased B-E junction lets carriers cross in *both* directions:

- **electrons** from the n-type emitter into the p-type base, J_{nE}. These are the useful carriers: they diffuse across the base and are collected;
- **holes** from the p-type base into the emitter, J_{pE}. They recombine in the emitter and never reach the collector: wasted base current.

The **emitter injection efficiency factor** is the fraction of the emitter current carried by the useful electrons (Neamen, Eq. 12.31a):

$$\gamma = \frac{J_{nE}}{J_{nE} + J_{pE}} = \frac{1}{1 + J_{pE}/J_{nE}}$$

We want γ as close to 1 as possible, since the common-base gain is $\alpha = \gamma \alpha_T \delta$ (Neamen, Eq. 12.30b).

2. How many carriers each side injects

By the law of the junction, each side injects minority carriers in proportion to its own **equilibrium minority concentration**: the base receives electrons in proportion to $n_{B0} = n_i^2/N_B$, the emitter receives holes in proportion to $p_{E0} = n_i^2/N_E$. So the ratio of hole to electron injection is set by $p_{E0}/n_{B0} = N_B/N_E$. A heavily doped base means many holes and few minority electrons, which is bad for γ.

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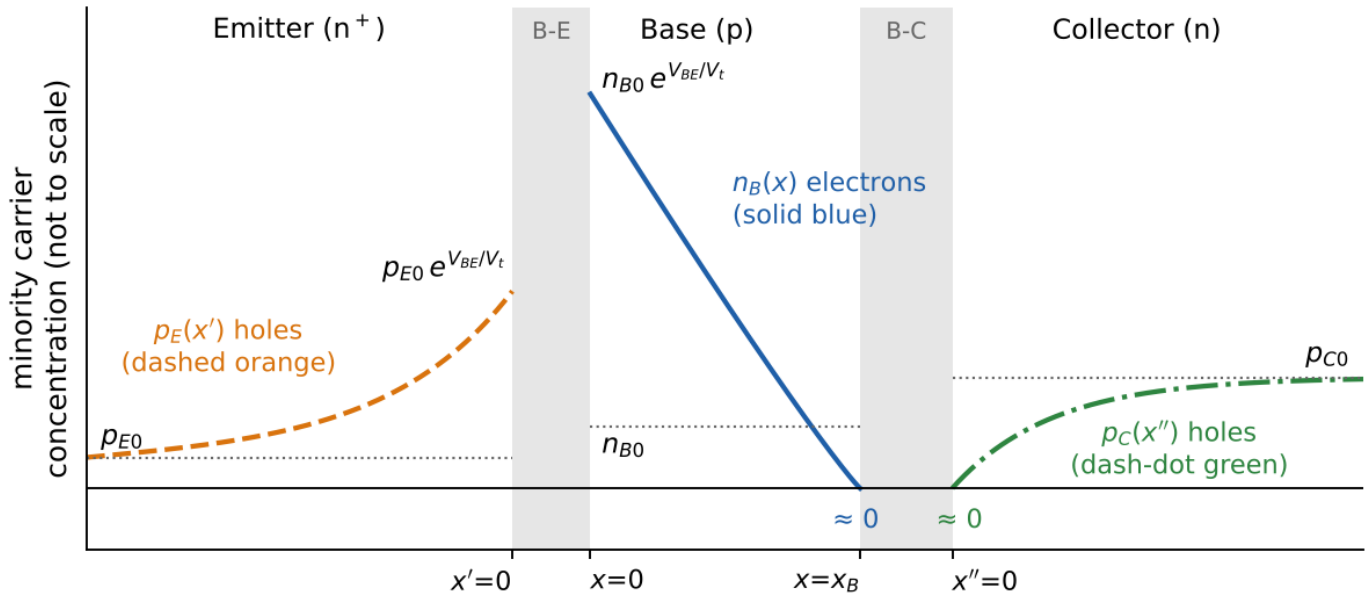


Figure 29: Minority carrier distributions in an NPN transistor in forward active mode (reused from the 2020-21 paper)

Reading the figure. In the base (centre) the solid blue line is the excess electron concentration, starting at $n_{B0}e^{V_{BE}/V_t}$ at the emitter edge and falling almost linearly to zero at the collector edge; its slope gives J_{nE}. In the emitter (left) the dashed orange curve is the injected hole concentration, starting at $p_{E0}e^{V_{BE}/V_t}$; its slope gives J_{pE}. Raising the base doping lowers n_{B0} (smaller blue line) and raises the hole injection relative to it.

2. Answer

Yes. The base doping directly sets the emitter injection efficiency: **the higher the base doping, the lower** γ. The reason, from the minority carrier currents (Neamen, Section 12.2.3):

Step 1: Definition

$$\gamma = \frac{J_{nE}}{J_{nE} + J_{pE}} = \frac{1}{1 + \dfrac{J_{pE}}{J_{nE}}}$$

Step 2: The two diffusion currents at the B-E junction

The electron current injected into the base and the hole current injected into the emitter are diffusion currents of minority carriers (Neamen, Eqs. 12.33, 12.34):

$$J_{nE} \approx \frac{eD_B n_{B0}}{L_B} \cdot \frac{1}{\tanh(x_B/L_B)} \left[e^{eV_{BE}/kT} - 1\right], \qquad J_{pE} = \frac{eD_E p_{E0}}{L_E} \cdot \frac{1}{\tanh(x_E/L_E)} \left[e^{eV_{BE}/kT} - 1\right]$$

(in J_{nE} the small term $1/\sinh(x_B/L_B)$ from the reverse-biased collector edge is dropped, since $e^{eV_{BE}/kT} \gg 1$ in forward active mode), where x_B, x_E are the neutral base and emitter widths and D, L the minority diffusion coefficients and lengths in each region.

Step 3: The ratio

The bias factor cancels (Neamen, Eq. 12.35a):

$$\gamma = \frac{1}{1 + \dfrac{p_{E0} D_E L_B}{n_{B0} D_B L_E} \cdot \dfrac{\tanh(x_B/L_B)}{\tanh(x_E/L_E)}}$$

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Step 4: Put in the doping

$p_{E0} = n_i^2/N_E$ and $n_{B0} = n_i^2/N_B$, so $p_{E0}/n_{B0} = N_B/N_E$. For short base and emitter ($x \ll L$, $\tanh u \approx u$) (Neamen, Eq. 12.35b):

$$\boxed{\gamma \approx \frac{1}{1 + \dfrac{N_B}{N_E} \cdot \dfrac{D_E}{D_B} \cdot \dfrac{x_B}{x_E}}}$$

Why: a higher base doping N_B lowers the equilibrium electron concentration in the base ($n_{B0} = n_i^2/N_B$), so fewer electrons are injected from the emitter, while the base, with more holes, injects relatively more holes back into the emitter. The ratio $J_{pE}/J_{nE} \propto N_B/N_E$ rises and γ falls.

Example

($D_E = D_B$, $x_E = x_B$, $N_E = 10^{19}$ cm^{-3}): $N_B = 10^{16}$ gives $\gamma = 0.9990$; $N_B = 10^{17}$ gives 0.9901; $N_B = 10^{18}$ gives 0.909.

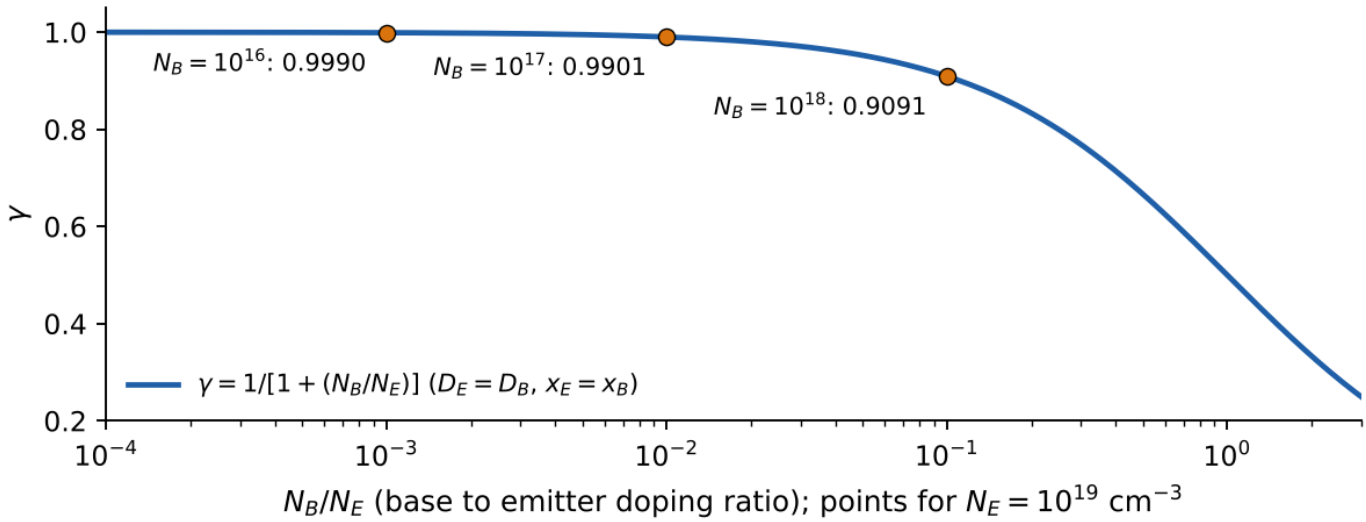


Figure 30: Emitter injection efficiency against the base-to-emitter doping ratio

Reading the figure. γ (solid blue) against N_B/N_E on a log axis, for $D_E = D_B$ and $x_E = x_B$. It stays near 1 while $N_B \ll N_E$ and falls steeply once N_B/N_E passes about 0.01. The dots are the three example points.

Conclusion: to keep $\gamma \approx 1$, the emitter is doped much more heavily than the base ($N_E \gg N_B$, the $n^{++}pn$ structure), and the base is kept thin.

3. Keyword Explanation

1. Emitter injection efficiency

$\gamma = J_{nE}/(J_{nE} + J_{pE})$: the fraction of emitter current made of carriers injected into the base.

2. Base transport factor

$\alpha_T = J_{nC}/J_{nE}$: the fraction of injected electrons that cross the base without recombining.

3. Common-base current gain

$\alpha = \gamma\alpha_T\delta$, and $\beta = \alpha/(1-\alpha)$.

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4. Notes and Remarks

- **Limits:** very heavy emitter doping causes **band-gap narrowing**, which raises p_{E0} and lowers γ again (Neamen, Section 12.4.2); this is why heterojunction bipolar transistors use a wide-gap emitter.
- A lower base doping also has costs: a higher base resistance and an earlier punch-through, so the base doping is a compromise.
- The equation sheet's $\delta n_B(x)$ and $\delta p_E(x')$ are the linear (short base and emitter) forms of the distributions in the figure above.

5. Final Concise Answer

(a) (Reading $N_A = 5 \times 10^{18}$ cm^{-3}.) $\phi_{fp} = 0.508$ V, $x_{dT} = 16.2$ nm, $|Q'_{SD}(max)| = 1.30 \times 10^{-6}$ C/cm^2; (i) $|Q_I| = \epsilon_{ox}E_{ox} - |Q'_{SD}| = 2.07 \times 10^{-6} - 1.30 \times 10^{-6} = 7.73 \times 10^{-7}$ C/cm^2; (ii) $|Q_I|LW/e \approx 1208$ electrons; (iii) $V_T = V_{FB}+2\phi_{fp}+|Q'_{SD}|/C_{ox} = 1+1.017+0.376 = 2.39$ V; (iv) $V_G = V_T+|Q_I|/C_{ox} = 2.62$ V.

(b) (i) $V_{BC} = -4.5$ V: forward active; (ii) $V_{BC} = +0.5$ V: saturation; (iii) both junctions forward: saturation (soft); (iv) B-E reverse, B-C forward: inverse active; (v) $I_C/I_B = 455 \approx \beta$: forward active; (vi) $I_B = 47$ mA, $I_C/I_B = 9.7 \ll \beta$: saturation.

(c) Insulated gate: very high input impedance, voltage control, almost no drive power; very low static power in CMOS; small, simple and dense, scaling to billions per chip; fast majority-carrier switching with no storage delay; no thermal runaway, easy paralleling; a good bidirectional switch without offset voltage.

(d) Yes:

$$\gamma = \frac{1}{1 + \dfrac{p_{E0}D_E L_B}{n_{B0}D_B L_E}\dfrac{\tanh(x_B/L_B)}{\tanh(x_E/L_E)}} \approx \frac{1}{1 + \dfrac{N_B}{N_E}\dfrac{D_E}{D_B}\dfrac{x_B}{x_E}}$$

Higher base doping lowers $n_{B0} = n_i^2/N_B$, so fewer electrons are injected relative to the holes injected back into the emitter, and γ falls; hence $N_E \gg N_B$.

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7. (a) Starting from charge neutrality condition, derive expressions for n_0 and p_0 in terms of N_a, N_d, and n_i. Marks: 5

(b) With proper equation and graph, explain variation of Fermi energy level in extrinsic semiconductor. Marks: 5

(c) A special semiconductor material is to be "designed". The semiconductor is to be n-type and doped with $1 \times 10^{15} cm^{-3}$ donor atoms. Assume complete ionization and assume $N_a = 0$. The effective density of states functions are given by $N_c = N_v = 1.5 \times 10^{19} cm^{-3}$ and are independent of temperature. A particular semiconductor device fabricated with this material requires the electron concentration to be no greater than $1.01 \times 10^{15} cm^{-3}$ at $T = 400\ K$. What is the minimum value of the bandgap energy? Marks: 5

(d) (i) Calculate the probability of an electron to be occupied at the lowest energy state of the conduction band of silicon when the Fermi energy is 0.3 eV below the conduction band at room temperature. (ii) Calculate the thermal electron concentration n_0. (iii) What is the thermal hole concentration p_0? (Room temperature N_c is $2.8 \times 10^{19}\ cm^{-3}$, n_i is $1.5 \times 10^{10}\ cm^{-3}$) Marks: 5

7.(a) Starting from charge neutrality condition, derive expressions for n_0 and p_0 in terms of N_a, N_d, and n_i. Marks: 5

1. Theory

1. Free carriers and fixed ions (Core Idea)

A piece of doped silicon holds two kinds of charge:

- **Mobile carriers:** free electrons in the conduction band (density n_0, charge $-e$ each) and holes in the valence band (density p_0, charge $+e$ each). The subscript 0 means thermal equilibrium.
- **Fixed ions:** a **donor** atom (P, As) that has given its extra electron to the conduction band is a fixed positive ion; there are N_d^+ of them per cm^3. An **acceptor** atom (B) that has taken an electron from the valence band (making a hole) is a fixed negative ion; there are N_a^- of them.

When both kinds of dopant are present, the material is **compensated** (Neamen, Section 4.5): some donor electrons simply drop into the empty acceptor states, so only the difference $|N_d - N_a|$ supplies free carriers.

2. Why the crystal is neutral

The crystal was built from neutral atoms, and in thermal equilibrium nothing has been added or taken away. So, in every uniform region, the negative charges exactly balance the positive charges. Counting them per cm^3 and cancelling the common factor e gives the **charge neutrality condition** (Neamen, Eq. 4.56):

$$n_0 + N_a^- = p_0 + N_d^+$$

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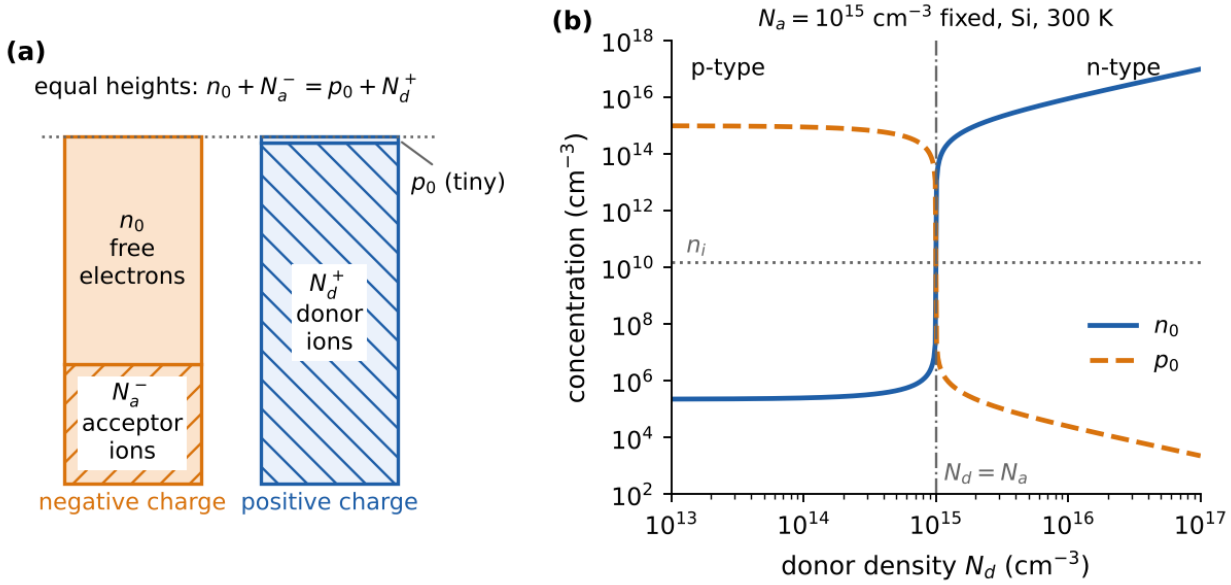


Figure 31: Charge balance in a compensated semiconductor, and n0 and p0 as the donor density is varied (reused from the 2020-21 paper)

Reading the figure. Panel (a) draws the two sides of the neutrality condition as stacks of equal height: acceptor ions plus free electrons (orange outline) on the negative side, donor ions plus the few holes (blue outline) on the positive side. Panel (b) is drawn from the result of this question for Si with $N_a = 10^{15}$ cm^{-3} fixed: the solid blue curve is n_0, the dashed orange curve p_0; left of $N_d = N_a$ (dash-dot line) the material is p-type, right of it n-type, and at $N_d = N_a$ both equal n_i (dotted grey).

3. Two facts the derivation uses

- **Complete ionization:** at room temperature in Si, essentially every donor and acceptor is ionized (the dopant levels are only about 0.045 eV from the band edges, less than $2kT$), so $N_d^+ = N_d$ and $N_a^- = N_a$ (Neamen, Section 4.4).
- **Mass-action law:** for a non-degenerate semiconductor in equilibrium, $n_0 = N_c e^{-(E_c - E_F)/kT}$ and $p_0 = N_v e^{-(E_F - E_v)/kT}$; in their product E_F cancels, so (Neamen, Eq. 4.43)

$$n_0 p_0 = N_c N_v \exp\left(-\frac{E_g}{kT}\right) = n_i^2$$

whatever the doping. This lets us write one carrier in terms of the other.

2. Answer

Assumptions: thermal equilibrium, uniform doping, complete ionization, non-degenerate material ($n_0 p_0 = n_i^2$).

Step 4 is the general quadratic formula and is marked (Basic); a student who knows it can skip it.

Step 1: Charge neutrality

Negative charge (electrons and ionized acceptors) equals positive charge (holes and ionized donors) (Neamen, Eq. 4.56):

$$n_0 + N_a^- = p_0 + N_d^+$$

With complete ionization, $N_a^- = N_a$ and $N_d^+ = N_d$ (Neamen, Eq. 4.58):

$$n_0 + N_a = p_0 + N_d$$

Step 2: Eliminate p_0 with the mass-action law

From $n_0 p_0 = n_i^2$, $p_0 = n_i^2 / n_0$ (Neamen, Eq. 4.59a):

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$$n_0 + N_a = \frac{n_i^2}{n_0} + N_d$$

Step 3: Form a quadratic in n_0

Multiply every term by n_0:

$$n_0^2 + N_a n_0 = n_i^2 + N_d n_0$$

Move everything to the left and collect the n_0 terms (Neamen, Eq. 4.59b):

$$n_0^2 - (N_d - N_a)\, n_0 - n_i^2 = 0$$

Step 4 (Basic): Solve with the quadratic formula

For $ax^2 + bx + c = 0$, completing the square gives $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. Here $a = 1$, $b = -(N_d - N_a)$ and $c = -n_i^2$, so $b^2 - 4ac = (N_d - N_a)^2 + 4n_i^2$:

$$n_0 = \frac{(N_d - N_a) \pm \sqrt{(N_d - N_a)^2 + 4n_i^2}}{2}$$

Divide each term by 2, taking the 2 inside the root as $\sqrt{X}/2 = \sqrt{X/4}$:

$$n_0 = \frac{N_d - N_a}{2} \pm \sqrt{\left(\frac{N_d - N_a}{2}\right)^2 + n_i^2}$$

Step 5: Keep the physical root

The square root is larger than $|N_d - N_a|/2$, so the minus sign would make n_0 negative. A concentration cannot be negative, so the plus sign is kept (Neamen, Eq. 4.60):

$$\boxed{n_0 = \frac{N_d - N_a}{2} + \sqrt{\left(\frac{N_d - N_a}{2}\right)^2 + n_i^2}}$$

Step 6: The hole concentration

Go back to Step 1 and eliminate n_0 instead, with $n_0 = n_i^2/p_0$ (Neamen, Eq. 4.61):

$$\frac{n_i^2}{p_0} + N_a = p_0 + N_d$$

Multiply by p_0: $n_i^2 + N_a p_0 = p_0^2 + N_d p_0$, so

$$p_0^2 - (N_a - N_d)\, p_0 - n_i^2 = 0$$

The quadratic formula ($a = 1$, $b = -(N_a - N_d)$, $c = -n_i^2$) and the positive root give (Neamen, Eq. 4.62)

$$\boxed{p_0 = \frac{N_a - N_d}{2} + \sqrt{\left(\frac{N_a - N_d}{2}\right)^2 + n_i^2}}$$

Steps to remember

Neutrality $n_0 + N_a^- = p_0 + N_d^+$ → complete ionization → $p_0 = n_i^2/n_0$ → multiply by n_0: quadratic → quadratic formula → keep the + root → repeat with $n_0 = n_i^2/p_0$ for p_0.

Special cases:

- n-type, $N_d - N_a \gg n_i$: $n_0 \approx N_d - N_a$, $p_0 = n_i^2/(N_d - N_a)$.

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- p-type, $N_a - N_d \gg n_i$: $p_0 \approx N_a - N_d$, $n_0 = n_i^2/(N_a - N_d)$.
- $N_d = N_a$ (fully compensated) or $N_d = N_a = 0$ (intrinsic): $n_0 = p_0 = n_i$.

3. Keyword Explanation

1. Charge neutrality

In a uniform semiconductor in equilibrium the total positive charge (holes and ionized donors) equals the total negative charge (electrons and ionized acceptors).

2. Complete ionization

Every donor has given up its electron and every acceptor has taken one, so $N_d^+ = N_d$, $N_a^- = N_a$.

3. Mass-action law

$n_0 p_0 = n_i^2$ in thermal equilibrium (non-degenerate), independent of the doping.

4. Notes and Remarks

- The same derivation is the answer to 2(a) of this paper's final; it is written in full again here, as each question stands on its own. It is also 3(a) of the 2020-21 paper (3(a), extra reading).
- **Example (Si, 300 K)**

  $N_d = 10^{16}$, $N_a = 3 \times 10^{15}$ cm^{-3} gives $n_0 = 7.0 \times 10^{15}$ and $p_0 = 3.2 \times 10^4$ cm^{-3}.
- The equation sheet of the final prints $\frac{(N_d - N_a)^2}{2}$ inside the root; the correct term is $\left(\frac{N_d - N_a}{2}\right)^2$, as Step 4 shows.

7.(b) With proper equation and graph, explain variation of Fermi energy level in extrinsic semiconductor. Marks: 5

1. Theory

1. What the Fermi level tells us (Core Idea)

The **Fermi level** E_F is the energy at which the Fermi-Dirac function equals $\frac{1}{2}$. Its position in the band gap sets how many electrons and holes there are (Neamen, Eqs. 4.11, 4.19):

$$n_0 = N_c \exp\left[-\frac{E_c - E_F}{kT}\right], \qquad p_0 = N_v \exp\left[-\frac{E_F - E_v}{kT}\right]$$

The closer E_F is to E_c, the more electrons; the closer to E_v, the more holes. In an **intrinsic** semiconductor $n_0 = p_0 = n_i$ and E_F is at the **intrinsic Fermi level** E_{Fi}, very near the middle of the gap. Using E_{Fi} as the reference, the same formulas read (Neamen, Eqs. 4.39, 4.40)

$$n_0 = n_i \exp\left[\frac{E_F - E_{Fi}}{kT}\right], \qquad p_0 = n_i \exp\left[-\frac{E_F - E_{Fi}}{kT}\right]$$

2. Extrinsic semiconductor

An **extrinsic** semiconductor is one whose carriers come mainly from dopants: n-type (donors, $n_0 \gg p_0$) or p-type (acceptors, $p_0 \gg n_0$). Doping does not change the bands; it moves E_F. Two things move it: the **doping concentration** and the **temperature** (through n_i).

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2. Answer

1. The equations

Taking the logarithm of the carrier equations above (Neamen, Eqs. 4.63 to 4.68):

- **n-type:**

$$E_c - E_F = kT \ln\left(\frac{N_c}{n_0}\right), \qquad E_F - E_{Fi} = kT \ln\left(\frac{n_0}{n_i}\right)$$

- **p-type:**

$$E_F - E_v = kT \ln\left(\frac{N_v}{p_0}\right), \qquad E_{Fi} - E_F = kT \ln\left(\frac{p_0}{n_i}\right)$$

where $n_0 = \frac{N_d - N_a}{2} + \sqrt{\left(\frac{N_d - N_a}{2}\right)^2 + n_i^2}$ (and p_0 in the same form), which reduce to $n_0 \approx N_d$ and $p_0 \approx N_a$ when the doping is much larger than n_i.

2. Variation with doping concentration

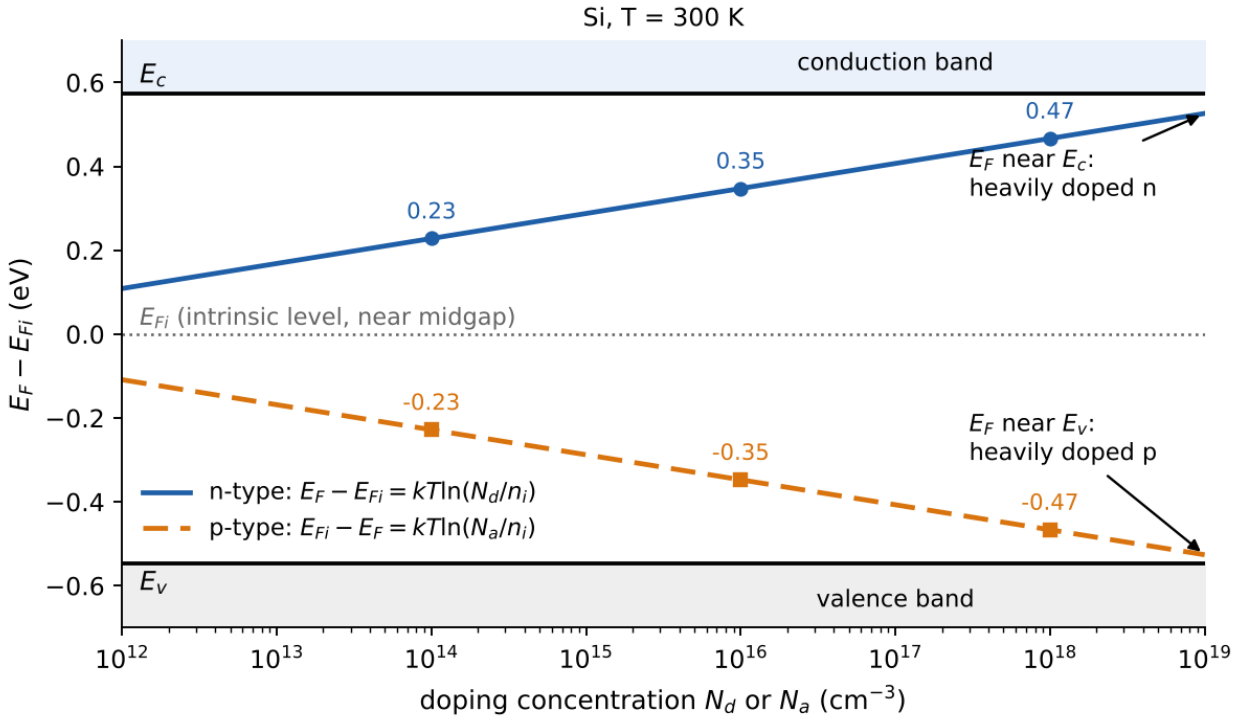


Figure 32: Position of the Fermi level in silicon at 300 K against donor and acceptor concentration (reused from the 2020-21 paper)

Reading the figure. The solid blue line is $E_F - E_{Fi}$ for n-type Si, the dashed orange line for p-type, against the doping on a log axis at 300 K. The marked points are 0.23, 0.35 and 0.47 eV at 10^{14}, 10^{16} and 10^{18} cm^{-3}. Each decade of doping moves E_F by $kT \ln 10 = 0.060$ eV: towards E_c for donors, towards E_v for acceptors.

- n-type: as N_d rises, $E_c - E_F = kT \ln(N_c/N_d)$ shrinks, so E_F moves up towards E_c.
- p-type: as N_a rises, $E_F - E_v = kT \ln(N_v/N_a)$ shrinks, so E_F moves down towards E_v.
- For a compensated material, the net doping $|N_d - N_a|$ replaces N_d or N_a.
- At very high doping ($\gtrsim 10^{18}$ to 10^{19} cm^{-3}) E_F reaches the band edge and the material becomes degenerate; the Boltzmann form above is then no longer accurate.

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3. Variation with temperature

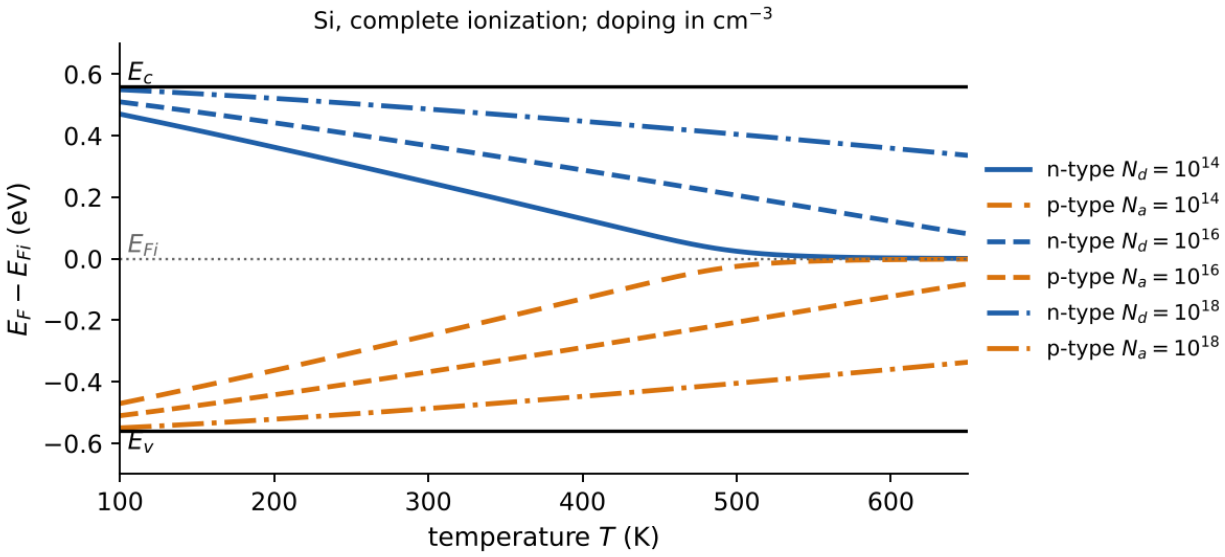


Figure 33: Fermi level against temperature for n- and p-type silicon at three dopings

Reading the figure. Blue curves are n-type, orange curves p-type, for 10^{14}, 10^{16} and 10^{18} cm^{-3} (blue: solid, dashed, dash-dot; orange: long dashes, short dashes, dash-dot), computed with $E_F - E_{Fi} = kT\ln(n_0/n_i)$ and the full n_0 formula. Every curve starts near its band edge at low temperature and bends towards E_{Fi} (dotted grey) as T rises; the lighter the doping, the sooner it reaches E_{Fi} (about 500 K for 10^{14} cm^{-3}).

- $n_i \propto T^{3/2}e^{-E_g/2kT}$ rises steeply with temperature (Neamen, Eq. 4.23). Since $E_F - E_{Fi} = kT\ln(n_0/n_i)$, a larger n_i pulls E_F towards E_{Fi}.
- At high temperature n_i becomes comparable with the doping, the material loses its extrinsic character and $E_F \to E_{Fi}$: it behaves as intrinsic.
- At very low temperature (freeze-out) the dopants are no longer ionized; E_F rises above E_d in n-type (falls below E_a in p-type), and at $T = 0$ K it lies between E_d and E_c (between E_v and E_a) (Neamen, Section 4.6.3).

Conclusion: E_F moves towards the majority-carrier band edge as the doping increases, and back towards the middle of the gap as the temperature increases.

3. Keyword Explanation

1. Extrinsic semiconductor

A semiconductor whose carrier concentrations are set by added impurity atoms rather than by thermal generation across the gap.

2. Intrinsic Fermi level

E_{Fi}: the Fermi level of the undoped material, near midgap (0.0128 eV below midgap in Si at 300 K).

3. Degenerate semiconductor

So heavily doped that E_F lies inside a band; the Boltzmann approximation fails.

4. Notes and Remarks

- The figure against temperature corresponds to Neamen's Figure 4.19; the figure against doping to his Figure 4.18.
- Numbers at 300 K for Si: $E_F - E_{Fi} = 0.228$, 0.288, 0.347, 0.407 and 0.467 eV for $N_d = 10^{14}$ to 10^{18} cm^{-3} (from `calc_2019-20.py`).
- The same picture holds for p-type with E_F below E_{Fi} and all signs reversed.

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7.(c) A special semiconductor material is to be "designed". The semiconductor is to be n-type and doped with $1 \times 10^{15} cm^{-3}$ donor atoms. Assume complete ionization and assume $N_a = 0$. The effective density of states functions are given by $N_c = N_v = 1.5 \times 10^{19} cm^{-3}$ and are independent of temperature. A particular semiconductor device fabricated with this material requires the electron concentration to be no greater than $1.01 \times 10^{15} cm^{-3}$ at $T = 400\ K$. What is the minimum value of the bandgap energy? Marks: 5

1. Theory

1. Why a device needs a minimum band gap (Core Idea)

An n-type device is designed to have $n_0 \approx N_d$: its electron concentration should be fixed by the doping, not by temperature. But electrons are also produced by thermal generation across the gap, and their number is the intrinsic concentration (Neamen, Eq. 4.23):

$$n_i^2 = N_c N_v \exp\left(-\frac{E_g}{kT}\right)$$

The smaller the gap or the hotter the device, the larger n_i. If n_i grows to be a noticeable fraction of N_d, the electron concentration rises above N_d and the device stops working as designed. So, for a given highest temperature, there is a **smallest band gap** that keeps n_i small enough.

2. The link between n_0, N_d and n_i

With complete ionization and $N_a = 0$, charge neutrality gives (Neamen, Eq. 4.60)

$$n_0 = \frac{N_d}{2} + \sqrt{\left(\frac{N_d}{2}\right)^2 + n_i^2}$$

The same relation can be written without square roots: $n_0 = N_d + p_0$ (neutrality) and $p_0 = n_i^2/n_0$ give

$$n_i^2 = n_0(n_0 - N_d)$$

which tells us directly the largest n_i that keeps n_0 at or below a limit.

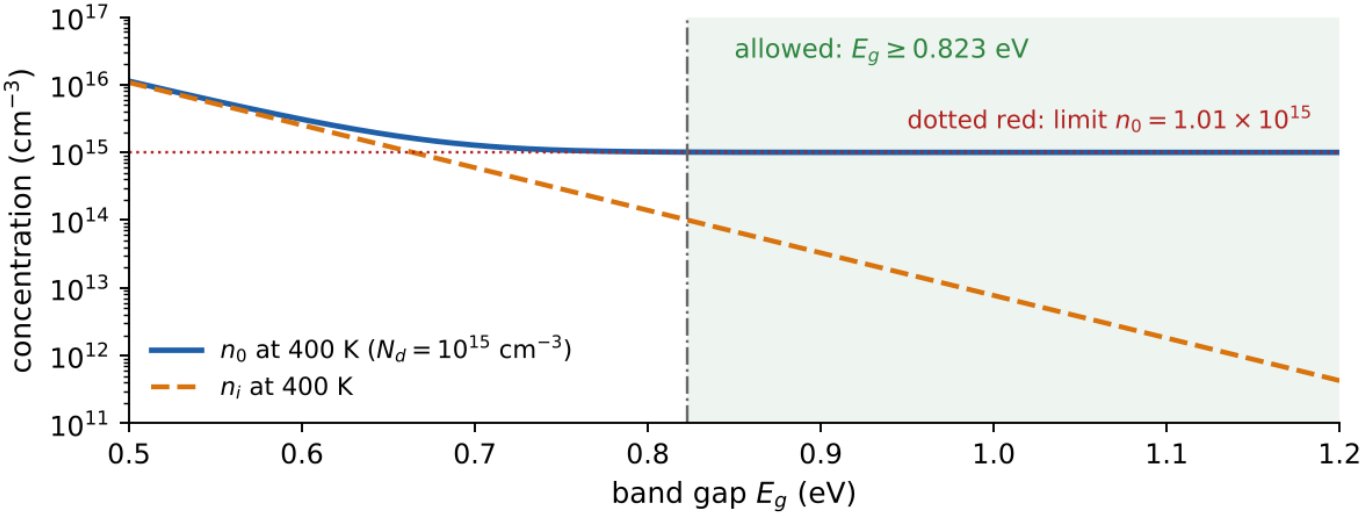


Figure 34: Electron concentration at 400 K against the band gap of the designed material

Reading the figure. For $N_d = 10^{15}$ cm^{-3} and $N_c = N_v = 1.5 \times 10^{19}$ cm^{-3} at 400 K, the dashed orange line is n_i and the solid blue curve n_0. For a small gap n_i is large and n_0 is well above N_d; as E_g grows, n_i falls exponentially and n_0 settles at N_d. The dotted red line is the limit 1.01×10^{15} cm^{-3}; it is met at $E_g = 0.823$ eV (dash-dot line), and every larger gap (green) is allowed.

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2. Answer

Given: n-type, $N_d = 10^{15}$ cm^{-3}, $N_a = 0$, complete ionization; $N_c = N_v = 1.5 \times 10^{19}$ cm^{-3} (independent of temperature); $n_0 \leq 1.01 \times 10^{15}$ cm^{-3} at $T = 400$ K.

$$kT = 0.0259 \times \frac{400}{300} = 0.034533 \text{ eV}$$

Step 1: Largest allowed n_i from charge neutrality

Neutrality with $N_a = 0$: $n_0 = N_d + p_0$, and $p_0 = n_i^2/n_0$. So $n_0 - N_d = n_i^2/n_0$, that is

$$n_i^2 = n_0(n_0 - N_d)$$

At the limit $n_0 = 1.01 \times 10^{15}$ cm^{-3}:

$$n_i^2 = (1.01 \times 10^{15})(1.01 \times 10^{15} - 10^{15}) = (1.01 \times 10^{15})(10^{13}) = 1.01 \times 10^{28} \text{ cm}^{-6}$$

$$n_i = 1.005 \times 10^{14} \text{ cm}^{-3}$$

(The hole concentration is then $p_0 = n_0 - N_d = 10^{13}$ cm^{-3}.)

Step 2: Band gap from $n_i^2 = N_c N_v e^{-E_g/kT}$

Solve for E_g:

$$E_g = kT \ln\left(\frac{N_c N_v}{n_i^2}\right)$$

$$\frac{N_c N_v}{n_i^2} = \frac{(1.5 \times 10^{19})^2}{1.01 \times 10^{28}} = \frac{2.25 \times 10^{38}}{1.01 \times 10^{28}} = 2.228 \times 10^{10}, \qquad \ln\left(2.228 \times 10^{10}\right) = 23.827$$

$$E_g = (0.034533)(23.827) = 0.823 \text{ eV}$$

Step 3: Minimum or maximum?

A larger E_g gives a smaller n_i and so a smaller n_0, which still meets the requirement. So 0.823 eV is the **minimum** band gap:

$$\boxed{E_{g,min} = 0.823 \text{ eV}}$$

Check: with $E_g = 0.823$ eV, $n_i = 1.5 \times 10^{19} e^{-0.823/(2\times 0.034533)} = 1.005 \times 10^{14}$ cm^{-3} and $n_0 = \frac{10^{15}}{2} + \sqrt{(5 \times 10^{14})^2 + (1.005 \times 10^{14})^2} = 1.0100 \times 10^{15}$ cm^{-3}. With $E_g = 0.80$ eV, $n_0 = 1.019 \times 10^{15}$ cm^{-3} (too high); with 0.90 eV, 1.001×10^{15} cm^{-3} (fine).

Final Answer: $E_{g,min} \approx 0.823$ eV (at 400 K, n_i must stay at or below 1.005×10^{14} cm^{-3}).

3. Keyword Explanation

1. Effective density of states

N_c (N_v): the number of states that, all placed at E_c (E_v), would hold the same number of electrons (holes); here taken as constant.

2. Intrinsic carrier concentration

$n_i = \sqrt{N_c N_v}\, e^{-E_g/2kT}$: the electron (and hole) concentration produced by thermal generation alone.

3. Complete ionization

All N_d donors have given their electrons to the conduction band.

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4. Notes and Remarks

- **Book check:** this is Neamen's Problem 4.61 in a changed form (the book's version is p-type with $N_a = 5\times10^{15}$ cm^{-3}, N_c, $N_v \propto T^2$ and 350 K). Here N_c, N_v are given as temperature independent, so no scaling is needed.
- **Value of kT:** with $kT = 0.0259 \times 400/300 = 0.034533$ eV (Neamen's room-temperature value scaled), $E_g = 0.8228$ eV; with $k = 8.617\times10^{-5}$ eV/K exactly, $kT = 0.034468$ eV and $E_g = 0.8213$ eV. Both round to 0.82 eV.
- The requirement means the thermally generated carriers may add at most 1 % to the doping-controlled electron concentration.

7.(d) (i) Calculate the probability of an electron to be occupied at the lowest energy state of the conduction band of silicon when the Fermi energy is 0.3 eV below the conduction band at room temperature. (ii) Calculate the thermal electron concentration n_0. (iii) What is the thermal hole concentration p_0? (Room temperature N_c is 2.8×10^{19} cm^{-3}, n_i is 1.5×10^{10} cm^{-3}) Marks: 5

1. Theory

1. The Fermi-Dirac function (Core Idea)

Electrons obey the Pauli principle: one electron per quantum state. In thermal equilibrium the probability that a state at energy E is occupied is the **Fermi-Dirac function** (Neamen, Eq. 3.79):

$$f_F(E) = \frac{1}{1+\exp\left(\dfrac{E-E_F}{kT}\right)}$$

At $E = E_F$ it is $\frac{1}{2}$. Several kT above E_F the 1 in the denominator is negligible and it becomes the **Boltzmann approximation** $f_F(E) \approx \exp[-(E-E_F)/kT]$.

The **lowest energy state of the conduction band** is the band edge $E = E_c$. "Room temperature" is $T = 300$ K, where $kT = 0.0259$ eV.

2. From probability to concentration

The number of electrons in the conduction band is the number of states times the probability that each is filled, summed over the band. Done with the density of states $g_c(E) \propto \sqrt{E-E_c}$ and the Boltzmann form, it gives (Neamen, Eq. 4.11)

$$n_0 = N_c \exp\left[-\frac{E_c - E_F}{kT}\right]$$

where N_c is the effective density of states (2.8×10^{19} cm^{-3} for Si at 300 K). The probability is tiny, but N_c is huge, so n_0 is a sensible number.

3. Holes from the mass-action law

In equilibrium $n_0 p_0 = n_i^2$ (Neamen, Eq. 4.43), so $p_0 = n_i^2/n_0$.

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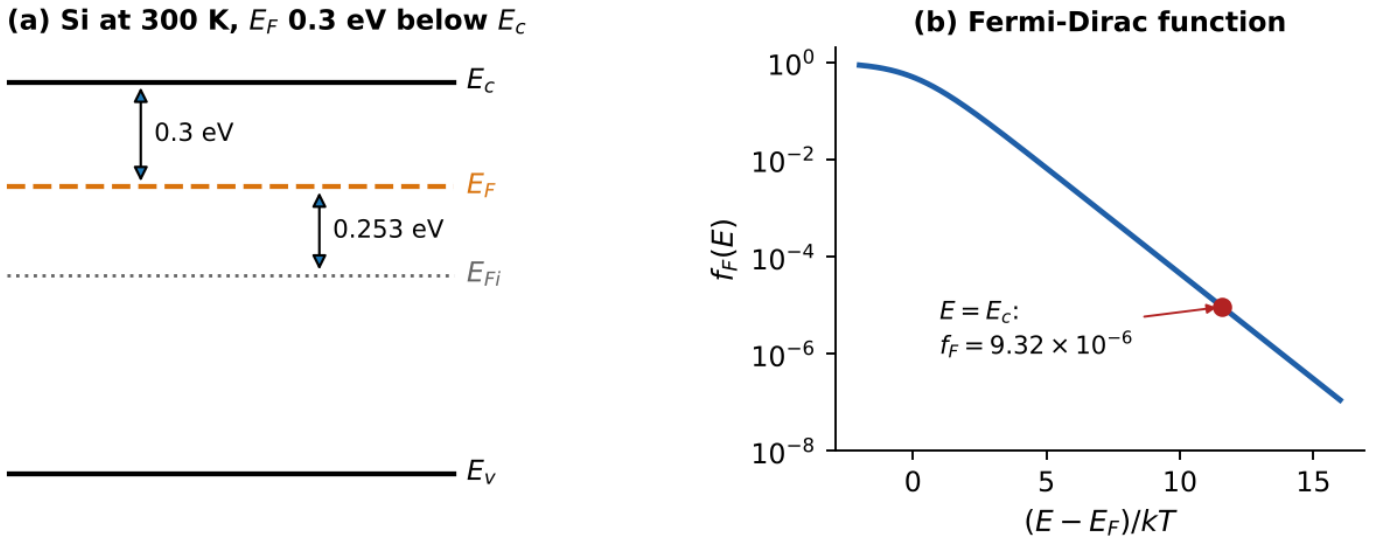


Figure 35: Band diagram of the silicon sample and the Fermi-Dirac function with the conduction band edge marked

Reading the figure. Panel (a): the Fermi level E_F (dashed orange) is 0.3 eV below E_c, and 0.253 eV above E_{Fi} (dotted grey), so the silicon is n-type. Panel (b): f_F against $(E - E_F)/kT$ on a log scale; the red dot is the conduction band edge, $(E_c - E_F)/kT = 11.58$, where $f_F = 9.32 \times 10^{-6}$.

2. Answer

Given: Si, $T = 300$ K, $kT = 0.0259$ eV; $E_c - E_F = 0.3$ eV; $N_c = 2.8 \times 10^{19}$ cm^{-3}; $n_i = 1.5 \times 10^{10}$ cm^{-3}.

Step 1 (i): Probability at $E = E_c$

$$\frac{E_c - E_F}{kT} = \frac{0.3}{0.0259} = 11.583, \qquad e^{11.583} = 1.0726 \times 10^5$$

$$f_F(E_c) = \frac{1}{1 + 1.0726 \times 10^5} = 9.32 \times 10^{-6}$$

$$\boxed{f_F(E_c) = 9.32 \times 10^{-6}}$$

The Boltzmann form gives the same, $e^{-11.583} = 9.32 \times 10^{-6}$, since $E_c - E_F > 3kT$.

Step 2 (ii): Thermal electron concentration

$$n_0 = N_c \exp\left[-\frac{E_c - E_F}{kT}\right] = (2.8 \times 10^{19})(9.323 \times 10^{-6}) = 2.61 \times 10^{14} \text{ cm}^{-3}$$

$$\boxed{n_0 = 2.61 \times 10^{14} \text{ cm}^{-3}}$$

Step 3 (iii): Thermal hole concentration

$$p_0 = \frac{n_i^2}{n_0} = \frac{(1.5 \times 10^{10})^2}{2.61 \times 10^{14}} = \frac{2.25 \times 10^{20}}{2.61 \times 10^{14}} = 8.62 \times 10^5 \text{ cm}^{-3}$$

$$\boxed{p_0 = 8.62 \times 10^5 \text{ cm}^{-3}}$$

Final Answer: (i) $f_F(E_c) = 9.32 \times 10^{-6}$; (ii) $n_0 = 2.61 \times 10^{14}$ cm^{-3}; (iii) $p_0 = 8.62 \times 10^5$ cm^{-3}. The sample is n-type with $E_F - E_{Fi} = kT \ln(n_0/n_i) = 0.253$ eV.

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3. Keyword Explanation

1. Fermi energy

The energy at which the probability of occupation is exactly $\frac{1}{2}$.

2. Thermal-equilibrium concentration

n_0, p_0: the carrier concentrations with no light, bias or injection.

3. Boltzmann approximation

$f_F \approx e^{-(E-E_F)/kT}$ when $E - E_F \gg kT$.

4. Notes and Remarks

- **Book check:** this follows Neamen's Example 4.1 (Fermi level 0.25 eV below E_c, f_F at $E_c + kT/2$, $n_0 = N_c e^{-(E_c-E_F)/kT}$); with 0.3 eV the method is the same.
- **Holes another way:** using $N_v = 1.04 \times 10^{19}$ cm^{-3} and $E_g = 1.12$ eV, $p_0 = N_v e^{-(E_g-0.3)/kT} = 1.85 \times 10^5$ cm^{-3}, not 8.62×10^5. The difference is because the given $n_i = 1.5 \times 10^{10}$ cm^{-3} is not the value that Neamen's N_c, N_v and E_g give (6.95×10^9 cm^{-3}). The question gives n_i, so $p_0 = n_i^2/n_0$ is the intended route.

5. Final Concise Answer

(a) Neutrality $n_0 + N_a = p_0 + N_d$ (complete ionization) with $p_0 = n_i^2/n_0$ gives $n_0^2 - (N_d - N_a)n_0 - n_i^2 = 0$, so $n_0 = \frac{N_d-N_a}{2} + \sqrt{\left(\frac{N_d-N_a}{2}\right)^2 + n_i^2}$; in the same way $p_0 = \frac{N_a-N_d}{2} + \sqrt{\left(\frac{N_a-N_d}{2}\right)^2 + n_i^2}$ (Neamen Eqs. 4.60, 4.62).
(b) $E_F - E_{Fi} = kT\ln(n_0/n_i)$ (n-type), $E_{Fi} - E_F = kT\ln(p_0/n_i)$ (p-type), or $E_c - E_F = kT\ln(N_c/N_d)$, $E_F - E_v = kT\ln(N_v/N_a)$: E_F moves 0.06 eV per decade of doping towards the majority band edge (0.23, 0.35, 0.47 eV above E_{Fi} for 10^{14}, 10^{16}, 10^{18} cm^{-3} n-type Si), and moves back towards E_{Fi} as temperature rises and n_i grows, reaching it when the material turns intrinsic.
(c) $n_i^2 = n_0(n_0 - N_d) = 1.01 \times 10^{28}$ cm^{-6}; $E_g = kT\ln(N_c N_v/n_i^2) = 0.034533\ln(2.228 \times 10^{10}) = 0.823$ eV minimum.
(d) $f_F(E_c) = 1/(1 + e^{0.3/0.0259}) = 9.32 \times 10^{-6}$; $n_0 = N_c f = 2.61 \times 10^{14}$ cm^{-3}; $p_0 = n_i^2/n_0 = 8.62 \times 10^5$ cm^{-3}.

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8. (a) Consider a p-type semiconductor that is homogeneous and infinite in extent. Assume a zero applied electric field. For a one-dimensional crystal, assume that excess carriers are being generated at $x = 0$ only. The excess carriers being generated at $x = 0$ will begin diffusing in both the $-x$ and $+x$ directions. Calculate the steady-state excess carrier concentration as a function of x. Marks: 6

(b) Define ambipolar transport. Find ambipolar transport equation. Hence show that diffuse and drift behavior of excess majority carrier is determined by excess minority carrier. Marks: 7

(c) Briefly describe different types of generation and recombination processes. Use appropriate diagram if necessary. Marks: 7

8.(a) Consider a p-type semiconductor that is homogeneous and infinite in extent. Assume a zero applied electric field. For a one-dimensional crystal, assume that excess carriers are being generated at $x = 0$ only. The excess carriers being generated at $x = 0$ will begin diffusing in both the $-x$ and $+x$ directions. Calculate the steady-state excess carrier concentration as a function of x. Marks: 6

1. Theory

1. Excess carriers spread and die (Core Idea)

Suppose light falls on a thin slice of an infinitely long p-type bar at $x = 0$ and keeps creating electron-hole pairs there. The extra (**excess**) electrons δn and holes δp are crowded at $x = 0$, so they **diffuse** outward, to the left and to the right. While they diffuse they **recombine**: an excess minority electron survives on average a time τ_{n0}, its lifetime. After a while a **steady state** is reached: at every point, the carriers arriving by diffusion are exactly matched by those lost to recombination. The concentration then no longer changes with time, only with distance.

In p-type material the excess electrons are **minority carriers**. By ambipolar transport (derived in 8(b), and in 4(b)) the excess holes follow them exactly ($\delta p = \delta n$), and the whole distribution moves with the minority carrier parameters D_n, μ_n, τ_{n0}.

2. The ambipolar transport equation for p-type material

At low injection in p-type material (Neamen, Eq. 6.55):

$$D_n \frac{\partial^2(\delta n)}{\partial x^2} + \mu_n E \frac{\partial(\delta n)}{\partial x} + g' - \frac{\delta n}{\tau_{n0}} = \frac{\partial(\delta n)}{\partial t}$$

The four terms are diffusion, drift, excess generation and recombination; the right side is the change in time.

3. The diffusion length

When the equation reduces to $D_n\,\delta n'' = \delta n/\tau_{n0}$, the combination

$$L_n = \sqrt{D_n \tau_{n0}}$$

has the unit of length ($\mathrm{cm}^2/\mathrm{s} \times \mathrm{s} = \mathrm{cm}^2$, then the square root). It is the **minority carrier diffusion length**: the average distance an excess electron diffuses before it recombines. In silicon it is typically tens of micrometres.

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2. Answer

Given: p-type, homogeneous, infinite, one-dimensional; $E = 0$; excess carriers generated only at $x = 0$; steady state.

Step 3 solves a general second-order equation and is marked (Basic); a student who knows it can skip it.

Step 1: The governing equation

The excess electrons are minority carriers in p-type material, so the ambipolar transport equation at low injection uses the electron parameters (Neamen, Eq. 6.55):

$$D_n \frac{\partial^2(\delta n)}{\partial x^2} + \mu_n E \frac{\partial(\delta n)}{\partial x} + g' - \frac{\delta n}{\tau_{n0}} = \frac{\partial(\delta n)}{\partial t}$$

Step 2: Apply the conditions

- zero applied field: $E = 0$, so the drift term vanishes;
- generation only at $x = 0$: $g' = 0$ for $x \neq 0$;
- steady state: $\partial(\delta n)/\partial t = 0$.

What is left (Neamen, Eq. 6.62):

$$D_n \frac{d^2(\delta n)}{dx^2} - \frac{\delta n}{\tau_{n0}} = 0$$

Divide by D_n and put $L_n^2 = D_n \tau_{n0}$ (Neamen, Eq. 6.63):

$$\frac{d^2(\delta n)}{dx^2} - \frac{\delta n}{L_n^2} = 0$$

Step 3 (Basic): Solve the second-order equation

Try $\delta n = e^{sx}$: then $s^2 e^{sx} - e^{sx}/L_n^2 = 0$, so $s^2 = 1/L_n^2$ and $s = \pm 1/L_n$. The general solution is the sum of both (Neamen, Eq. 6.64):

$$\delta n(x) = A e^{-x/L_n} + B e^{+x/L_n}$$

Step 4: Boundary conditions for $x > 0$

Far away the excess must vanish: $\delta n \to 0$ as $x \to +\infty$. The term $B e^{+x/L_n}$ would grow without limit, so $B = 0$ for $x \geq 0$. At $x = 0$, $\delta n = \delta n(0)$, so $A = \delta n(0)$ (Neamen, Eq. 6.65a):

$$\delta n(x) = \delta n(0)\, e^{-x/L_n}, \qquad x \geq 0$$

Step 5: Boundary conditions for $x < 0$

Now $\delta n \to 0$ as $x \to -\infty$, where e^{-x/L_n} would grow, so $A = 0$ and $B = \delta n(0)$ (Neamen, Eq. 6.65b):

$$\delta n(x) = \delta n(0)\, e^{+x/L_n}, \qquad x \leq 0$$

Step 6: The complete result

Both halves join at $x = 0$ with the same value:

$$\boxed{\delta n(x) = \delta p(x) = \delta n(0) \exp\left(-\frac{|x|}{L_n}\right), \qquad L_n = \sqrt{D_n \tau_{n0}}}$$

The excess holes equal the excess electrons at every point (ambipolar transport, charge neutrality).

Steps to remember

Ambipolar equation with minority (electron) parameters → $E = 0$, $g' = 0$ for $x \neq 0$, steady state → $\delta n'' - \delta n/L_n^2 = 0$ → $A e^{-x/L_n} + B e^{x/L_n}$ → finite at $\pm\infty$ → $\delta n(0) e^{-|x|/L_n}$.

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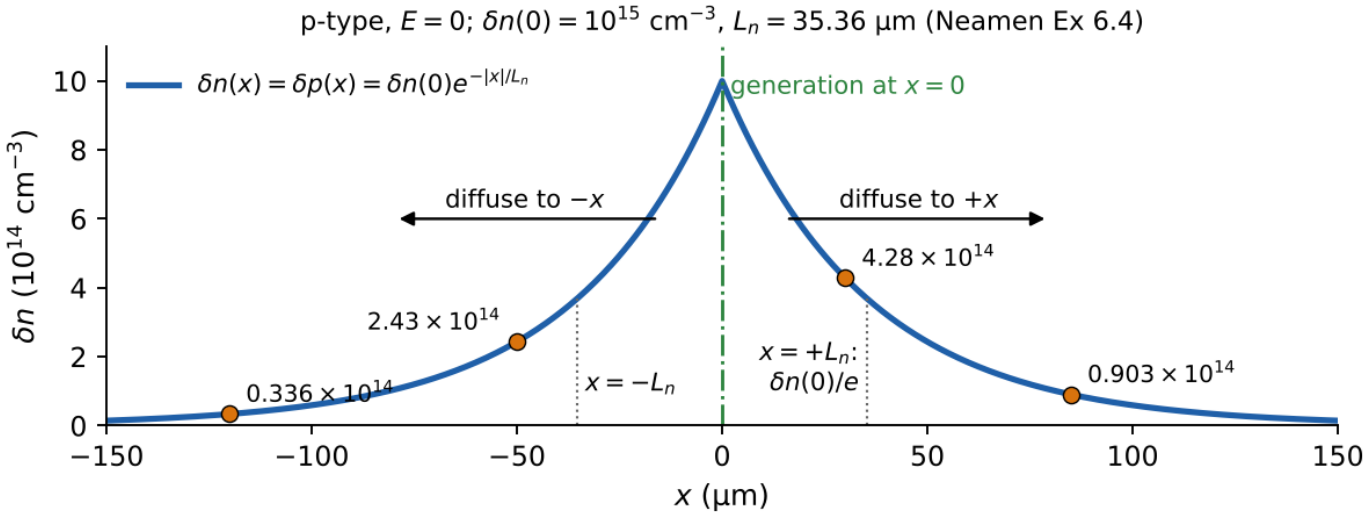


Figure 36: Steady-state excess carrier concentration with generation at x = 0 in an infinite p-type crystal

Reading the figure. The solid blue curve is $\delta n(x) = \delta p(x)$ for Neamen's Exercise Ex 6.4 values ($\delta n(0) = 10^{15}$ cm^{-3}, $D_n = 25$ cm^2/s, $\tau_{n0} = 5 \times 10^{-7}$ s, so $L_n = 35.36$ μm). It peaks at the generation plane $x = 0$ and decays symmetrically on both sides; at $x = \pm L_n$ it is $\delta n(0)/e$ (dotted lines). The dots mark the book's check points at +30, −50, +85 and −120 μm.

Example (Neamen, Ex 6.4)

with $L_n = \sqrt{(25)(5 \times 10^{-7})} = 3.536 \times 10^{-3}$ cm $= 35.36$ μm and $\delta n(0) = 10^{15}$ cm^{-3}: $\delta n = 4.28 \times 10^{14}$ at +30 μm, 2.43×10^{14} at −50 μm, 9.03×10^{13} at +85 μm and 3.36×10^{13} cm^{-3} at −120 μm.

Final Answer: $\delta n(x) = \delta p(x) = \delta n(0)e^{-|x|/L_n}$, i.e. $\delta n(0)e^{-x/L_n}$ for $x \geq 0$ and $\delta n(0)e^{+x/L_n}$ for $x \leq 0$, with $L_n = \sqrt{D_n \tau_{n0}}$; it falls to $1/e$ of its peak one diffusion length from the source.

3. Keyword Explanation

1. Steady state

A condition in which nothing changes with time: $\partial/\partial t = 0$.

2. Minority carrier diffusion length

$L_n = \sqrt{D_n \tau_{n0}}$: the mean distance an excess electron diffuses in p-type material before it recombines.

3. Excess carriers

Carriers above the equilibrium values: $\delta n = n - n_0$, $\delta p = p - p_0$.

4. Notes and Remarks

- **Book check:** the question is Neamen's Example 6.4 word for word. The book's answer is the same, $\delta n(x) = \delta n(0)e^{-x/L_n}$ for $x \geq 0$ and $\delta n(0)e^{+x/L_n}$ for $x \leq 0$, and its comment that the excess falls to $1/e$ at $x = L_n$. The example numbers above are its Exercise Ex 6.4, whose answers ($L_n = 35.36$ μm; 4.28×10^{14}, 2.43×10^{14}, 9.04×10^{13}, 3.36×10^{13} cm^{-3}) match.
- **The pen notes on the scan** (in the right margin beside this question) write $E = 0$, $Ae^{-x/L_n} + Be^{x/L_n}$ and a boxed $L_n = \sqrt{D\tau}$: the same steps as here.
- The total electron concentration is $n = n_0 + \delta n$ and the hole concentration $p = p_0 + \delta n \approx p_0$ at low injection.
- The same question is 5(a) of this session's final (5(a)), solved in full there too. The 2020-21 paper's 3(d) solves the one-sided case ($x > 0$) with numbers (extra reading).

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8.(b) Define ambipolar transport. Find ambipolar transport equation. Hence show that diffuse and drift behavior of excess majority carrier is determined by excess minority carrier. Marks: 7

1. Theory

1. Excess carriers and why they move together (Core Idea)

When light (or injection) creates extra electron-hole pairs in a semiconductor, the extra electrons δn and extra holes δp are called **excess carriers**. Each kind would like to diffuse and drift on its own: electrons with D_n, μ_n, holes with D_p, μ_p, and in an applied field in opposite directions.

But as soon as the electron cloud and the hole cloud separate even slightly, a net charge appears, and with it an **internal electric field** E_{int} that pulls them back together (Poisson's equation, Neamen Eq. 6.32):

$$\frac{\partial E_{int}}{\partial x} = \frac{e(\delta p - \delta n)}{\epsilon_s}$$

A tiny imbalance makes a large field (the imbalance relaxes in the dielectric relaxation time, about a picosecond, 3(c)). So the excess electrons and holes **diffuse and drift together**, as one neutral cloud with $\delta n \approx \delta p$, with a single effective diffusion coefficient and a single effective mobility. This is called **ambipolar transport** (Neamen, Section 6.3).

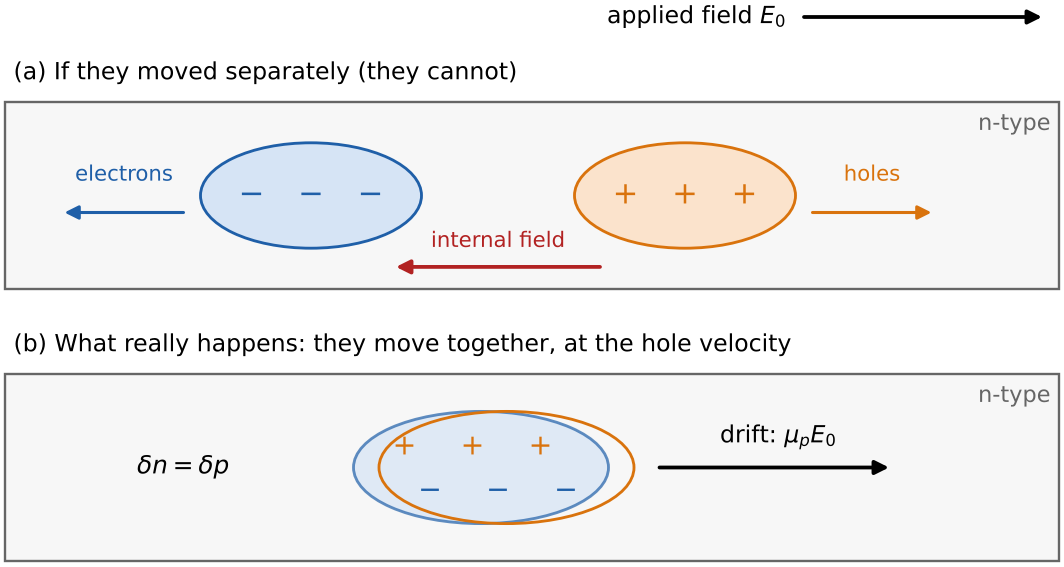


Figure 37: Excess electrons and holes are held together by their internal field and drift as one cloud (reused from the 2020-21 paper)

Reading the figure. Panel (a) shows what would happen if the clouds moved separately in an applied field E_0: electrons (blue, −) to the left, holes (orange, +) to the right, which would build a strong internal field (red arrow) opposing the separation. Panel (b) shows what really happens: the two clouds stay together ($\delta n = \delta p$) and drift as one, in n-type material at the hole (minority carrier) velocity $\mu_p E_0$.

2. The continuity equation (bookkeeping of carriers)

In a thin slice of width dx, the number of holes changes because of three things (Neamen, Section 6.2.1): holes flowing in minus holes flowing out (a flux F_p^+, holes per cm^2 per s), generation g_p, and recombination p/τ_{pt}. This bookkeeping gives the **continuity equation** (Neamen, Eq. 6.18):

$$\frac{\partial p}{\partial t} = -\frac{\partial F_p^+}{\partial x} + g_p - \frac{p}{\tau_{pt}}$$

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The same for electrons, with flux F_n^-. The flux comes from drift and diffusion. Putting the current equations into the continuity equations gives one **time-dependent diffusion equation** for each carrier. Ambipolar transport combines the two.

2. Answer

Definition. **Ambipolar transport** is the motion of excess electrons and excess holes **together**, as one neutral distribution ($\delta n \approx \delta p$), with a single ambipolar diffusion coefficient D' and ambipolar mobility μ'. The internal electric field created by any small separation of the two charged clouds holds them together, so they diffuse and drift as one (Neamen, Section 6.3).

Derivation of the ambipolar transport equation (one dimension; Neamen, Sections 6.2 and 6.3). Steps 1 and 4 build general tools (the continuity bookkeeping and the product rule) and are marked (Basic); a student who knows them can skip them.

Step 1 (Basic): Continuity equation for holes

Take a slice of area A between x and $x + dx$. The hole flux (holes per unit area per second) entering at x is $F_p^+(x)$ and leaving at $x + dx$ is $F_p^+(x + dx) = F_p^+(x) + \frac{\partial F_p^+}{\partial x} dx$ (Taylor expansion, Neamen Eq. 6.15). The rate of change of the number of holes in the slice is inflow minus outflow, plus generation, minus recombination:

$$\frac{\partial p}{\partial t} A\, dx = \left[F_p^+(x) - F_p^+(x + dx)\right] A + g_p A\, dx - \frac{p}{\tau_{pt}} A\, dx$$

Divide by $A\, dx$ (Neamen, Eq. 6.18):

$$\frac{\partial p}{\partial t} = -\frac{\partial F_p^+}{\partial x} + g_p - \frac{p}{\tau_{pt}}$$

and for electrons, in the same way (Neamen, Eq. 6.19):

$$\frac{\partial n}{\partial t} = -\frac{\partial F_n^-}{\partial x} + g_n - \frac{n}{\tau_{nt}}$$

where g is the generation rate and τ_{pt}, τ_{nt} include thermal-equilibrium generation and recombination.

Step 2: The fluxes from drift and diffusion

The hole and electron current densities are drift plus diffusion (Neamen, Eqs. 6.20, 6.21):

$$J_p = e\mu_p pE - eD_p \frac{\partial p}{\partial x}, \qquad J_n = e\mu_n nE + eD_n \frac{\partial n}{\partial x}$$

A flux is a current divided by the charge of the carrier (Neamen, Eqs. 6.22, 6.23):

$$F_p^+ = \frac{J_p}{(+e)} = \mu_p pE - D_p \frac{\partial p}{\partial x}, \qquad F_n^- = \frac{J_n}{(-e)} = -\mu_n nE - D_n \frac{\partial n}{\partial x}$$

Step 3: Substitute into the continuity equations

$$\frac{\partial p}{\partial t} = -\mu_p \frac{\partial (pE)}{\partial x} + D_p \frac{\partial^2 p}{\partial x^2} + g_p - \frac{p}{\tau_{pt}}$$

$$\frac{\partial n}{\partial t} = +\mu_n \frac{\partial (nE)}{\partial x} + D_n \frac{\partial^2 n}{\partial x^2} + g_n - \frac{n}{\tau_{nt}}$$

(Neamen, Eqs. 6.24, 6.25.)

Step 4 (Basic): Product rule on the drift term

$\frac{\partial (pE)}{\partial x} = E\frac{\partial p}{\partial x} + p\frac{\partial E}{\partial x}$ (Neamen, Eq. 6.26). So, rearranged:

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$$D_p\frac{\partial^2 p}{\partial x^2} - \mu_p\left(E\frac{\partial p}{\partial x} + p\frac{\partial E}{\partial x}\right) + g_p - \frac{p}{\tau_{pt}} = \frac{\partial p}{\partial t}$$

$$D_n\frac{\partial^2 n}{\partial x^2} + \mu_n\left(E\frac{\partial n}{\partial x} + n\frac{\partial E}{\partial x}\right) + g_n - \frac{n}{\tau_{nt}} = \frac{\partial n}{\partial t}$$

These are the **time-dependent diffusion equations** (Neamen, Eqs. 6.27, 6.28; the first is on the paper's equation sheet).

Step 5: Write them for the excess carriers

Put $p = p_0 + \delta p$, $n = n_0 + \delta n$ with p_0, n_0 uniform in space and constant in time, so their derivatives vanish: $\partial^2 p/\partial x^2 = \partial^2(\delta p)/\partial x^2$, $\partial p/\partial x = \partial(\delta p)/\partial x$, $\partial p/\partial t = \partial(\delta p)/\partial t$. The same for electrons.

Step 6: Charge neutrality and equal rates

- The internal field holds the clouds together, so $\delta n = \delta p$ everywhere (the small charge difference is still kept in $\partial E/\partial x$, which is not zero).
- Electrons and holes are generated in pairs and recombine in pairs (Neamen, Eqs. 6.34, 6.35): $g_n = g_p \equiv g$ and $n/\tau_{nt} = p/\tau_{pt} \equiv R$.

Writing δn for both excess concentrations (Neamen, Eqs. 6.36, 6.37):

$$D_p\frac{\partial^2(\delta n)}{\partial x^2} - \mu_p\left(E\frac{\partial(\delta n)}{\partial x} + p\frac{\partial E}{\partial x}\right) + g - R = \frac{\partial(\delta n)}{\partial t} \qquad \text{(i)}$$

$$D_n\frac{\partial^2(\delta n)}{\partial x^2} + \mu_n\left(E\frac{\partial(\delta n)}{\partial x} + n\frac{\partial E}{\partial x}\right) + g - R = \frac{\partial(\delta n)}{\partial t} \qquad \text{(ii)}$$

Step 7: Eliminate $\partial E/\partial x$

Multiply (i) by $\mu_n n$ and (ii) by $\mu_p p$ and add. The field-gradient terms are $-\mu_n\mu_p np\,\partial E/\partial x$ and $+\mu_p\mu_n pn\,\partial E/\partial x$: they cancel. The rest gives (Neamen, Eq. 6.38)

$$(\mu_n n D_p + \mu_p p D_n)\frac{\partial^2(\delta n)}{\partial x^2} + \mu_n\mu_p(p-n)E\frac{\partial(\delta n)}{\partial x} + (\mu_n n + \mu_p p)(g - R) = (\mu_n n + \mu_p p)\frac{\partial(\delta n)}{\partial t}$$

Step 8: Divide by $(\mu_n n + \mu_p p)$: the ambipolar transport equation

$$\boxed{D'\frac{\partial^2(\delta n)}{\partial x^2} + \mu' E\frac{\partial(\delta n)}{\partial x} + g - R = \frac{\partial(\delta n)}{\partial t}}$$

(Neamen, Eq. 6.39), with the **ambipolar diffusion coefficient** and **ambipolar mobility** (Neamen, Eqs. 6.40, 6.41):

$$D' = \frac{\mu_n n D_p + \mu_p p D_n}{\mu_n n + \mu_p p}, \qquad \mu' = \frac{\mu_n\mu_p(p-n)}{\mu_n n + \mu_p p}$$

With the Einstein relation $\mu_n/D_n = \mu_p/D_p = e/kT$ (Neamen, Eq. 6.42), D' becomes (Neamen, Eq. 6.43)

$$D' = \frac{D_n D_p(n+p)}{D_n n + D_p p}$$

Hence: the excess majority carrier follows the minority carrier.

Step 9: p-type material at low injection

Low injection means $\delta n \ll p_0$, and in p-type $n_0 \ll p_0$. So $n = n_0 + \delta n \ll p = p_0 + \delta n \approx p_0$. In D' and μ' drop n next to p:

$$D' = \frac{D_n D_p p}{D_p p} = D_n, \qquad \mu' = \frac{\mu_n\mu_p p}{\mu_p p} = \mu_n$$

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(Neamen, Eqs. 6.45, 6.46.) The rates: $g - R = g' - \delta n/\tau_{n0}$ (Neamen, Eq. 6.51), with g' the excess generation rate and τ_{n0} the minority electron lifetime. The ambipolar equation becomes (Neamen, Eq. 6.55)

$$D_n \frac{\partial^2(\delta n)}{\partial x^2} + \mu_n E \frac{\partial(\delta n)}{\partial x} + g' - \frac{\delta n}{\tau_{n0}} = \frac{\partial(\delta n)}{\partial t}$$

Step 10: n-type material at low injection

Now $p \ll n \approx n_0$: $D' = D_p$ and $\mu' = -\mu_p$ (Neamen, Eqs. 6.47, 6.48), and (Neamen, Eq. 6.56)

$$D_p \frac{\partial^2(\delta p)}{\partial x^2} - \mu_p E \frac{\partial(\delta p)}{\partial x} + g' - \frac{\delta p}{\tau_{p0}} = \frac{\partial(\delta p)}{\partial t}$$

Steps to remember

Continuity for p and n → drift + diffusion fluxes → time-dependent diffusion equations → excess form with $\delta n = \delta p$, $g_n = g_p$, $R_n = R_p$ → multiply by $\mu_n n$ and $\mu_p p$, add: $\partial E/\partial x$ cancels → divide: D', μ' → low injection: D' = minority D, μ' = minority μ.

Conclusion. In an extrinsic semiconductor at low injection, D', μ' and the lifetime in the ambipolar equation are those of the **minority carrier** (electrons in p-type, holes in n-type), and μ' even has the minority carrier's sign (direction of drift). Since $\delta p = \delta n$, the excess **majority** carriers diffuse and drift with exactly the same distribution: their diffusion and drift behaviour is determined by the excess minority carriers. Physically, the majority carriers are so many that a tiny redistribution of them follows the minority cloud and neutralizes it, while the slow, rare minority carriers set the pace.

3. Keyword Explanation

1. Excess carriers

Electrons and holes above their thermal-equilibrium values, $\delta n = n - n_0$, $\delta p = p - p_0$.

2. Low-level injection

The excess carrier concentration is much smaller than the thermal-equilibrium majority carrier concentration.

3. Ambipolar diffusion coefficient and mobility

D' and μ': the single diffusion coefficient and mobility with which the neutral excess electron-hole distribution moves.

4. Continuity equation

The particle balance of a small volume: change = net inflow + generation − recombination.

4. Notes and Remarks

- **Intrinsic material:** with $n = p$, $\mu' = 0$ (the cloud does not drift at all) and $D' = 2D_nD_p/(D_n + D_p)$, between the two diffusion coefficients.
- The equation sheet of the paper carries the two time-dependent diffusion equations of Step 4 (in two forms) and the solution for a pulse; see 4(d).
- The same question is 4(b) of this session's final (4(b)), solved in full there too.
- **The pen note on the scan** beside this question reads "$\mu' = \mu_n$ for p[-type]" (cut off at the page edge): the same result as Step 9.

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8.(c) Briefly describe different types of generation and recombination processes. Use appropriate diagram if necessary. Marks: 7

1. Theory

1. Generation and recombination (Core Idea)

- **Generation** is a process that **creates** an electron and a hole: an electron is lifted from the valence band to the conduction band, leaving a hole behind. It needs energy, from heat (lattice vibrations) or from light.
- **Recombination** is the reverse: a conduction-band electron falls back into an empty valence-band state, so an electron and a hole **disappear** together. The energy they release leaves as light (a photon) or as heat (phonons).

Their rates are written per unit volume per second ($\text{cm}^{-3}\text{s}^{-1}$): G for generation and R for recombination (Neamen, Section 6.1).

2. Equilibrium and excess carriers

In **thermal equilibrium** thermal generation goes on all the time, but it is exactly balanced by recombination, so n_0 and p_0 stay constant (Neamen, Eqs. 6.1 to 6.3):

$$G_{n0} = G_{p0} = R_{n0} = R_{p0}$$

When light or an injected current adds **excess** carriers, an extra generation rate g' appears ($g'_n = g'_p$, since carriers are made in pairs, Neamen Eq. 6.4), and recombination speeds up to remove them, with $R'_n = R'_p$ (Neamen, Eq. 6.6). At low injection the excess recombination rate is proportional to the excess concentration (Neamen, Eqs. 6.13, 6.14):

$$R' = \frac{\delta n}{\tau_{n0}} \text{ (p-type)}, \qquad R' = \frac{\delta p}{\tau_{p0}} \text{ (n-type)}$$

where τ_{n0}, τ_{p0} are the **minority carrier lifetimes**.

3. What decides the path

An electron and a hole can meet in three ways: **directly** across the gap, through an **intermediate energy state** (a trap) inside the gap, or with the help of a **third carrier** (Auger). Which one dominates depends on the material: direct-gap GaAs favours direct (radiative) recombination; indirect-gap Si recombines mostly through traps.

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2. Answer

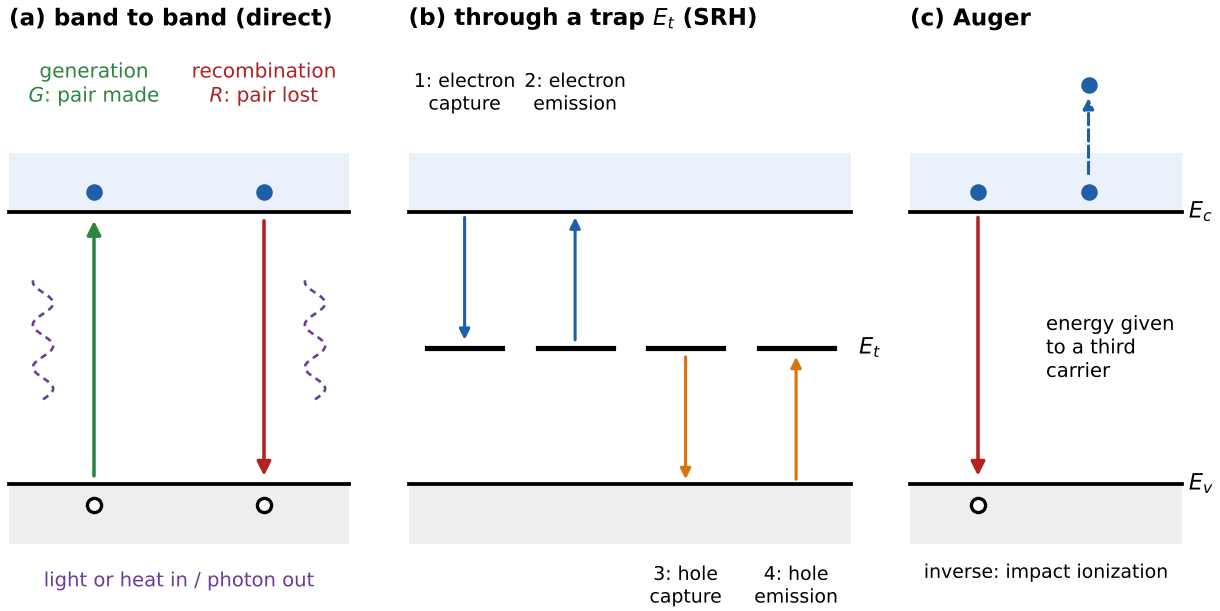


Figure 38: Generation and recombination processes: band to band, through a trap (Shockley-Read-Hall), and Auger

Reading the figure. Panel (a), band to band: the green upward arrow is generation (an electron lifted across the gap by heat or light, leaving a hole); the red downward arrow is recombination (the electron falls into the hole, giving out a photon, wavy purple line). Panel (b), through a trap level E_t: the four processes 1 electron capture, 2 electron emission (blue arrows, between E_c and E_t), 3 hole capture and 4 hole emission (orange arrows, between E_t and E_v). Panel (c), Auger: the electron-hole recombination (red) hands its energy to a second conduction electron, which is kicked high into the band (dashed blue) and then loses the energy as heat.

1. Direct (band-to-band) generation and recombination (Neamen, Section 6.1.2)

- **Generation:** an electron in the valence band absorbs energy at least E_g (a photon of $h\nu \geq E_g$, or thermal energy) and jumps to the conduction band: one electron-hole pair is created.
- **Recombination:** a conduction electron drops straight into a hole; the energy $\approx E_g$ is emitted as a photon (radiative recombination). The rate is proportional to the product of the concentrations (Neamen, Eq. 6.7):

$$\frac{dn(t)}{dt} = \alpha_r \left[n_i^2 - n(t)p(t)\right]$$

the first term being thermal generation and the second recombination. At low injection in p-type material it gives $\delta n(t) = \delta n(0) e^{-t/\tau_{n0}}$ with $\tau_{n0} = 1/(\alpha_r p_0)$ (Neamen, Eqs. 6.10, 6.11).

- Important in **direct band-gap** materials (GaAs): it is the basis of LEDs and lasers; light absorption is the basis of photodetectors and solar cells.

2. Recombination and generation through traps (Shockley-Read-Hall, SRH) (Neamen, Section 6.5.1)

Defects and impurities create allowed states E_t inside the gap, called **traps** or recombination centres. Four processes take place (panel (b)):

1. **Electron capture:** a conduction electron is captured by an empty trap.
2. **Electron emission:** an electron in a filled trap is emitted to the conduction band.
3. **Hole capture:** a filled trap gives its electron to the valence band (captures a hole).
4. **Hole emission:** an empty trap takes an electron from the valence band (emits a hole).

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Recombination is process 1 followed by 3; generation is 4 followed by 2. The net rate is (Neamen, Eq. 6.99)

$$R = \frac{C_n C_p N_t (np - n_i^2)}{C_n(n + n') + C_p(p + p')}$$

which is positive (net recombination) when $np > n_i^2$ and negative (net generation) when $np < n_i^2$, as in a reverse-biased depletion region. Traps near midgap are the most effective. This is the main process in **silicon** (indirect gap); the energy goes to the lattice as heat (non-radiative).

3. Auger recombination and impact ionization

- **Auger recombination:** an electron and a hole recombine and give the energy to a **third carrier** (another electron or hole), which is raised high in its band and then loses the energy as heat (Neamen, Section 14.4.1, Figure 14.21c). Its rate grows as n^2p or np^2, so it matters at **high doping or high injection**. Non-radiative.
- **Impact ionization** is the reverse, a generation process: a carrier accelerated by a strong field hits the lattice hard enough to create a new electron-hole pair. It causes avalanche breakdown of pn junctions.

4. Surface recombination

At a surface the crystal ends and dangling bonds give many traps in the gap, so SRH recombination is much faster there; it is described by a surface recombination velocity (Neamen, Section 6.6).

Summary:

Process	Generation	Recombination	Energy	Main where
Band to band	absorption of $h\nu \geq E_g$ or heat	electron falls into hole	photon (radiative)	GaAs, LEDs, lasers
Through traps (SRH)	trap emits electron and hole	trap captures electron then hole	heat	Si, defects, surfaces
Auger / impact	impact ionization (high field)	energy to a third carrier	heat	heavy doping, high injection

Table 10: Summary

3. Keyword Explanation

1. Generation rate

The number of electron-hole pairs created per cm^3 per second.

2. Recombination rate

The number of electron-hole pairs destroyed per cm^3 per second.

3. Trap (recombination centre)

An allowed energy level inside the band gap, from defects or impurities, that can capture and emit electrons and holes.

4. Minority carrier lifetime

The mean time an excess minority carrier exists before it recombines; $R' = \delta n/\tau_{n0}$.

4. Notes and Remarks

- **Direct and indirect gap:** in silicon the bottom of the conduction band and the top of the valence band are at different crystal momenta, so a direct transition must also involve a phonon and is slow; that is why Si recombines mainly through traps and is a poor light emitter.

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- The SRH result at low injection in n-type material reduces to $R = \delta p/\tau_{p0}$ with $\tau_{p0} = 1/(C_p N_t)$ (Neamen, Eqs. 6.100 to 6.103): the lifetime is set by the trap density.

5. Final Concise Answer

(a) In p-type material the ambipolar equation uses the minority (electron) parameters; with $E = 0$, $g' = 0$ for $x \neq 0$ and steady state it becomes $D_n\, d^2\delta n/dx^2 - \delta n/\tau_{n0} = 0$, solved by $Ae^{-x/L_n} + Be^{x/L_n}$; finiteness at $\pm\infty$ gives $\delta n(x) = \delta p(x) = \delta n(0)e^{-|x|/L_n}$ with $L_n = \sqrt{D_n\tau_{n0}}$ (Neamen Example 6.4).

(b) Ambipolar transport: excess electrons and holes drift and diffuse together ($\delta n = \delta p$), held by their internal field. Continuity plus drift-diffusion fluxes, with $\partial E/\partial x$ eliminated, give $D'\partial^2\delta n/\partial x^2 + \mu' E\, \partial\delta n/\partial x + g - R = \partial\delta n/\partial t$, $D' = \frac{D_n D_p(n+p)}{D_n n + D_p p}$, $\mu' = \frac{\mu_n\mu_p(p-n)}{\mu_n n + \mu_p p}$; at low injection D', μ' and τ become the minority carrier's (D_n, μ_n in p-type; D_p, $-\mu_p$ in n-type), so the excess majority carriers move with the minority carriers.

(c) Band-to-band: absorption of $h\nu \geq E_g$ (or heat) creates pairs, an electron falling into a hole emits a photon, rate $\alpha_r(np - n_i^2)$ (GaAs). Shockley-Read-Hall: traps in the gap capture and emit electrons and holes (four processes), net rate $C_n C_p N_t(np - n_i^2)/[C_n(n + n') + C_p(p + p')]$, non-radiative, dominant in Si. Auger: the energy goes to a third carrier (heavy doping); its inverse, impact ionization, generates pairs in high fields. Surfaces recombine fast through dangling-bond traps.

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FORMULA SHEET

EEE421

Solid State Devices

Formula Sheet

Session 2019–20 paper

Every formula behind the 4th Year 1st Semester Final Examination, 2023, and its two term tests

TOPICS

1 Crystals and thermal motion — 7 formulas

2 Carriers in equilibrium — 10 formulas

3 Dopants and charge neutrality — 4 formulas

4 Transport — 8 formulas

5 Excess carriers — 9 formulas

6 PN junction — 8 formulas

7 MOSFET and BJT — 7 formulas

Each topic is one table: **Title** · **Equation** · **Notation** · **Reference**.

Prepared by Nur Mohammad

November 2026

BACK OF THE COVER

Before you use the sheet

How to read a row

★ **Atoms per fcc cell**	$n = 8 \times \frac{1}{8} + 6 \times \frac{1}{2} = 4$	corner atoms shared by 8 cells, face atoms by 2; sc: 1, bcc: 2	1(b)

the name you search for

the formula, exactly as derived

every new symbol with its unit

link: opens the derivation

In the exam

- Write the formula first, then put the numbers in: the formula alone often earns a mark.
- Read the symbols in the Notation column before you substitute, and check the units.
- Not sure where a formula comes from? Its Reference opens the derivation.
- The **Key formulas** page at the end gathers the rows marked ★.
- Lost a symbol? The **Symbol index** at the end links to the row where it first appears.

Contents

1 Crystals and thermal motion 4

2 Carriers in equilibrium 4

3 Dopants and charge neutrality 5

4 Transport 5

5 Excess carriers 6

6 PN junction 7

7 MOSFET and BJT 7

8 Key formulas 9

9 Symbol index 11

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Conventions (Neamen): e is the magnitude of the electron charge, $kT = 0.0259$ eV and $V_t = kT/e = 0.0259$ V at 300 K; $\epsilon_s = \epsilon_r\epsilon_0$ with $\epsilon_0 = 8.85 \times 10^{-14}$ F/cm. Si at 300 K: $N_c = 2.8 \times 10^{19}$ cm^{-3}, $N_v = 1.04 \times 10^{19}$ cm^{-3}, $n_i = 1.5 \times 10^{10}$ cm^{-3}, $E_g = 1.12$ eV, $\epsilon_r = 11.7$. References point to the solutions of the 2019–20 paper.

Columns: Title · Equation · Notation · Ref. (reference).

1. Crystals and thermal motion:

Title	Equation	Notation	Ref.
★ **Atoms per fcc cell**	$n = 8 \times \frac{1}{8} + 6 \times \frac{1}{2} = 4$	corner atoms shared by 8 cells, face atoms by 2; sc: 1, bcc: 2	1(b)
★ **Volume density**	density $= \frac{n}{a^3}$	a = lattice constant in cm $a = 0.425$ nm: 5.21×10^{22} cm^{-3}	1(b)
Maxwell-Boltzmann speeds	$f(v) = 4\pi \left(\frac{m}{2\pi kT}\right)^{3/2} v^2 \exp\left(-\frac{mv^2}{2kT}\right)$	m = effective mass of the carrier	1(c)
★ **Mean speed**	$\bar{v} = \sqrt{\frac{8kT}{\pi m^*}}$	$0.1m_0$: 3.40×10^5 m/s; $1.5m_0$: 8.78×10^4 m/s (300 K)	1(c)
rms (thermal) speed	$v_{rms} = \sqrt{\frac{3kT}{m^*}}$	from $\frac{1}{2}m^*\overline{v^2} = \frac{3}{2}kT$; 3.69×10^5 and 9.53×10^4 m/s	1(c)
Schottky barrier	$\phi_{B0} = \phi_m - \chi$, $V_{bi} = \phi_{B0} - \phi_n$	ϕ_m = metal work function, χ = electron affinity (4.01 V for Si), $e\phi_n = E_c - E_F$	1(d)
Thermionic emission	$J = J_{sT}\left[\exp\left(\frac{eV_a}{kT}\right) - 1\right]$, $J_{sT} = A^*T^2 \exp\left(-\frac{e\phi_{Bn}}{kT}\right)$	A^* = effective Richardson constant	1(d)

Table 1: Crystals and thermal motion

2. Carriers in equilibrium:

Title	Equation	Notation	Ref.
★ **Fermi-Dirac function**	$f_F(E) = \frac{1}{1 + \exp\left(\frac{E - E_F}{kT}\right)}$	$E_F - E_c = -0.3$ eV: $f_F(E_c) = 9.32 \times 10^{-6}$	3(a), 7(d)
★ **Density of states**	$g_c(E) = \frac{4\pi(2m_n^*)^{3/2}}{h^3}\sqrt{E - E_c}$	g_v the same with m_p^* and $\sqrt{E_v - E}$	3(a)
Carrier distributions	$n(E) = g_c(E)f_F(E)$, $p(E) = g_v(E)[1 - f_F(E)]$	peak at $kT/2$ inside the band; areas n_0, p_0	3(a)
★ **Electron concentration**	$n_0 = N_c \exp\left[-\frac{E_c - E_F}{kT}\right]$	$E_c - E_F = 0.3$ eV: $n_0 = 2.61 \times 10^{14}$ cm^{-3}	3(a), 7(d)
Hole concentration	$p_0 = N_v \exp\left[-\frac{E_F - E_v}{kT}\right]$		3(a)

Title	Equation	Notation	Ref.
★ **Mass-action law**	$n_0 p_0 = n_i^2 = N_c N_v \exp\left(-\frac{E_g}{kT}\right)$	$p_0 = n_i^2/n_0 = 8.62 \times 10^5$ cm^{-3} for 7(d)	2(a), 7(a), 7(d)
★ **Band gap from** n_i	$E_g = kT \ln\left(\frac{N_c N_v}{n_i^2}\right)$, $n_i^2 = n_0(n_0 - N_d)$	400 K, $n_0 \leq 1.01 \times 10^{15}$: $E_{g,min} = 0.823$ eV	7(c)
★ **Fermi level**	$E_F - E_{Fi} = kT \ln \frac{n_0}{n_i}$, $E_{Fi} - E_F = kT \ln \frac{p_0}{n_i}$	0.06 eV per decade of doping; $\to 0$ at high T	7(b)
Fermi level from the band edges	$E_c - E_F = kT \ln \frac{N_c}{n_0}$, $E_F - E_v = kT \ln \frac{N_v}{p_0}$		7(b)
n_i at another temperature	$n_i(T) = n_i(300)\left(\frac{T}{300}\right)^{3/2} \exp\left[-\frac{E_g}{2k}\left(\frac{1}{T} - \frac{1}{300}\right)\right]$	80 K, $n_i(300) = 2 \times 10^{10}$: 3.70×10^{-17} cm^{-3}	2(c), 4(c)

Table 2: Carriers in equilibrium

3. Dopants and charge neutrality:

Title	Equation	Notation	Ref.
★ **Charge neutrality**	$n_0 + N_a^- = p_0 + N_d^+$	complete ionization: $N_a^- = N_a$, $N_d^+ = N_d$	2(a), 7(a)
★ **Electrons**	$n_0 = \frac{N_d - N_a}{2} + \sqrt{\left(\frac{N_d - N_a}{2}\right)^2 + n_i^2}$	$\approx N_d - N_a$ when $N_d - N_a \gg n_i$	2(a), 7(a)
★ **Holes**	$p_0 = \frac{N_a - N_d}{2} + \sqrt{\left(\frac{N_a - N_d}{2}\right)^2 + n_i^2}$	$\approx N_a - N_d$ when $N_a - N_d \gg n_i$	2(a), 7(a)
Electrons left on donors	$\frac{n_d}{n_d + n_0} = \frac{1}{1 + \frac{N_c}{2N_d}\exp\left[-\frac{E_c - E_d}{kT}\right]}$	near 1 in freeze-out	2(c)

Table 3: Dopants and charge neutrality

4. Transport:

Title	Equation	Notation	Ref.
★ **Mobility**	$\mu = \frac{e\tau}{m^*}$, $v_d = \mu E$	τ = mean time between collisions	4(a)
★ **Two scattering mechanisms**	$\mu_L \propto T^{-3/2}$, $\mu_I \propto \frac{T^{3/2}}{N_I}$, $\frac{1}{\mu} = \frac{1}{\mu_L} + \frac{1}{\mu_I}$	$N_I = N_d^+ + N_a^-$ (total ionized impurity)	4(a), 3(d)

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Title	Equation	Notation	Ref.
★ **Conductivity**	$\sigma = e(\mu_n n + \mu_p p)$	compensated n-type: $\sigma = e\mu_n(N_I)(N_d - N_a)$; $\sigma = 16$ S/cm, $N_a = 10^{17}$: $N_d \approx 3.5 \times 10^{17}$, $\mu_n \approx 400$	3(d)
★ **Hall voltage**	$V_H = E_H W$, $V_H = -\dfrac{I_x B_z}{ned}$	$V_H < 0$: n-type; W across the field, d along B_z	2(b)
Hall concentration and mobility	$n = -\dfrac{I_x B_z}{edV_H}$, $\mu_n = \dfrac{I_x L}{enV_x W d}$	$V_H = -0.825$ mV, $n = 4.92 \times 10^{15}$ cm^{-3}, $\mu_n = 1015$ cm^2/V s	2(b)
★ **Dielectric relaxation**	$\rho(t) = \rho(0)e^{-t/\tau_d}$, $\tau_d = \dfrac{\epsilon}{\sigma}$	from $\nabla \cdot J = -\partial\rho/\partial t$, $J = \sigma E$, $\nabla \cdot E = \rho/\epsilon$; $N_d = 10^{16}$: 0.539 ps	3(c)
Current density	$J = en\mu_n E + ep\mu_p E + eD_n \dfrac{dn}{dx} - eD_p \dfrac{dp}{dx}$	drift plus diffusion	3(b), 4(b)
Einstein relation	$\dfrac{\mu_n}{D_n} = \dfrac{\mu_p}{D_p} = \dfrac{e}{kT}$		4(b)

Table 4: Transport

5. Excess carriers:

Title	Equation	Notation	Ref.
Continuity	$\dfrac{\partial p}{\partial t} = -\dfrac{\partial F_p^+}{\partial x} + g_p - \dfrac{p}{\tau_{pt}}$	F_p^+ = hole flux	4(b), 8(b)
★ **Ambipolar transport equation**	$D' \dfrac{\partial^2(\delta n)}{\partial x^2} + \mu' E \dfrac{\partial(\delta n)}{\partial x} + g - R = \dfrac{\partial(\delta n)}{\partial t}$	$\delta n = \delta p$	4(b), 8(b)
★ **Ambipolar coefficients**	$D' = \dfrac{D_n D_p(n+p)}{D_n n + D_p p}$, $\mu' = \dfrac{\mu_n \mu_p (p-n)}{\mu_n n + \mu_p p}$	low injection: p-type D_n, μ_n; n-type D_p, $-\mu_p$	4(b), 8(b)
★ **Uniform decay and build-up**	$\delta p(t) = \delta p(0)e^{-t/\tau_{p0}}$, $\delta p(t) = g'\tau_{p0}(1 - e^{-t/\tau_{p0}})$	τ_{p0} = minority carrier lifetime	4(d)
★ **Steady state, source at $x = 0$**	$\delta n(x) = \delta n(0)e^{-\lvert x\rvert/L_n}$, $L_n = \sqrt{D_n \tau_{n0}}$	from $D_n \delta n'' - \delta n/\tau_{n0} = 0$; Ex 6.4: $L_n = 35.36$ μm	5(a), 8(a), 4(d)
Pulse	$\delta p(x,t) = \dfrac{e^{-t/\tau_{p0}}}{(4\pi D_p t)^{1/2}} \exp\left[-\dfrac{(x - \mu_p E_0 t)^2}{4D_p t}\right]$	spreads, decays and drifts at $\mu_p E_0$	4(d)
★ **Quasi-Fermi levels**	$E_{Fn} - E_{Fi} = kT \ln \dfrac{n_0 + \delta n}{n_i}$, $E_{Fi} - E_{Fp} = kT \ln \dfrac{p_0 + \delta p}{n_i}$	$\delta n = 10^{13}$: 0.2984 eV and 0.179 eV	5(c)
Direct recombination	$\dfrac{dn}{dt} = \alpha_r(n_i^2 - np)$, $R' = \dfrac{\delta n}{\tau_{n0}}$	$\tau_{n0} = 1/(\alpha_r p_0)$ in p-type	8(c)

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Title	Equation	Notation	Ref.
SRH recombination	$R = \dfrac{C_n C_p N_t (np - n_i^2)}{C_n(n+n') + C_p(p+p')}$	N_t = trap density; $R < 0$ means net generation	8(c)

Table 5: Excess carriers

6. PN junction:

Title	Equation	Notation	Ref.
★ **Built-in potential**	$V_{bi} = \dfrac{kT}{e} \ln\left(\dfrac{N_a N_d}{n_i^2}\right)$	4(c): 0.820 V (300 K), 1.069 V (80 K); 5(d): 0.671 V	3(b), 4(c), 5(d)
Barrier and minority carriers	$n_{p0} = n_{n0} \exp\left(-\dfrac{eV_{bi}}{kT}\right)$	only this fraction climbs the barrier	3(b), 5(b)
★ **Depletion width**	$W = \left\{\dfrac{2\epsilon_s(V_{bi}+V_R)}{e}\left[\dfrac{N_a+N_d}{N_a N_d}\right]\right\}^{1/2}$	4(c): 0.267 μm, 0.305 μm at 80 K; 5(d): 0.691 μm (0 V), 2.48 μm (8 V)	4(c), 5(d)
Split of W	$x_n = W\dfrac{N_a}{N_a+N_d}$, $x_p = W\dfrac{N_d}{N_a+N_d}$	$N_a x_p = N_d x_n$	4(c), 5(d)
★ **Peak field**	$\lvert E_{max}\rvert = \dfrac{eN_d x_n}{\epsilon_s} = \dfrac{2(V_{bi}+V_R)}{W}$	4(c): 6.13×10^4 V/cm; 5(d): 1.94 and 6.98×10^4 V/cm	4(c), 5(d)
★ **Law of the junction**	$p_n(x_n) = p_{n0}e^{eV_a/kT}$, $n_p(-x_p) = n_{p0}e^{eV_a/kT}$		5(b)
★ **Ideal diode**	$J = \left[\dfrac{eD_p p_{n0}}{L_p} + \dfrac{eD_n n_{p0}}{L_n}\right]\left[\exp\left(\dfrac{eV_a}{kT}\right) - 1\right]$	$L_p = \sqrt{D_p\tau_{p0}}$, $L_n = \sqrt{D_n\tau_{n0}}$	5(b)
Current components	$J_p(x) = J_p(x_n)e^{(x_n-x)/L_p}$, $J_n(x) = J - J_p(x)$	minority current decays, majority current makes up the rest	5(b)

Table 6: PN junction

7. MOSFET and BJT:

Title	Equation	Notation	Ref.
Fermi potential	$\phi_{fp} = V_t \ln\dfrac{N_a}{n_i}$	$N_A = 5\times10^{18}$: 0.508 V	6(a)
★ **Maximum depletion**	$x_{dT} = \left(\dfrac{4\epsilon_s\phi_{fp}}{eN_a}\right)^{1/2}$, $\lvert Q'_{SD}(max)\rvert = eN_a x_{dT}$	16.2 nm, 1.30×10^{-6} C/cm^2	6(a)
★ **Oxide field limit**	$\epsilon_{ox}E_{ox} = \lvert Q'_{SD}\rvert + \lvert Q'_I\rvert$	$\epsilon_{ox} = 3.9\epsilon_0$; $\lvert Q'_I\rvert = 7.73\times10^{-7}$ C/cm^2, 1208 electrons	6(a)

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Title	Equation	Notation	Ref.
★ **Threshold voltage**	$V_T = V_{FB} + 2\phi_{fp} + \frac{\lvert Q'_{SD}(max)\rvert}{C_{ox}}$, $C_{ox} = \frac{\epsilon_{ox}}{t_{ox}}$	$V_T = 2.39$ V	6(a)
Gate voltage above threshold	$V_G = V_T + \frac{\lvert Q'_I\rvert}{C_{ox}} = V_{FB} + 2\phi_{fp} + E_{ox}t_{ox}$	$V_G = 2.62$ V	6(a)
★ **BJT modes**	active: $V_{BE} > 0$, $V_{BC} < 0$; saturation: both > 0; cutoff: both < 0; inverse: $V_{BE} < 0$, $V_{BC} > 0$	$V_{BC} = V_{BE} - V_{CE}$; saturation also when $I_C < \beta I_B$	6(b)
★ **Emitter injection efficiency**	$\gamma = \frac{J_{nE}}{J_{nE} + J_{pE}} \approx \frac{1}{1 + \frac{N_B}{N_E}\frac{D_E}{D_B}\frac{x_B}{x_E}}$	higher base doping lowers γ	6(d)

Table 7: MOSFET and BJT

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8. Key formulas:

The formulas marked ★ in the sheet, on one page. Each title links back to its row.

Title	Equation
Atoms per fcc cell	$n = 8 \times \frac{1}{8} + 6 \times \frac{1}{2} = 4$
Volume density	$\text{density} = \frac{n}{a^3}$
Mean speed	$\bar{v} = \sqrt{\frac{8kT}{\pi m^*}}$
Fermi-Dirac function	$f_F(E) = \frac{1}{1 + \exp\left(\frac{E - E_F}{kT}\right)}$
Density of states	$g_c(E) = \frac{4\pi(2m_n^*)^{3/2}}{h^3}\sqrt{E - E_c}$
Electron concentration	$n_0 = N_c \exp\left[-\frac{E_c - E_F}{kT}\right]$
Mass-action law	$n_0 p_0 = n_i^2 = N_c N_v \exp\left(-\frac{E_g}{kT}\right)$
Band gap from n_i	$E_g = kT \ln\left(\frac{N_c N_v}{n_i^2}\right)$, $n_i^2 = n_0(n_0 - N_d)$
Fermi level	$E_F - E_{Fi} = kT \ln\frac{n_0}{n_i}$, $E_{Fi} - E_F = kT \ln\frac{p_0}{n_i}$
Charge neutrality	$n_0 + N_a^- = p_0 + N_d^+$
Electrons	$n_0 = \frac{N_d - N_a}{2} + \sqrt{\left(\frac{N_d - N_a}{2}\right)^2 + n_i^2}$
Holes	$p_0 = \frac{N_a - N_d}{2} + \sqrt{\left(\frac{N_a - N_d}{2}\right)^2 + n_i^2}$
Mobility	$\mu = \frac{e\tau}{m^*}$, $v_d = \mu E$
Two scattering mechanisms	$\mu_L \propto T^{-3/2}$, $\mu_I \propto \frac{T^{3/2}}{N_I}$, $\frac{1}{\mu} = \frac{1}{\mu_L} + \frac{1}{\mu_I}$
Conductivity	$\sigma = e(\mu_n n + \mu_p p)$
Hall voltage	$V_H = E_H W$, $V_H = -\frac{I_x B_z}{ned}$
Dielectric relaxation	$\rho(t) = \rho(0)e^{-t/\tau_d}$, $\tau_d = \frac{\epsilon}{\sigma}$
Ambipolar transport equation	$D'\frac{\partial^2(\delta n)}{\partial x^2} + \mu' E\frac{\partial(\delta n)}{\partial x} + g - R = \frac{\partial(\delta n)}{\partial t}$
Ambipolar coefficients	$D' = \frac{D_n D_p(n + p)}{D_n n + D_p p}$, $\mu' = \frac{\mu_n \mu_p (p - n)}{\mu_n n + \mu_p p}$
Uniform decay and build-up	$\delta p(t) = \delta p(0)e^{-t/\tau_{p0}}$, $\delta p(t) = g'\tau_{p0}(1 - e^{-t/\tau_{p0}})$

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Title	Equation
Steady state, source at $x = 0$	$\delta n(x) = \delta n(0) e^{-\lvert x\rvert/L_n}$, $L_n = \sqrt{D_n \tau_{n0}}$
Quasi-Fermi levels	$E_{Fn} - E_{Fi} = kT \ln \dfrac{n_0 + \delta n}{n_i}$, $E_{Fi} - E_{Fp} = kT \ln \dfrac{p_0 + \delta p}{n_i}$
Built-in potential	$V_{bi} = \dfrac{kT}{e} \ln \left(\dfrac{N_a N_d}{n_i^2} \right)$
Depletion width	$W = \left\{ \dfrac{2\epsilon_s (V_{bi} + V_R)}{e} \left[\dfrac{N_a + N_d}{N_a N_d} \right] \right\}^{1/2}$
Peak field	$\lvert E_{max} \rvert = \dfrac{e N_d x_n}{\epsilon_s} = \dfrac{2(V_{bi} + V_R)}{W}$
Law of the junction	$p_n(x_n) = p_{n0} e^{eV_a/kT}$, $n_p(-x_p) = n_{p0} e^{eV_a/kT}$
Ideal diode	$J = \left[\dfrac{e D_p p_{n0}}{L_p} + \dfrac{e D_n n_{p0}}{L_n} \right] \left[\exp \left(\dfrac{eV_a}{kT} \right) - 1 \right]$
Maximum depletion	$x_{dT} = \left(\dfrac{4\epsilon_s \phi_{fp}}{e N_a} \right)^{1/2}$, $\lvert Q'_{SD}(max) \rvert = e N_a x_{dT}$
Oxide field limit	$\epsilon_{ox} E_{ox} = \lvert Q'_{SD} \rvert + \lvert Q'_I \rvert$
Threshold voltage	$V_T = V_{FB} + 2\phi_{fp} + \dfrac{\lvert Q'_{SD}(max) \rvert}{C_{ox}}$, $C_{ox} = \dfrac{\epsilon_{ox}}{t_{ox}}$
BJT modes	active: $V_{BE} > 0$, $V_{BC} < 0$; saturation: both > 0; cutoff: both < 0; inverse: $V_{BE} < 0$, $V_{BC} > 0$
Emitter injection efficiency	$\gamma = \dfrac{J_{nE}}{J_{nE} + J_{pE}} \approx \dfrac{1}{1 + \dfrac{N_B}{N_E} \dfrac{D_E}{D_B} \dfrac{x_B}{x_E}}$

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9. Symbol index:

Every symbol the sheet defines, in the order it first appears. The last column links to that row.

Symbol	Meaning	First used in
a	lattice constant in cm	Volume density
m	effective mass of the carrier	Maxwell-Boltzmann speeds
ϕ_m	metal work function, χ = electron affinity (4.01 V for Si), $e\phi_n = E_c - E_F$	Schottky barrier
A^*	effective Richardson constant	Thermionic emission
τ	mean time between collisions	Mobility
F_p^+	hole flux	Continuity
τ_{p0}	minority carrier lifetime	Uniform decay and build-up
N_t	trap density; $R < 0$ means net generation	SRH recombination

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